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Curso de ROS 2

Apuntes de clase

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List of Acronyms and Special Terms

Acronyms

A

AMR Autonomous Mobile Robot. — Pages: [117–119](#), [125](#), [170](#), [171](#), [173](#), [174](#), [176](#), [183](#), [184](#), [188](#), [192](#).

D

DDS Data Distribution Service. — Pages: [207–210](#), [212–215](#).

R

ROS Robot Operating System. — Pages: [1–6](#), [8](#), [9](#), [13–43](#), [45–49](#), [51–64](#), [66–68](#), [70](#), [81–86](#), [89](#), [90](#), [93](#), [94](#), [96](#), [97](#), [99](#), [100](#), [102–105](#), [107–109](#), [120](#), [126](#), [130](#), [135–138](#), [141](#), [142](#), [145–147](#), [149](#), [150](#), [156](#), [157](#), [159](#), [162](#), [164](#), [165](#), [168–173](#), [175–177](#), [180](#), [187](#), [188](#), [190–194](#), [196–198](#), [200](#), [201](#), [203](#), [207–209](#), [212](#).

RViz ROS Visualization. — Pages: [107](#), [108](#), [119–125](#), [135](#), [136](#), [138](#), [141](#), [142](#), [169–171](#), [173](#), [174](#), [176](#), [178](#), [184](#), [186–189](#), [192](#), [206](#).

S

SLAM Simultaneous Localization and Mapping. — Pages: [175–177](#), [186–192](#), [203](#), [206](#).

T

TF Transform. — Pages: [141](#), [180](#), [182](#), [187](#), [189](#), [190](#).

U

UAV Unmanned Aerial Vehicle. — Pages: [117](#), [125](#), [149](#), [156](#), [170](#), [171](#), [173](#), [174](#).

UDP User Datagram Protocol. — Pages: [208](#), [212](#), [214](#).

URDF Unified Robot Description Format. — Pages: [107–121](#), [134](#), [136](#), [137](#), [156](#), [164](#), [171](#), [178](#), [183](#), [184](#), [189](#), [190](#).

V

VS Code Visual Studio Code. — Pages: [7](#), [8](#), [107](#), [109](#), [126](#), [130](#).

CHAPTER

1

ROS 2 Environment Setup and Development Tools

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🌱 **Abstract** - This chapter provides an introduction to the [Robot Operating System \(ROS\) 2](#), covering its basic concepts, installation process, and configuration of the development environment. It also includes practice exercises to help students familiarize themselves with [ROS 2](#) and verify their setup by running some example nodes. By the end of this chapter, students will have a functional [ROS 2](#) environment ready for development and experimentation in robotics applications.

1.1 Introduction to ROS 2

1.1.1 Basic Concepts

[ROS](#) is not an actual operating system but a flexible *framework* for writing robot software. It provides a collection of software libraries, tools, and conventions to simplify the task of creating complex and robust robot applications. From drivers and state-of-the-art algorithms to powerful developer tools, [ROS](#) has the open source tools you need for your next robotics project. Some of its main advantages include:

- **Multi-language:** While [ROS](#) supports multiple programming languages, the primary supported languages are C++ and Python. Support for other languages like Java exists but is less common.

- **Free and open-source:** ROS 1 and ROS 2 are available on Ubuntu, with ROS 2 also supporting Windows and macOS. However, the availability and stability can vary across different versions and platforms.
- **Support:** Since ROS was started in 2007, a lot has changed in the robotics and ROS community. The ROS 2 project began in 2015 to address the evolving needs of the robotics community, building upon the strengths of ROS 1 and introducing improvements for better performance, security, and support for real-time systems.

1.1.2 Some ROS Distributions

ROS is released as distributions (or "distros"), with several versions supported concurrently. Some releases come with long-term support (LTS), meaning they are more stable and have undergone extensive testing, while other distributions are newer, have shorter lifetimes, and support more recent platforms and package versions. Generally, a new ROS distro is released every year on [World Turtle Day](#), with LTS releases appearing in even-numbered years. ROS is available for different versions of Ubuntu and other operating systems. Some of the notable distributions include:

- **Indigo Igloo (ROS 1):** Ubuntu 14.04 (Trusty Tahr)
- **Kinetic Kame (ROS 1):** Ubuntu 16.04 (Xenial Xerus)
- **Melodic Morenia (ROS 1):** Ubuntu 18.04 (Bionic Beaver)
- **Noetic Ninjemys (ROS 1):** Ubuntu 20.04 (Focal Fossa)
- **Foxy Fitzroy (ROS 2):** Ubuntu 20.04 (Focal Fossa)
- **Humble Hawksbill (ROS 2):** Ubuntu 22.04 (Jammy Jellyfish)
- **Jazzy Jalisco (ROS 2):** Ubuntu 24.04 (Noble Numbat)
- **Kilted Kaiju (ROS 2):** Ubuntu 24.04 (Noble Numbat)

For more details, see the [ROS Distributions list](#). Currently, the Humble Hawksbill, Jazzy Jalisco, and Kilted Kaiju distributions are actively supported, according to the [ROS 2 release schedule](#).

1.1.3 ROS Distribution for the Course



Figure 1.1: ROS 2 Humble Hawksbill Logo.

For this course, we will use **ROS 2 Humble Hawksbill**, a long-term support (LTS) release that offers enhanced performance, security, and real-time capabilities. Its logo, shown in Figure 1.1, reflects its distinctive branding. ROS 2 Humble Hawksbill is compatible with Ubuntu 22.04 (Jammy Jellyfish) and Windows 10, making it a versatile choice for both simulation and deployment on physical hardware. This distribution will serve as the foundation for all class exercises and, in particular, for project development.

1.2 Installation of Ubuntu 22 and ROS 2 Humble Hawksbill

1.2.1 Installing Ubuntu 22.04.5 LTS (Jammy Jellyfish)

1. **Download the Universal USB Installer**

Visit the [Uptodown website](#) to download the [Universal USB Installer](#), version **2.0.1.4**, on [Windows](#).

2. **Download the Ubuntu ISO**

Download the Ubuntu 22.04.5 LTS (Jammy Jellyfish) ISO - *64-bit PC (AMD64) desktop image*, from the official release page: [Ubuntu releases](#).

3. **Create a Bootable USB Drive**

Insert a USB drive (minimum 8 GB) and use the Universal USB Installer tool with the downloaded ISO to create a bootable USB stick. Follow the on-screen prompts to properly set up the drive. You may also follow the instructions provided by your instructor, the official guide: [Universal USB Installer](#), or refer to this tutorial: [YouTube Guide](#).

4. **Prepare Your Machine (if dual-booting with Windows)**

If you are keeping Windows and installing Ubuntu alongside it, you should free some unallocated disk space before installation:

- Defragment the **C:** drive.
- Additionally, you can shrink the Windows partition to create at least 32 GB of unallocated space (advanced installation).
- If the volume reduction fails, delete old restore points to free up additional space. See, for example, this tutorial: [Windows - Shrinking hard disk volume to create new partition](#).

5. **Install Ubuntu on Your Machine**

Boot the target computer from the USB drive. Follow the Ubuntu installation wizard to install Ubuntu 22.04 LTS and configure your system settings as needed:

- Select language and keyboard layout.
- Connect to a network (optional during installation).
- Choose installation type (*Normal* or *Minimal*).
- Allocate disk space (at least 64 GB) if you want to *Install Ubuntu alongside Windows* or select *Erase disk and install Ubuntu*.
- Create a user account, password, and set the hostname.

1.2.2 First Steps After Installing Ubuntu 22

Open a terminal from the application menu by typing `terminal` or by pressing `Ctrl+Alt+T`, and then:

- Update the system:

```
$ sudo apt update && sudo apt upgrade -y
```

- Check the installed Python version:

```
$ python3 --version
```

1.2.3 Installing ROS 2 Humble Hawksbill

Recommended Method: Using Debian Packages

The official [ROS 2 Humble Hawksbill documentation](#) recommends installing from deb packages for a stable and straightforward setup. Follow the installation instructions on the official [ROS 2 Humble Hawksbill Installation Guide](#). This method installs pre-built binaries and generally covers all core dependencies needed for running ROS 2.

Instructions:

Set locale

Make sure you have a locale which supports UTF-8. The following settings are tested, although any UTF-8 supported locale should work.

```
$ locale # check for UTF-8

$ sudo apt update && sudo apt install locales
$ sudo locale-gen en_US en_US.UTF-8
$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
$ export LANG=en_US.UTF-8

$ locale # verify settings
```

Setup Sources

You will need to add the [ROS 2 apt repository](#) to your system. First, ensure that the Ubuntu Universe repository is enabled.

```
$ sudo apt install software-properties-common
$ sudo add-apt-repository universe
```

Now, add the [ROS 2 GPG key](#) with apt:

```
$ sudo apt update && sudo apt install curl -y
$ sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
  ↳ /usr/share/keyrings/ros-archive-keyring.gpg
```



Then, add the repository to your sources list:

```
$ echo "deb [arch=$(dpkg --print-architecture)
→ signed-by=/usr/share/keyrings/ros-archive-keyring.gpg]
→ http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME)
→ main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
```

Install ROS 2 Packages

Update your apt repository caches after setting up the repositories:

```
$ sudo apt update
```

ROS 2 packages are built on frequently updated Ubuntu systems. It is always recommended to ensure your system is up to date before installing new packages:

```
$ sudo apt upgrade
```

⚠ Warning:

Due to early updates in Ubuntu 22.04, it is important that `systemd` and `udev`-related packages are updated before installing ROS 2. Installing ROS 2 dependencies on a freshly installed system without upgrading can trigger the removal of critical system packages. Please refer to [ros2/ros2#1272](#) and [Launchpad #1974196](#) for more information.

Desktop Install (Recommended), including ROS, RViz, demos, and tutorials:

```
$ sudo apt install ros-humble-desktop
```

Development Tools, including compilers and other tools to build ROS packages:

```
$ sudo apt install ros-dev-tools
```

Environment Setup – Sourcing the Setup Script

Set up your environment by sourcing the following file (replace `.bash` with your shell if not using bash):

```
$ source /opt/ros/humble/setup.bash
```

Alternative: Building from Source

If you require customizations or intend to develop complex packages, you can also install ROS 2 Humble Hawksbill from source. Note that building from source may require additional dependency installations (similar to ROS 1). For most beginners and for the purposes of this course, the deb package installation is recommended.

Sourcing the Setup Script

After installing ROS 2, source the setup script to ensure your environment is correctly configured:

```
$ source /opt/ros/humble/setup.bash
```



Optionally, add the sourcing command to your shell's initialization file (e.g., `.bashrc`) so that the ROS environment variables are automatically loaded every time a new terminal session is opened (replace `.bash` with your shell if not using bash):

```
$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
$ source ~/.bashrc
```

1.3 Try Some Examples

1.3.1 Talker-Listener

If you installed `ros-humble-desktop`, try some examples. In one terminal, source the setup file and run a C++ talker:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_cpp talker
```

In another terminal, source the setup file and run a Python listener:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_py listener
```

You should see the talker publishing messages and the listener confirming reception, verifying that both the C++ and Python APIs are working properly.

1.3.2 Turtlesim: Publish and Move a Turtle

Another example you can try is the `turtlesim` package, which demonstrates basic ROS 2 communication using a simulated turtle.

First, ensure that the package is installed:

```
$ sudo apt install ros-humble-turtlesim
```

Then, in a new terminal, launch the turtlesim node:

```
$ ros2 run turtlesim turtlesim_node
```

Next, open another terminal, use the `teleop` node to control the turtle with your keyboard:

```
$ ros2 run turtlesim turtle_teleop_key
```

Now, pressing the arrow keys will move the turtle in the corresponding direction. This example demonstrates publishing velocity commands to a topic that the turtlesim node subscribes to, allowing interaction with the simulated environment.

1.4 colcon Configuration

`colcon` is the command-line tool used in ROS 2 to build sets of packages in a workspace. It automatically detects packages (via `package.xml`), resolves dependencies, and builds them (often in parallel) to simplify the development process. `colcon` also supports options like `--symlink-install` for faster iterative development.



1.4.1 Installing `colcon`

Update your package list and ensure that the common `colcon` extensions are installed. Although these packages are usually installed as part of the `ros-humble-desktop` and `ros-dev-tools` installations, it is good practice to verify their presence by running the following commands:

```
$ sudo apt update
$ sudo apt install python3-colcon-common-extensions
```

1.4.2 Enabling `colcon` Argument Completion

To simplify command usage with `colcon`, add the argument completion script to your shell initialization file (e.g., `.bashrc`). Execute the following commands:

```
$ echo "source /usr/share/colcon_argcomplete/hook/colcon_argcomplete.bash" >> ~/.bashrc
$ source ~/.bashrc
```

You can verify that the sourcing command was added by viewing your `.bashrc` file:

```
$ cat ~/.bashrc
```

1.5 Configuring the Development Environment

With the following tools and extensions, your development environment will be well-equipped for Python projects.

1.5.1 Installing Visual Studio Code

Visual Studio Code (VS Code) is a versatile code editor that supports various programming languages and development environments. To install it on Ubuntu 22.04, follow these steps:

1. Using the Ubuntu Software Center

- Open the *Ubuntu Software* application from the application menu.
- In the search bar, type **Visual Studio Code**.
- Locate **Visual Studio Code** in the search results and click *Install*.

2. Using the Official `.deb` Package

- Visit the official **VS Code** download page: code.visualstudio.com.
- Click on the `.deb` package suitable for Debian/Ubuntu.
- Once downloaded, open a terminal (**Ctrl+Alt+T**) and navigate to the directory containing the downloaded file.
- Install the package using:

```
$ sudo apt install ./<file>.deb
```

Replace `<file>` with the actual filename.

These methods ensure that **VS Code** is added to your system repositories and receives updates automatically.



1.5.2 Installing Recommended VS Code Extensions

Enhance your development experience by installing the following [VS Code](#) extensions:

- **C++** by *Microsoft* – Offers C++ IntelliSense, debugging, and code browsing.
- **CMake** by *twxs* – Simplifies working with CMake projects.
- **CMake Tools** by *Microsoft* – Provides CMake project integration.
- **Error Lens** by *Alexander* – Displays inline error messages in the code editor.
- **Indent-Rainbow** by *oderwat* – Highlights indentation levels with different colors.
- **Python** by *Microsoft* – Provides rich support for Python, including features such as IntelliSense, linting, and debugging.
- **Robotics Developer Environment** by *Ranch Hand Robotics LLC* – Helps build robotics solutions targeting the [ROS 2](#).
- **URDF** by *smilerobotics* – Adds syntax highlighting and support for URDF files.
- **URDF Visualizer** by *morningfrog* – Allows to preview URDF files within the editor.
- **vscode-icons** by *VSCode Icons Team* – Adds file icons for better visual identification.
- **XML** by *Red Hat* – Enhances XML editing capabilities.
- **XML Tools** by *Josh Johnson* – Offers additional functionalities for XML files.

To install these extensions:

1. Open VS Code from the application menu by typing `code`.
2. Navigate to the *Extensions* view by clicking on the square icon (📦) in the sidebar or pressing `Ctrl+Shift+X`.
3. Search for each extension by name and click *Install*.

1.5.3 Installing Terminator for Multiple Terminals

Terminator is a terminal emulator that allows splitting the window into multiple terminals, facilitating simultaneous operations. To install Terminator:

1. Open a terminal.
2. Update the package list:

```
$ sudo apt update
```

3. Install Terminator:

```
$ sudo apt install terminator
```

4. Launch Terminator from the applications menu, by typing `terminator` in the terminal, or by pressing `Ctrl+Alt+T`.

Installing Terminator may have set it as the default, so when you press `Ctrl+Alt+T` it opens Terminator instead of GNOME Terminal (or your original terminal). To achieve your goal, you can change the default and then assign Terminator to a different shortcut, as described in the next.



Reset the Default Terminal Emulator

Ubuntu uses the alternative system for the “x-terminal-emulator.” You can reconfigure it so that pressing **Ctrl+Alt+T** launches your original terminal (for example, GNOME Terminal):

1. **Open a terminal** (if Terminator is the only one available, you can temporarily launch it).
2. Run the command:

```
$ sudo update-alternatives --config x-terminal-emulator
```

3. You will see a list of installed terminal emulators. Type the number corresponding to your original terminal (e.g., `/usr/bin/gnome-terminal.wrapper`) and press **Enter**.

This sets the default terminal emulator that the system shortcut uses.

Create a Custom Shortcut for Terminator

Next, if you want a separate shortcut (like **Ctrl+Alt+E**) that launches Terminator:

1. **Open Settings** in your Ubuntu desktop.
2. Navigate to **Keyboard** (or **Keyboard Shortcuts**).
3. Scroll down to or click on **Custom Shortcuts**.
4. Click the **Add Shortcut** button.
5. Set a name (e.g., "Terminator") and for the command enter:

```
terminator
```

6. Assign the new shortcut to **Ctrl+Alt+E** (click the shortcut key field and press the keys).
7. Save your new shortcut.

Now:

- **Ctrl+Alt+T** will launch your default terminal (GNOME Terminal),
- **Ctrl+Alt+E** will launch Terminator.

This way, you keep your usual terminal for everyday tasks and have Terminator available on a separate key combination for when you need its advanced features.

1.6 Practice Assignment: Running ROS 2 Examples

In this assignment, you will run the ROS 2 examples presented in Section 1.3 on your computer and capture evidence of successful execution using a screenshot. Submit your evidence in the designated assignment area on Canvas (within the ROS 2 module) as a PNG file. The filename must follow this format:

`FirstnameLastname_evidence_sX.png`,

where `sX` indicates the session number, for example:

`EduardoDavila_evidence_s1.png`.



1.6.1 Examples to Try

Talker-Listener

After installing `ros-humble-desktop` (and optionally `Terminator`), try the following:

- In **Terminal 1**, source the setup file and run a C++ talker:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_cpp talker
```

- In **Terminal 2**, source the setup file and run a Python listener:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_py listener
```

You should observe that the talker publishes messages and the listener confirms their reception.

Turtlesim: Publish and Move a Turtle

Another example is provided by the `turtlesim` package:

- In **Terminal 3**, run the turtlesim node:

```
$ ros2 run turtlesim turtlesim_node
```

- In **Terminal 4**, run the teleoperation node to control the turtle:

```
$ ros2 run turtlesim turtle_teleop_key
```

Use the arrow keys to move the turtle. This example demonstrates publishing velocity commands to control the simulated turtle.

1.6.2 Submission Instructions

1. Run the examples as described above.
2. Capture a **screenshot showing the terminal output** where the examples are running successfully (see Figure 1.2 for an example).
3. Save the screenshot in PNG format.
4. Name the file according to the format: `FirstnameLastname_evidence_sX.png` (replace `X` with the session number).
5. Submit your screenshot file as instructed by the course guidelines.

Note: Your machine's username and device name must be visible in the screenshot to verify the authenticity of the submission, as shown in Figure 1.2. Evidence that appears copied, unclear, or altered will be considered invalid and may result in a score of 0 for this assignment.



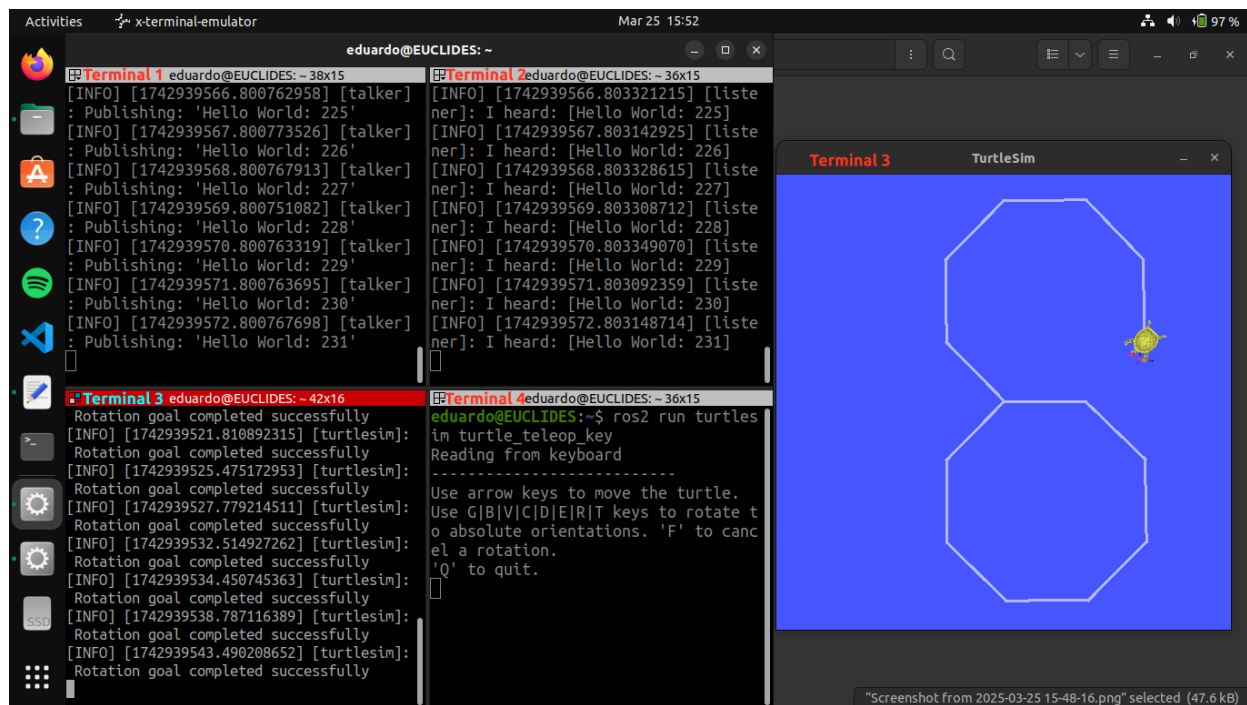


Figure 1.2: Example screenshot showing terminal outputs from both the Talker-Listener example and the Turtlesim node example in action.

CHAPTER

2

ROS 2 Workspace, Package, and Node Development: Publisher and Subscriber

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✳ **Abstract** - In this chapter, we will cover the fundamental concepts of [Robot Operating System \(ROS\) 2](#), including the creation and architecture of workspaces and packages, as well as the development of basic publisher nodes in both C++ and Python. We will explore the structure of [ROS 2](#) packages, the role of nodes, and how they communicate through topics. By following step-by-step instructions, you will learn how to set up your development environment, create a workspace, and implement simple publisher nodes that can send messages to other nodes in the [ROS 2](#) ecosystem. This chapter serves as a foundation for building more complex robotic applications using [ROS 2](#). All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) , located in the `ros2_ws/src` folder, corresponding to the [ROS 2](#) packages with the `s2` prefix.

2.1 Fundamental Concepts of ROS 2

[ROS 2](#) is a [middleware](#) based on a strongly-typed, anonymous publish/subscribe mechanism that enables message passing between different processes. At the core of any [ROS 2](#) system is the [ROS graph](#), which represents the network of nodes and the connections through which they communicate. This section introduces the fundamental concepts necessary to understand the basics of [ROS 2](#).

2.1.1 Workspace

A [ROS 2 workspace](#) is a directory that contains one or more [ROS 2](#) packages. It serves as the working environment for building, developing, and testing [ROS 2](#) applications.

Best practices recommend creating a separate directory for each workspace, with a meaningful name that reflects its purpose. Inside the workspace, it is common to include a `src` folder to store [ROS 2](#) packages, ensuring an organized structure.

2.1.2 Packages

A [ROS 2 package](#) is the fundamental unit of software organization in [ROS 2](#). It provides a structured way to manage code, facilitating its distribution, installation, and reuse. A package may contain:

- Nodes
- Libraries
- Configuration files
- Launch files
- Other resources necessary for execution

A single workspace can contain multiple packages, even if they use different build types (e.g., CMake, Python). However, packages cannot be nested within each other.

[ROS 2](#) uses `ament` as its build system and `colcon` as its build tool. Packages can be created using either CMake or Python, each with minimum required contents, as detailed in Sections [2.3](#) and [2.4](#), respectively.

2.1.3 Nodes

A [ROS 2 node](#) is a fundamental component that represents a single process performing computation. Nodes communicate with each other through:

- [Topics](#) for continuous data exchange.
- [Services](#) for request-response interactions.
- [Actions](#) for long-running tasks with feedback.

Each node is part of the [ROS graph](#) and can communicate with other nodes within the same process, across different processes, or even across multiple machines. Nodes are designed to be modular, meaning that each node should focus on a single logical task. A node can simultaneously act as:

- A [publisher](#) for sending messages.
- A [subscriber](#) for receiving messages.
- A [service client](#) for requesting a computation.
- A [service server](#) for providing a computation.
- An [action client](#) for initiating long-running tasks with feedback.
- An [action server](#) for executing long-running tasks with periodic feedback.

Connections between nodes are established dynamically, allowing for a modular and scalable system design.



2.1.4 Topics

Topics are a core mechanism in ROS 2 for message-based communication between nodes. They are used for continuous data exchange, such as transmitting sensor readings, robot state updates, or other real-time information.

ROS 2 follows a **publish/subscribe** model for topics, meaning:

- **Publishers** send data to a topic.
- **Subscribers** receive data from a topic.
- Multiple publishers and subscribers can exist on the same topic.
- Communication is **anonymous**, so subscribers can receive messages without knowing which publisher transmitted the data.

Publish/Subscribe Model

The publish/subscribe system allows for flexible and decoupled communication:

- Publishers and subscribers communicate via a shared **topic name**.
- Multiple publishers and subscribers can exist on a topic simultaneously.
- When a publisher sends data, all subscribers of that topic receive it.

This architecture is often compared to a **bus system** in electrical engineering, where multiple devices can share a common communication channel.

Anonymous Communication

ROS 2 implements **anonymous communication**, meaning:

- A subscriber does not need to know which publisher sent a message.
- Publishers and subscribers can be dynamically replaced without affecting the system.
- Debugging and monitoring tools (e.g., **ros2bagrecord**) can subscribe to topics without interrupting existing communication.

Strongly-Typed Messages

ROS 2 enforces **strong typing** in its publish/subscribe system to ensure data consistency and integrity. This guarantees:

1. **Strict Data Types:** Each field in a message adheres to a predefined data type.

```
uint32 field1
string field2
```

In this example, **field1** must always be an unsigned 32-bit integer, and **field2** must always be a string.

2. **Well-Defined Semantics:** Message contents adhere to clear conventions. For example, an IMU message includes a 3-dimensional angular velocity vector, where each component is explicitly defined in radians per second. This ensures consistency and prevents misinterpretation of the transmitted data.

2.2 Workspace Creation and Setup

A ROS 2 workspace is a directory that contains one or more ROS 2 packages, as described in Section 2.1. When you build your workspace using `colcon`, it automatically creates the `build`, `install`, and `log` directories. This structure helps manage dependencies and package configurations efficiently, ensuring a clean and organized development environment.

2.2.1 Creating the Workspace

After installing ROS 2, set up your workspace by creating a directory (e.g., `~/ros2_ws`) and adding a `src` folder:

```
$ mkdir -p ros2_ws/src
```

The best practice is to create and organize your packages inside the `src` folder to keep the workspace structure clean and maintainable.

2.2.2 Building the Workspace

Navigate to the workspace directory:

```
$ cd ros2_ws
```

Build the workspace using `colcon`:

```
$ colcon build
```

For development purposes, you might consider using the `--symlink-install` option. This allows changes in source files to take effect immediately without requiring a full rebuild:

```
$ colcon build --symlink-install
```

If the build completes successfully, you will see the following confirmation message: `'colcon build' successful`, and the `build`, `install`, and `log` directories will be created within your workspace.

Note:

The `--symlink-install` option only creates symbolic links to the source files from Python packages. Therefore, changes to source files from C++ packages will still require a rebuild to reflect in the installed binaries.

2.2.3 Sourcing the Workspace Environment

To overlay your workspace on top of the system installation and load the environment variables for your newly built packages, run:

```
$ source ./install/setup.bash
```

Note: Unlike the ROS 2 and `colcon` setup scripts, this sourcing command is not permanently added to your shell's initialization file. For learning purposes, you must run it in each new terminal session where you work with your ROS 2 workspace.

For more details on `colcon` and workspace management, refer to the [ROS 2 Colcon](#) tutorial.



2.3 Creating a ROS 2 Package for C++ Nodes

2.3.1 Navigating to the `src` Directory

Begin by opening a terminal and navigating to the `src` directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ~/ros2_ws/src
```

2.3.2 Creating the C++ Package

Use the `ros2 pkg create` command to create a new C++ package. Replace `s2_cpp_pubsub` with your desired package name and provide an appropriate description:

```
$ ros2 pkg create s2_cpp_pubsub --build-type ament_cmake --dependencies rclcpp std_msgs
↳ --license Apache-2.0 --description "Your package description here"
```

Let's break down the components of this command:

- `s2_cpp_pubsub`: Specifies the name of the new package.
- `--build-type ament_cmake`: Indicates that the package uses C++ and will be built with `ament_cmake`.
- `--dependencies rclcpp std_msgs`: Lists the package dependencies, in this case, `rclcpp` (the ROS 2 C++ API) and `std_msgs` (standard message definitions).
- `--license Apache-2.0`: Sets the license type for the package.
- `--description "Your package description here"`: Provides a brief description of the package.

2.3.3 Building the Package

After creating the package, navigate back to the root of your workspace and build the package using `colcon`:

```
$ cd ~/ros2_ws
$ colcon build --packages-select s2_cpp_pubsub
```

Alternatively, to build all packages in the workspace:

```
$ colcon build
```

2.3.4 Examining the Package Contents

Navigate to the newly created package directory, `s2_cpp_pubsub`, and list its contents:

```
$ cd src/s2_cpp_pubsub
$ ls
```

You should see the following key files and directories:

- `CMakeLists.txt`: Contains instructions for building the code within the package using CMake.



- `include/`: Directory containing the public header files for the package.
- `package.xml`: Defines metadata about the package, including its name, version, description, licenses, maintainers, authors, and dependencies.
- `src/`: Directory containing the source code files for the package.

2.3.5 Detailed Examination of Key Package Files

`package.xml`

The `package.xml` file contains essential metadata about the ROS 2 package. Key elements include:

- `<name>`: Specifies the package's name.
- `<version>`: Indicates the current version of the package.
- `<description>`: Provides a brief overview of the package's purpose.
- `<maintainer>`: Contains contact information for the package maintainer.
- `<license>`: Details the licensing information.
- `<depend>`: Lists build-time and runtime dependencies required by the package.

To examine the `package.xml` file, use the following command:

```
$ code package.xml
```

Ensure that the dependencies listed here match those specified during package creation.

`CMakeLists.txt`

The `CMakeLists.txt` file contains instructions for building the package using CMake. It specifies details such as the minimum required CMake version, project name, dependencies, include directories, source files, and targets to be built.

To inspect the `CMakeLists.txt` file, execute:

```
$ code CMakeLists.txt
```

2.3.6 Ensuring Consistency Between `package.xml` and `CMakeLists.txt`

It is crucial that the information in `package.xml` and `CMakeLists.txt` is consistent. Discrepancies between these files can lead to build or runtime issues. Ensure that details like the package name, version, description, maintainer, license, and dependencies align across both files.

By following these steps, you have created a C++-based ROS 2 package, built it, and examined its structure and configuration files. This foundational setup is essential for developing and organizing your ROS 2 C++ nodes effectively.

For more details on package creation and management, refer to the official [Creating Your First ROS 2 Package](#) tutorial.



2.4 Creating a ROS 2 Package for Python Nodes

2.4.1 Navigating to the `src` Directory

Begin by opening a terminal and navigating to the `src` directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ~/ros2_ws/src
```

2.4.2 Creating the Python Package

Use the `ros2 pkg create` command to create a new Python package. Replace `s2_py_pubsub` with your desired package name and provide an appropriate description:

```
$ ros2 pkg create s2_py_pubsub --build-type ament_python --dependencies rclpy std_msgs  
→ --license Apache-2.0 --description "Your package description here"
```

Let's break down the components of this command:

- `s2_py_pubsub`: Specifies the name of the new package.
- `--build-type ament_python`: Indicates that the package uses Python and will be built with `ament_python`.
- `--dependencies rclpy std_msgs`: Lists the package dependencies, in this case, `rclpy` (the ROS 2 Python API) and `std_msgs` (standard message definitions).
- `--license Apache-2.0`: Sets the license type for the package.
- `--description "Your package description here"`: Provides a brief description of the package.

2.4.3 Building the Package

After creating the package, navigate back to the root of your workspace and build the package using `colcon`:

```
$ cd ~/ros2_ws  
$ colcon build --packages-select s2_py_pubsub
```

Alternatively, to build all packages in the workspace:

```
$ colcon build
```

2.4.4 Examining the Package Contents

Navigate to the newly created package directory, `s2_py_pubsub`, and list its contents:

```
$ cd src/s2_py_pubsub  
$ ls
```

You should see the following key files and directories:

- `s2_py_pubsub/`: Directory with the same name as your package, used by ROS 2 tools to find your package, contains `__init__.py`.



- `package.xml`: Defines metadata about the package, including its name, version, description, licenses, maintainers, authors, and dependencies.
- `setup.py`: Script for installing and setting up the package using Python's `setuptools`.
- `setup.cfg`: Configuration file for `setuptools`.
- `resource/`: Directory containing resource files.
- `test/`: Directory designated for test files.

2.4.5 Detailed Examination of Key Package Files

`package.xml`

The `package.xml` file contains essential metadata about the ROS 2 package. Key elements include:

- `<name>`: Specifies the package's name.
- `<version>`: Indicates the current version of the package.
- `<description>`: Provides a brief overview of the package's purpose.
- `<maintainer>`: Contains contact information for the package maintainer.
- `<license>`: Details the licensing information.
- `<depend>`: Lists build-time and runtime dependencies required by the package.

To examine the `package.xml` file, use the following command:

```
$ code package.xml
```

Ensure that the dependencies listed here match those specified during package creation.

`setup.py`

The `setup.py` script utilizes `setuptools` to configure how the package is installed and structured. It specifies details such as the package name, version, author information, license, and included packages or modules.

To inspect the `setup.py` file, execute:

```
$ code setup.py
```

2.4.6 Ensuring Consistency Between `package.xml` and `setup.py`

It is crucial that the information in `package.xml` and `setup.py` is consistent. Inconsistencies may cause build failures or unexpected runtime behavior. Ensure that details like the package name, version, description, maintainer, license, and dependencies align across both files.

By following these steps, you have created a Python-based ROS 2 package, built it, and examined its structure and configuration files. This foundational setup is essential for developing and organizing your ROS 2 Python nodes effectively.

For more details on package creation and management, refer to the official [Creating a ROS 2 Package](#) tutorial.



2.5 Creating a ROS 2 Publisher Node with C++

2.5.1 Setting Up the C++ Node

Navigating to the Package Directory

Open a terminal and navigate to your C++ package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s2_cpp_pubsub/src
```

Replace `s2_cpp_pubsub` with your actual package name.

Creating the C++ Node File

Create a new C++ file for your publisher node, for example, `publisher.cpp`:

```
$ touch publisher.cpp
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `publisher.cpp` file.

2.5.2 Editing the C++ Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `publisher.cpp` file for editing.

Implementing the Node Code

Below is an example implementation of a simple ROS 2 publisher node in C++:

```
#include <chrono>
#include <functional>
#include <memory>
#include <string>

#include "rclcpp/rclcpp.hpp"
#include "std_msgs/msg/string.hpp"

using namespace std::chrono_literals;

/* This example creates a subclass of Node and uses std::bind() to register a
 * member function as a callback from the timer. */
```

```

class CppPublisher : public rclcpp::Node {
public:
    CppPublisher():
        Node("cpp_publisher_node"), count_(0) {
        RCLCPP_INFO(this->get_logger(), "C++ Publisher node has been started");
        publisher_ = this->create_publisher<std_msgs::msg::String>("cpp_str_topic",
↪ 10);
        timer_ = this->create_wall_timer(
            500ms, std::bind(&CppPublisher::timer_callback, this));
    }

private:
    void timer_callback() {
        std_msgs::msg::String message = std_msgs::msg::String();
        message.data = "Hello, this is Eduardo from the C++ Publisher: " +
↪ std::to_string(count_++);
        RCLCPP_INFO(this->get_logger(), "Publishing: '%s'", message.data.c_str());
        publisher_>publish(message);
    }
    rclcpp::TimerBase::SharedPtr timer_;
    rclcpp::Publisher<std_msgs::msg::String>::SharedPtr publisher_;
    size_t count_;
};

int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppPublisher>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```

This code defines a node that publishes a "Hello, World" message along with a counter to the topic `cpp_str_topic` every 0.5 seconds.

Breaking Down the Publisher Node Code

We now examine each part of the code in detail to understand how the ROS 2 C++ publisher node works.

1. Includes and Namespace

```

#include <chrono>
#include <functional>
#include <memory>
#include <string>

#include "rclcpp/rclcpp.hpp"
#include "std_msgs/msg/string.hpp"

using namespace std::chrono_literals;

```



- `<chrono>` and `<memory>` are used for time management and smart pointers.
- `<functional>` and `<string>` support function binding and string operations.
- `rclcpp/rclcpp.hpp` provides the ROS 2 C++ API.
- `std_msgs/msg/string.hpp` defines the standard String message type.
- The `using namespace std::chrono_literals;` statement allows the use of time literals (e.g., `500ms`) for time durations.

2. Defining the Node Class

```
class CppPublisher : public rclcpp::Node {
public:
    CppPublisher():
        Node("cpp_publisher_node"), count_(0) {
        RCLCPP_INFO(this->get_logger(), "C++ Publisher node has been started");
        publisher_ = this->create_publisher<std_msgs::msg::String>("cpp_str_topic",
→ 10);
        timer_ = this->create_wall_timer(
            500ms, std::bind(&CppPublisher::timer_callback, this));
    }
};
```

- The `CppPublisher` class inherits from `rclcpp::Node` to create a new ROS 2 node.
- `public`: This access specifier indicates that all members declared after it (until another access specifier is encountered) are publicly accessible. This means the constructor and any public methods can be accessed from outside the class.
- The constructor `CppPublisher()` initializes the node with the name `"cpp_publisher_node"` and sets the counter `count_` to 0.
- `RCLCPP_INFO(this->get_logger(), "C++ Publisher node has been started");` logs an informational message.
- A publisher is created to send `std_msgs::msg::String` messages on the `"cpp_str_topic"` topic with a queue size of 10.
- A wall timer is set up to call the `timer_callback` method every 500 milliseconds.

3. Timer Callback Method

```
private:
    void timer_callback() {
        std_msgs::msg::String message = std_msgs::msg::String();
        message.data = "Hello, this is Eduardo from the C++ Publisher: " +
→ std::to_string(count_++);
        RCLCPP_INFO(this->get_logger(), "Publishing: '%s'", message.data.c_str());
        publisher_->publish(message);
    }
    rclcpp::TimerBase::SharedPtr timer_;
    rclcpp::Publisher<std_msgs::msg::String>::SharedPtr publisher_;
    size_t count_;
};
```

- `private`: This access specifier indicates that all members declared after it are accessible only within the class itself. This encapsulation prevents external access to these members, ensuring they can only be modified or called by member functions of the class.



- The `timer_callback` method is invoked periodically by the timer. It creates a new message, sets its `data` field to include the current counter value, logs the message, and publishes it on the topic `cpp_str_topic`.
- `rclcpp::TimerBase::SharedPtrtimer_;` is a shared pointer to a timer object that triggers the callback at regular intervals.
- `rclcpp::Publisher<std_msgs::msg::String>::SharedPtrpublisher_;` is a shared pointer to a publisher that sends messages of type `std_msgs::msg::String`.
- `size_tcount_;` is a counter used to generate a unique message each time the callback is invoked.

4. The Main Function

```
int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppPublisher>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

- `rclcpp::init(argc, argv)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- `node=std::make_shared<CppPublisher>()` creates an instance of the `CppPublisher` node.
- `rclcpp::spin(node)` keeps the node active, processing callbacks (such as `timer_callback()`) until a shutdown signal is received.
- `node.reset()` releases the shared pointer to the node so that its resources can be deallocated and the destructor can be called before ROS 2 shuts down.
- `rclcpp::shutdown()` gracefully shuts down ROS 2, releasing all allocated resources when the node stops.
- `return 0;` indicates successful program termination.

This detailed breakdown explains each component of the ROS 2 C++ publisher node. For further details and additional context, refer to the [Creating ROS 2 Nodes \(C++\)](#) tutorial.

2.5.3 Integrating the Publisher Node into the Package

Modifying `CMakeLists.txt`

To enable ROS 2 to build and run your C++ node, modify the `CMakeLists.txt` file of your package. Open `CMakeLists.txt` in VS Code and add or verify the following lines:




```
# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(std_msgs REQUIRED)

add_executable(publisher_exe src/publisher.cpp)
ament_target_dependencies(publisher_exe rclcpp std_msgs)

install(TARGETS
  publisher_exe
  DESTINATION lib/${PROJECT_NAME}
)
```

Breaking down each line:

- `find_package(ament_cmake REQUIRED)` locates the ament build system.
- `find_package(rclcpp REQUIRED)` and `find_package(std_msgs REQUIRED)` ensure that the required dependencies are found.
- `add_executable(publisher_exe src/publisher.cpp)` creates an executable from your source file.
- `ament_target_dependencies(publisher_exe rclcpp std_msgs)` links the required libraries.
- The `install` command specifies where the executable should be installed.

Building the Package

After saving the changes to `CMakeLists.txt`, build your package. In the current terminal session, from the root of the workspace, compile with:

```
$ colcon build
```

Alternatively, to build only the C++ package:

```
$ colcon build --packages-select s2_cpp_pubsub
```

If the build completes successfully, you will see the following confirmation message: `'colcon build' successful`.

2.5.4 Running the Publisher Node as a Standalone Executable (Optional)

You can run your C++ publisher node outside of the ROS 2 environment. Open a new terminal (or use an additional session in Terminator) and navigate to the location where the executable was installed. This could be either:

```
$ cd build/s2_cpp_pubsub
```

or:

```
$ cd install/s2_cpp_pubsub/lib/s2_cpp_pubsub
```

as the `install` command specified, in the `CMakeLists.txt` file, where the executable should be installed. Then, run the executable:

```
$ ./publisher_exe
```



as the `add_executable` command specified, in the `CMakeLists.txt` file, how the executable should be named.

Sourcing the Environment

Before running the node, in the first (or current) terminal session from the root of the workspace, source the workspace's setup file to load the necessary environment variables:

```
$ source ./install/setup.bash
```

This step ensures that ROS 2 and the built packages are properly configured.

2.5.5 Running the ROS 2 Publisher Node with ROS 2

Execute your C++ publisher node using the `ros2run` command:

```
$ ros2 run s2_cpp_pubsub publisher_exe
```

At this point, if everything is set up correctly, your node should be actively publishing messages to the `cpp_str_topic` topic.

2.5.6 Summary of Key Identifiers

At the end of the creation and configuration of the ROS 2 C++ publisher node, it is important to distinguish between the following identifiers used:

1. `publisher.cpp`: The filename of your C++ source file.
2. `cpp_publisher_node`: The name of the node as defined in the constructor of your C++ class.
3. `publisher_exe`: The name of the executable specified in `CMakeLists.txt` under the `add_executable` command.

Ensuring these identifiers are correctly set and consistently used is crucial for the proper functioning of your ROS 2 C++ nodes.

For more details on node creation and setup in C++, refer to the official [Creating ROS 2 Nodes \(C++\)](#) tutorial.

2.6 Creating a ROS 2 Subscriber Node with C++

2.6.1 Setting Up the C++ Node

Navigating to the Package Directory

Open a terminal and navigate to your C++ package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s2_cpp_pubsub/src
```

Replace `s2_cpp_pubsub` with your actual package name.



Creating the C++ Node File

Create a new C++ file for your subscriber node, for example, `subscriber.cpp`:

```
$ touch subscriber.cpp
```

Then, list the folder contents to verify the file has been created:

```
$ ls
```

You should now see the `subscriber.cpp` file.

2.6.2 Editing the C++ Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `subscriber.cpp` file for editing.

Implementing the Node Code

Below is an example implementation of a simple ROS 2 subscriber node in C++:

```
#include <memory>
#include "rclcpp/rclcpp.hpp"
#include "std_msgs/msg/string.hpp"

using std::placeholders::_1;

class CppSubscriber : public rclcpp::Node
{
public:
    CppSubscriber()
    : Node("cpp_subscriber_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Subscriber node has been started");
        subscription_ = this->create_subscription<std_msgs::msg::String>(
            "py_str_topic", 10, std::bind(&CppSubscriber::topic_callback, this, _1));
    }

private:
    void topic_callback(const std_msgs::msg::String & msg) {
        RCLCPP_INFO(this->get_logger(), "I heard: '%s'", msg.data.c_str());
    }
    rclcpp::Subscription<std_msgs::msg::String>::SharedPtr subscription_;
};
```

```
int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppSubscriber>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

This code defines a node that subscribes to the topic `py_str_topic` and logs the messages received.

Breaking Down the Subscriber Node Code

1. Includes and Namespace

```
#include <memory>
#include "rclcpp/rclcpp.hpp"
#include "std_msgs/msg/string.hpp"

using std::placeholders::_1;
```

- `<memory>` is used for smart pointers.
- `rclcpp/rclcpp.hpp` provides the ROS 2 C++ API.
- `std_msgs/msg/string.hpp` defines the standard String message type.
- `using std::placeholders::_1;` simplifies binding the callback function.

2. Defining the Node Class

```
class CppSubscriber : public rclcpp::Node
{
public:
    CppSubscriber()
    : Node("cpp_subscriber_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Subscriber node has been started");
        subscription_ = this->create_subscription<std_msgs::msg::String>(
            "py_str_topic", 10, std::bind(&CppSubscriber::topic_callback, this, _1));
    }
}
```

- The `CppSubscriber` class inherits from `rclcpp::Node` to create a new ROS 2 node.
- `public:` makes the constructor accessible outside the class.
- The constructor `CppSubscriber()` initializes the node with the name `"cpp_subscriber_node"` and creates a subscription to the `"py_str_topic"` topic with a queue size of 10.
- The subscription's callback is bound using `std::bind`.



3. Listener Callback Method

```
private:
    void topic_callback(const std_msgs::msg::String & msg) {
        RCLCPP_INFO(this->get_logger(), "I heard: '%s'", msg.data.c_str());
    }
    rclcpp::Subscription<std_msgs::msg::String>::SharedPtr subscription_;
};
```

- `private:` restricts access to the callback and the subscription pointer, encapsulating internal details.
- The `topic_callback` method is invoked when a new message is received. It logs the content of the message.

4. The Main Function

```
int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppSubscriber>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

- `rclcpp::init(argc, argv)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- `node=std::make_shared<CppSubscriber>()` creates an instance of the `CppSubscriber` node.
- `rclcpp::spin(node)` keeps the node active, processing callbacks (such as `topic_callback()`) until a shutdown signal is received.
- `node.reset()` releases the shared pointer to the node so that its resources can be deallocated and the destructor can be called before ROS 2 shuts down.
- `rclcpp::shutdown()` gracefully shuts down ROS 2, releasing all allocated resources when the node stops.
- `return 0;` indicates successful program termination.

This detailed breakdown explains each component of the ROS 2 C++ subscriber node. For further details and additional context, refer to the [Creating ROS 2 Nodes \(C++\)](#) tutorial.

2.6.3 Integrating the Subscriber Node into the Package

Modifying `CMakeLists.txt`

To build and run your C++ subscriber node, modify the `CMakeLists.txt` file of your package. Open `CMakeLists.txt` in VS Code and add or verify the following lines, keeping any other existing ROS 2 node setup (e.g., `publisher_exe`):



```
# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(std_msgs REQUIRED)

add_executable(publisher_exe src/publisher.cpp)
ament_target_dependencies(publisher_exe rclcpp std_msgs)

add_executable(subscriber_exe src/subscriber.cpp)
ament_target_dependencies(subscriber_exe rclcpp std_msgs)

install(TARGETS
  publisher_exe
  subscriber_exe
  DESTINATION lib/${PROJECT_NAME})
)
```

Breaking down each line:

- `find_package(ament_cmakeREQUIRED)` locates the ament build system.
- `find_package(rclcppREQUIRED)` and `find_package(std_msgsREQUIRED)` ensure that the required dependencies are found.
- `add_executable(subscriber_exe src/subscriber.cpp)` creates an executable from your source file.
- `ament_target_dependencies(subscriber_exe rclcpp std_msgs)` links the required libraries.
- The `install` command specifies where the executable should be installed.

Building the Package

After updating `CMakeLists.txt`, build your package. In the current terminal session from the root of your workspace, compile with:

```
$ colcon build
```

Alternatively, to build only the C++ package:

```
$ colcon build --packages-select s2_cpp_pubsub
```

If the build completes successfully, you will see the following confirmation message: `'colcon build' successful`.

2.6.4 Running the Subscriber Node as a Standalone Executable (Optional)

You can run your C++ subscriber node outside of the ROS 2 environment. Open a new terminal (or use an additional session in Terminator) and navigate to the directory where the executable was installed. This could be either:

```
$ cd build/s2_cpp_pubsub
```

or:

```
$ cd install/s2_cpp_pubsub/lib/s2_cpp_pubsub
```



as the `install` command specified, in the `CMakeLists.txt` file, where the executable should be installed. Then, run the executable:

```
$ ./subscriber_exe
```

as the `add_executable` command specified, in the `CMakeLists.txt` file, how the executable should be named.

Sourcing the Environment

Before running the node with ROS 2, in the first (or current) terminal session from the root of your workspace, source the workspace's setup file to load the necessary environment variables:

```
$ source ./install/setup.bash
```

This ensures that ROS 2 and your built packages are properly configured.

2.6.5 Running the ROS 2 Subscriber Node with ROS 2

Execute your C++ subscriber node using the `ros2run` command:

```
$ ros2 run s2_cpp_pubsub subscriber_exe
```

At this point, if everything is set up correctly and your Python publisher node is running, your subscriber node should be actively receiving and logging messages published from the `py_str_topic` topic.

2.6.6 Summary of Key Identifiers

At the end of the creation and configuration of the ROS 2 C++ subscriber node, it is important to distinguish between the following identifiers used:

1. `subscriber.cpp`: The filename of your C++ source file.
2. `cpp_subscriber_node`: The name of the node as defined in the constructor of your C++ class.
3. `subscriber_exe`: The name of the executable specified in `CMakeLists.txt` under the `add_executable` command.

Ensuring these identifiers are correctly set and consistently used is crucial for the proper functioning of your ROS 2 C++ nodes.

For more details on node creation and setup in C++, refer to the official [Creating ROS 2 Nodes \(C++\)](#) tutorial.

2.7 Creating a ROS 2 Publisher Node with Python

2.7.1 Setting Up the Python Node

Navigating to the Package Directory

Open a terminal and navigate to your Python package directory within the ROS 2 workspace (e.g., `~/ros2_ws`):



```
$ cd ros2_ws/src/s2_py_pubsub/s2_py_pubsub
```

Replace `s2_py_pubsub` with your actual package name.

Creating the Python Node File

Create a new Python file for your publisher node, for example, `publisher.py`:

```
$ touch publisher.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see both the `__init__.py` and `publisher.py` files.

Making the Python File Executable

Ensure the new Python file is executable:

```
$ chmod +x publisher.py
```

Examine the folder contents again to confirm that `publisher.py` now appears in a different color, indicating that it is executable.

2.7.2 Editing the Python Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `publisher.py` file for editing.

Implementing the Node Code

Below is an example implementation of a simple ROS 2 publisher node in Python:

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from std_msgs.msg import String

class PyPublisher(Node):
    def __init__(self):
        super().__init__('py_publisher_node')
        self.get_logger().info("Python Publisher node has been started")
        self.publisher_ = self.create_publisher(String, 'py_str_topic', 10)
        timer_period = 0.5 # seconds
        self.timer = self.create_timer(timer_period, self.timer_callback)
        self.i = 0
```




```
def timer_callback(self):
    msg = String()
    msg.data = 'Hello, this is Eduardo from the Python Publisher: %d' % self.i
    self.get_logger().info('Publishing: "%s"' % msg.data)
    self.publisher_.publish(msg)
    self.i += 1

# To ensure that resources are properly cleaned up and that
# all intended statements are executed upon shutdown,
# you should handle the KeyboardInterrupt exception.
def main(args=None):
    rclpy.init(args=args)
    node = PyPublisher()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.destroy_node()
    finally:
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

This script defines a node that publishes a "Hello, World" message along with a counter to the topic named `py_str_topic` every 0.5 seconds.

Breaking Down the Publisher Node Code

We now examine each part of the code in detail to understand how the ROS 2 Python publisher node works.

1. Shebang and Imports

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from std_msgs.msg import String
```

- The shebang line (`#!/usr/bin/env python3`) ensures that the script is executed with Python 3.
- `rclpy` is the ROS 2 Python client library, which provides the tools needed to write ROS 2 nodes.
- `Node` is the base class for creating ROS 2 nodes.
- `String` is the message type imported from the standard messages package.

2. Defining the Node Class

```
class PyPublisher(Node):
    def __init__(self):
        super().__init__('py_publisher_node')
        self.get_logger().info("Python Publisher node has been started")
        self.publisher_ = self.create_publisher(String, 'py_str_topic', 10)
        timer_period = 0.5 # seconds
        self.timer = self.create_timer(timer_period, self.timer_callback)
        self.i = 0
```



- The `PyPublisher` class inherits from `Node` to create a new ROS 2 node.
- In the `__init__` method:
 - `super().__init__()` initializes the node with the name `py_publisher_node`.
 - `self.get_logger().info("PythonPublisherNodehasbeenstarted")` logs an informational message when the node starts.
 - `self.create_publisher(String, 'py_str_topic', 10)` creates a publisher that will publish messages of type `String` to the topic `py_str_topic` with a queue size of 10.
 - `self.create_timer(timer_period, self.timer_callback)` sets up a timer to call the `timer_callback` method every 0.5 seconds.
 - `self.i=0` initializes a counter for use in the published messages.

3. Timer Callback Method

```
def timer_callback(self):
    msg = String()
    msg.data = 'Hello, this is Eduardo from the Python Publisher: %d' % self.i
    self.publisher_.publish(msg)
    self.get_logger().info('Publishing: "%s"' % msg.data)
    self.i += 1
```

- The `timer_callback` method is called periodically by the timer.
- A new `String` message is created, and its `data` field is set to include the current counter value.
- The message is published to `py_str_topic`.
- An informational log displays the published message.
- The counter `self.i` is incremented for the next callback.

4. The Main Function

```
# To ensure that resources are properly cleaned up and that
# all intended statements are executed upon shutdown,
# you should handle the KeyboardInterrupt exception.
def main(args=None):
    rclpy.init(args=args)
    node = PyPublisher()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.destroy_node()
    finally:
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

- `rclpy.init(args=args)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- An instance of the `PyPublisher` node is created.



- `rclpy.spin(node)` keeps the node active, processing callbacks until a shutdown signal is received.
- The `try-except-finally` block ensures graceful shutdown:
 - `try`: Runs the spinning loop to keep the node active.
 - `except KeyboardInterrupt`: Catches the interrupt signal (e.g., `Ctrl+C`) and explicitly destroys the node to release resources.
 - `finally`: Ensures that `rclpy.shutdown()` is called to clean up resources, regardless of how the spin loop is exited.
- `if __name__ == '__main__':` Ensures that the `main()` function is called only when the script is executed directly.

This detailed breakdown explains each component of the ROS 2 Python publisher node. For further details and additional context, refer to the [Creating ROS 2 Nodes \(Python\)](#) tutorial.

2.7.3 Running the Publisher Node Standalone (Optional)

After editing the file, you can run your Python publisher node out of ROS 2 environment. For this, open a new terminal (or use an additional session in Terminator) and navigate to the location of the Python file:

```
$ cd src/s2_py_pubsub/s2_py_pubsub
```

Then, test the Python file by running it with one of the following commands:

```
$ ./publisher.py
```

to run it as an executable, or

```
$ python3 publisher.py
```

to run it as a Python script.

2.7.4 Integrating the Publisher Node into the Package

Modifying `setup.py`

To enable ROS 2 to recognize and execute your publisher node, you need to specify it in the `setup.py` file of your package. Open `setup.py` in VS Code and locate the `entry_points` field. Then, add your node as follows:

```
entry_points={
    'console_scripts': [
        'publisher_exe = s2_py_pubsub.publisher:main',
    ],
},
```

Breaking down the components of this specification:

- `publisher_exe`: Specifies the command-line executable name that you will use to run the node.
- `s2_py_pubsub`: Indicates the name of your package.



- **publisher**: Specifies the Python file (without the `.py` extension) containing the node.
- **main**: Refers to the function that will be executed when the node runs.

Building the Package

After saving the changes to `setup.py`, build your package. In the first terminal session, from the root of the workspace, compile with:

```
$ colcon build
```

Alternatively, to build only the Python package:

```
$ colcon build --packages-select s2_py_pubsub
```

For development purposes, consider using the `--symlink-install` option so that changes in the source files are immediately reflected without requiring a full rebuild:

```
$ colcon build --symlink-install
```

or, more specifically:

```
$ colcon build --packages-select s2_py_pubsub --symlink-install
```

If the build completes successfully, you will see the following confirmation message: `'colcon build' successful`.

Sourcing the Environment

Before running the node, in the first (or current) terminal session from the root of the workspace, source the workspace's setup file to load the necessary environment variables:

```
$ source ./install/setup.bash
```

This step ensures that ROS 2 and the built packages are properly configured.

2.7.5 Running the ROS 2 Publisher Node with ROS 2

Execute your Python publisher node using the `ros2run` command:

```
$ ros2 run s2_py_pubsub publisher_exe
```

At this point, if everything is set up correctly, your node should be actively publishing messages to the `py_str_topic` topic.

2.7.6 Summary of Key Identifiers

At the end of the creation and configuration of the ROS 2 publisher node, it is important to distinguish between the different identifiers used:

1. **publisher.py**: The filename of your Python script.
2. **py_publisher_node**: The name of the node as defined in the `super().__init__()` call within your script.
3. **publisher_exe**: The command-line executable name specified in `setup.py` under the `entry_points` field.



Ensuring these identifiers are correctly set and consistently used is crucial for the proper functioning of your ROS 2 Python nodes.

For more details on node creation and setup in Python, refer to the official [Creating ROS 2 Nodes \(Python\)](#) tutorial.

2.8 Creating a ROS 2 Subscriber Node with Python

2.8.1 Setting Up the Python Node

Navigating to the Package Directory

Open a terminal and navigate to your Python package directory within the ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s2_py_pubsub/s2_py_pubsub
```

Replace `s2_py_pubsub` with your actual package name.

Creating the Python Node File

Create a new Python file for your subscriber node, for example, `subscriber.py`:

```
$ touch subscriber.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see both the `__init__.py` and `subscriber.py` files.

Making the Python File Executable

Ensure the new Python file is executable:

```
$ chmod +x subscriber.py
```

Check the folder contents again to confirm that `subscriber.py` now appears in a different color, indicating that it is executable.

2.8.2 Editing the Python Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `subscriber.py` file for editing.



Implementing the Node Code

Below is an example implementation of a simple ROS 2 subscriber node in Python:

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from std_msgs.msg import String

class PySubscriber(Node):

    def __init__(self):
        super().__init__('py_subscriber_node')
        self.get_logger().info("Python Subscriber node has been started")
        self.subscription = self.create_subscription(
            String,
            'cpp_str_topic',
            self.listener_callback,
            10)
        self.subscription  # prevent unused variable warning

    def listener_callback(self, msg):
        self.get_logger().info('I heard: "%s"' % msg.data)

# To ensure that resources are properly cleaned up and that
# all intended statements are executed upon shutdown,
# you should handle the KeyboardInterrupt exception.
def main(args=None):
    rclpy.init(args=args)
    node = PySubscriber()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.destroy_node()
    finally:
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

This script defines a node that subscribes to the topic `cpp_str_topic` and logs the messages received.

2.8.3 Breaking Down the Subscriber Node Code

1. Shebang and Imports

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from std_msgs.msg import String
```

- The shebang ensures the script runs with Python 3.
- `rclpy` is the ROS 2 Python client library.



- `Node` is the base class for creating ROS 2 nodes.
- `String` is the message type imported from the standard messages package.

2. Defining the Node Class

```
class PySubscriber(Node):  
  
    def __init__(self):  
        super().__init__('py_subscriber_node')  
        self.get_logger().info("Python Subscriber node has been started")  
        self.subscription = self.create_subscription(  
            String,  
            'cpp_str_topic',  
            self.listener_callback,  
            10)  
        self.subscription  # prevent unused variable warning
```

- The `PySubscriber` class inherits from `Node` to create a new ROS 2 node.
- The node is initialized with the name `py_subscriber_node`.
- `self.get_logger().info("PythonSubscribernodehasbeenstarted")` logs an informational message when the node starts.
- `self.create_subscription` creates a subscription to the topic `cpp_str_topic` for messages of type `String` with a queue size of 10.
- `self.listener_callback` is set as the callback function to be invoked upon receiving a message.

3. Listener Callback Method

```
def listener_callback(self, msg):  
    self.get_logger().info('I heard: "%s"' % msg.data)
```

- `listener_callback` is executed every time a message is received.
- An informational log displays the data of the received message.

4. The Main Function

```
# To ensure that resources are properly cleaned up and that  
# all intended statements are executed upon shutdown,  
# you should handle the KeyboardInterrupt exception.  
def main(args=None):  
    rclpy.init(args=args)  
    node = PySubscriber()  
    try:  
        rclpy.spin(node)  
    except KeyboardInterrupt:  
        node.destroy_node()  
    finally:  
        rclpy.shutdown()  
  
if __name__ == '__main__':  
    main()
```



- `rclpy.init(args=args)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- An instance of the `PySubscriber` node is created.
- `rclpy.spin(node)` keeps the node active, processing callbacks until a shutdown signal is received.
- The `try-except-finally` block ensures graceful shutdown:
 - `try`: Runs the spinning loop to keep the node active.
 - `except KeyboardInterrupt`: Catches the interrupt signal (e.g., `Ctrl+C`) and explicitly destroys the node to release resources.
 - `finally`: Ensures that `rclpy.shutdown()` is called to clean up resources, regardless of how the spin loop is exited.
- `if __name__ == '__main__':` Ensures that the `main()` function is called only when the script is executed directly.

This detailed breakdown explains each component of the ROS 2 Python subscriber node. For further details and additional context, refer to the [Creating ROS 2 Nodes \(Python\)](#) tutorial.

2.8.4 Running the Subscriber Node Standalone (Optional)

After editing the file, you can run your Python subscriber node out of ROS 2 environment. For this, open a new terminal session (or use an additional session in Terminator) and navigate to the location of the Python file:

```
$ cd src/s2_py_pubsub/s2_py_pubsub
```

Then, test the Python file by running it with one of the following commands:

```
$ ./subscriber.py
```

to run it as an executable, or

```
$ python3 subscriber.py
```

to run it as a Python script.

2.8.5 Integrating the Subscriber Node into the Package

Modifying `setup.py`

To allow ROS 2 to recognize and execute your subscriber node, specify it in the `setup.py` file of your package. Open `setup.py` in VS Code and locate the `entry_points` field. Then, add your node entry immediately after any existing ROS 2 node entries (e.g., `publisher_exe`):

```
entry_points={
    'console_scripts': [
        'publisher_exe = s2_py_pubsub.publisher:main',
        'subscriber_exe = s2_py_pubsub.subscriber:main',
    ],
}
```



Breaking down the components of this specification:

- `subscriber_exe`: Specifies the command-line executable name that you will use to run the subscriber node.
- `s2_py_pubsub`: Indicates the name of your package.
- `subscriber`: Specifies the Python file (without the `.py` extension) that contains the node.
- `main`: Refers to the function that will be executed when the node is run.

Building the Package

After updating `setup.py`, build your package. In the first terminal session, from the root of the workspace, compile with:

```
$ colcon build
```

Alternatively, to build only the Python package:

```
$ colcon build --packages-select s2_py_pubsub
```

For development purposes, you might consider using the `--symlink-install` option so that changes in the source files are immediately reflected without requiring a full rebuild:

```
$ colcon build --symlink-install
```

or, more specifically:

```
$ colcon build --packages-select s2_py_pubsub --symlink-install
```

If the build completes successfully, you will see the following confirmation message: `'colcon build' successful`.

Sourcing the Environment

Before running the node, in the first (or current) terminal session from the root of the workspace, source the workspace's setup file to load the necessary environment variables:

```
$ source ./install/setup.bash
```

This ensures that ROS 2 and your built packages are properly configured.

2.8.6 Running the ROS 2 Subscriber Node with ROS 2

Execute your subscriber node using the `ros2run` command:

```
$ ros2 run s2_py_pubsub subscriber_exe
```

At this point, if everything is set up correctly and your C++ publisher node is running, your subscriber node should be actively receiving and logging messages published from the `cpp_str_topic` topic.

2.8.7 Summary of Key Identifiers

It is important to distinguish between the following identifiers in your ROS 2 subscriber node:

1. `subscriber.py`: The filename of your Python script.



2. `py_subscriber_node`: The name of the node as defined in the `super().__init__()` call within your script.
3. `subscriber_exe`: The command-line executable name specified in `setup.py` under the `entry_points` field.

Ensuring that these identifiers are correctly set and consistently used is crucial for the proper functioning of your ROS 2 Python nodes.

For more details on creating ROS 2 nodes in Python, refer to the official [Creating ROS 2 Nodes \(Python\)](#) tutorial.

2.9 Practice Assignment: Running ROS 2 Nodes

In this assignment, you will run ROS 2 publisher and subscriber nodes implemented in both C++ and Python, as described in Sections 2.5, 2.6, 2.7, and 2.8. You will then capture evidence of successful execution using a screenshot and submit it in the designated assignment area on Canvas (within the ROS 2 module) as a PNG file. The filename must follow this format:

`FirstnameLastname_evidence_sX.png`,

where `sX` indicates the session number. For example, in this case, the correct filename would be:

`EduardoDavila_evidence_s2.png`.

2.9.1 Executing ROS 2 Publishers and Subscribers

Running the Python Publisher and Subscriber

After implementing and saving your Python nodes and updating `package.xml` and `setup.py` files, build your package and run the nodes. In two separate terminal sessions (or using the Terminator emulator), execute the following commands from the root of your workspace:

- In **Terminal 1**, build your package, source the setup file, and run the Python publisher node:

```
$ colcon build --packages-select s2_py_pubsub
$ source install/setup.bash
$ ros2 run s2_py_pubsub publisher_exe
```

- In **Terminal 2**, source the setup file and run the Python subscriber node:

```
$ source install/setup.bash
$ ros2 run s2_py_pubsub subscriber_exe
```

You should observe that the `py_publisher` node sends a personalized message (e.g., displaying your name), and the `py_subscriber` node awaits the reception of the messages.



Running the C++ Publisher and Subscriber

After implementing and saving your C++ nodes and updating `package.xml` and `CMakeLists.txt` files, build your package and run the nodes. In two additional terminal sessions (or another two sessions in **Terminator**), execute the following commands from the root of your workspace:

- In **Terminal 3**, build your package, source the setup file, and run the C++ publisher node:

```
$ colcon build --packages-select s2_cpp_pubsub
$ source install/setup.bash
$ ros2 run s2_cpp_pubsub publisher_exe
```

- In **Terminal 4**, source the setup file and run the C++ subscriber node:

```
$ source install/setup.bash
$ ros2 run s2_cpp_pubsub subscriber_exe
```

Here, you should observe that the `py_publisher` node publishes your personalized message (e.g., displaying your name), whereas the `cpp_subscriber` node confirms its reception. Additionally, the `cpp_publisher` node publishes your other personalized message (e.g., displaying your name), whereas the `py_subscriber` node confirms its reception.

Visualizing the ROS Graph with `rqt_graph`

To ensure that your publisher and subscriber nodes are correctly connected, use the `rqt_graph` tool. Then, in **Terminal 5**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_graph rqt_graph
```

or simply:

```
$ rqt_graph
```

This visualization should show the topics connecting your publisher and subscriber nodes as described above.

2.9.2 Submission Instructions

1. Run the ROS 2 nodes as described above.
2. Capture a **screenshot showing the terminal outputs** from all four nodes (Python and C++ publishers and subscribers) along with the visualization of the ROS graph using the `rqt_graph` tool. See Figure 2.1 for reference.
3. Save the screenshot as a PNG file.
4. Name the file following this format: `FirstnameLastname_evidence_sX.png` (replace `X` with the week number and `Y` with the session number).
5. Submit your file as instructed by the course guidelines on Canvas.



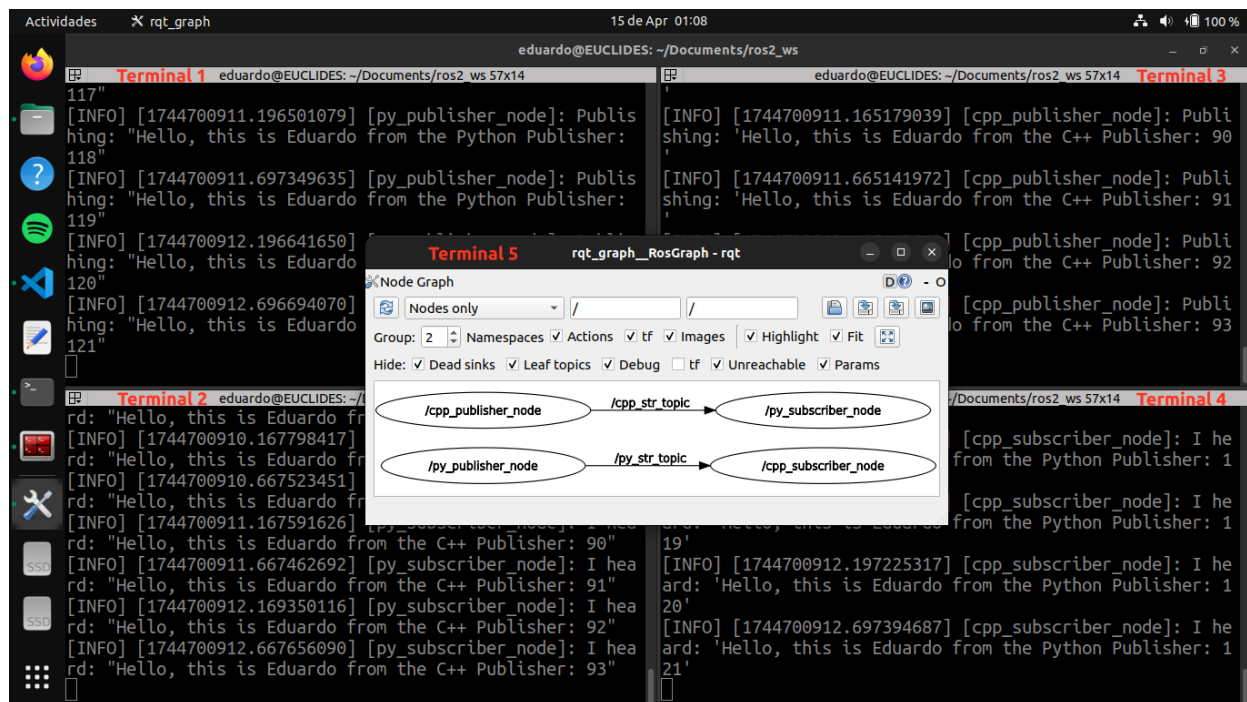


Figure 2.1: Example screenshot showing terminal outputs of the publisher and subscriber nodes, in C++ and Python, in action.

Note: Your machine's username and device name must be visible in the screenshot to verify the authenticity of the submission, as shown in Figure 2.1. Any submission that appears copied, unclear, or altered will be considered invalid and may result in a score of 0 for this assignment.

CHAPTER

3

ROS 2 Service and Client

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✳ **Abstract** - This chapter introduces the services and clients in the [Robot Operating System \(ROS\)](#) 2 environment, providing step-by-step instructions for creating both C++ and Python service and client nodes, integrating them into ROS 2 packages, and testing their functionality. By the end of this chapter, you will have a solid understanding of how to implement and use services in ROS 2 to enable more complex interactions between nodes.

All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) ↗, located in the `ros2_ws/src` folder, corresponding to the ROS 2 packages with the `s3` prefix.

3.1 Introduction to ROS 2 Services and Clients

Within the [ROS](#) 2 framework, a service consists of two parts: a [server](#), which provides a functionality, and a [client](#), which calls that functionality. Unlike topics, which support one-to-many communication, services allow for a synchronous or asynchronous request-response interaction between nodes. Asynchronous service calls are generally recommended to avoid blocking the client node while waiting for a response.

3.1.1 Concepts: Service and Client

Service/Server: A [ROS](#) 2 service provides a mechanism for a node to offer a specific functionality that other nodes can call. The service operates on a request/response model, where the server waits for a request and then sends back a response.

Client: A ROS 2 client calls a service provided by a server. This interaction can be synchronous, meaning the client blocks until the response is received (or a timeout occurs). However, in our examples, we use asynchronous service call APIs: `async_send_request()` in C++ and `call_async()` in Python. These asynchronous APIs allow the client node to continue processing other tasks while waiting for the service response, which is generally recommended over synchronous calls. (For more details, refer to the guide on [Synchronous vs. Asynchronous Clients](#).)

3.1.2 Prerequisites

Before you begin, ensure that you are familiar with workspace and package creation, as described in Sections 2.2, 2.3, and 2.4. Open a terminal and navigate to the `src` directory of your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.
Next, create the following packages in your workspace:

- C++ package for the service and client nodes:

```
$ ros2 pkg create s3_cpp_srvcli --build-type ament_cmake --dependencies rclcpp  
→ example_interfaces --license Apache-2.0 --description "Your package description  
→ here"
```

- Python package for the service and client nodes:

```
$ ros2 pkg create s3_py_srvcli --build-type ament_python --dependencies rclpy  
→ example_interfaces --license Apache-2.0 --description "Your package description  
→ here"
```

You may replace `s3_cpp_srvcli` and `s3_py_srvcli` with package names of your preference.

3.2 Creating a ROS 2 Service Node in C++

3.2.1 Setting Up the C++ Service Node

Open a terminal and navigate to the `src` directory of your C++ package (e.g., `s3_cpp_srvcli`). Then, create a new C++ file for the service node, (e.g., `server.cpp`):

```
$ cd ros2_ws/src/s3_cpp_srvcli/src  
$ touch server.cpp
```

3.2.2 Editing the C++ Service Node

Open your workspace in VS Code:

```
$ cd ~/ros2_ws  
$ code .
```

In VS Code, locate and open `server.cpp` and implement your service node code (code to be provided later).



Implementing the C++ Service Node Code

Below is an example implementation of a simple ROS 2 service node in C++:

```
#include "rclcpp/rclcpp.hpp"
#include "example_interfaces/srv/add_two_ints.hpp"

class CppService : public rclcpp::Node {
public:
    CppService()
    : Node("cpp_server_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Server node has been started");
        service_ = this->create_service<example_interfaces::srv::AddTwoInts>(
            "cpp_add_two_ints_service",
            std::bind(&CppService::AddTwoInts_callback, this, std::placeholders::_1,
                std::placeholders::_2)
        );
        RCLCPP_INFO(this->get_logger(), "Service 'cpp_add_two_ints_service' is ready
        to receive requests");
    }

private:
    void AddTwoInts_callback(
        const std::shared_ptr<example_interfaces::srv::AddTwoInts::Request> request,
        std::shared_ptr<example_interfaces::srv::AddTwoInts::Response> response) {
        response->sum = request->a + request->b;
        RCLCPP_INFO(this->get_logger(), "Received request:\na: %ld, b: %ld",
            request->a, request->b);
        RCLCPP_INFO(this->get_logger(), "Sending back response: sum = %ld",
            response->sum);
    }

    rclcpp::Service<example_interfaces::srv::AddTwoInts>::SharedPtr service_;
};

int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppService>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

This code defines a server node that offers a service to add two integers using the service name `cpp_add_two_ints_service`.

Breaking Down the Service Node Code

We now examine each part of the code in detail to understand how the C++ service node works.

1. Includes and Setup

```
#include "rclcpp/rclcpp.hpp"
#include "example_interfaces/srv/add_two_ints.hpp"
```



- `rclcpp/rclcpp.hpp` provides the ROS 2 C++ API.
- `example_interfaces/srv/add_two_ints.hpp` defines the `AddTwoInts` service message type.

2. Defining the Node Class

```
class CppService : public rclcpp::Node {
public:
    CppService()
    : Node("cpp_server_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Server node has been started");
        service_ = this->create_service<example_interfaces::srv::AddTwoInts>(
            "cpp_add_two_ints_service",
            std::bind(&CppService::AddTwoInts_callback, this, std::placeholders::_1,
                std::placeholders::_2)
        );
        RCLCPP_INFO(this->get_logger(), "Service 'cpp_add_two_ints_service' is ready
        to receive requests");
    }
};
```

- The `CppService` class inherits from `rclcpp::Node`, creating a new ROS 2 node.
- `public`: This access specifier indicates that all members declared after it (until another access specifier is encountered) are publicly accessible.
- The `CppService()` constructor initializes the node with the name `"cpp_server_node"`.
- `RCLCPP_INFO(this->get_logger(), "C++Servernodehasbeenstarted")` logs an informational message when the node starts.
- The node registers a service named `"cpp_add_two_ints_service"` using the `AddTwoInts` service type. The callback function, `AddTwoInts_callback`, is bound to handle incoming requests using `std::bind()`, ensuring that the correct instance and parameters are forwarded. Additionally, `std::placeholders::_1` and `_2` are used to represent the request and response objects passed to the callback at runtime.
- `RCLCPP_INFO(this->get_logger(), "Service'cpp_add_two_ints_service'isreadytoreceiverequests")` logs an informational message to indicate service is ready.

3. Request Callback Method

```
private:
    void AddTwoInts_callback(
        const std::shared_ptr<example_interfaces::srv::AddTwoInts::Request> request,
        std::shared_ptr<example_interfaces::srv::AddTwoInts::Response> response) {
        response->sum = request->a + request->b;
        RCLCPP_INFO(this->get_logger(), "Received request:\na: %ld, b: %ld",
            request->a, request->b);
        RCLCPP_INFO(this->get_logger(), "Sending back response: sum = %ld",
            response->sum);
    }

    rclcpp::Service<example_interfaces::srv::AddTwoInts>::SharedPtr service_;
};
```



- `private`: This access specifier indicates that all members declared after it are accessible only within the class itself.
- The `AddTwoInts_callback` method is invoked when the service receives a request. It calculates the sum of `a` and `b`, logs the request and the response, and then sends back the result.
- `service_`: A shared pointer to the service object that handles incoming requests.

4. The Main Function

```
int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppService>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

- `rclcpp::init(argc, argv)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- `node=std::make_shared<CppService>()` creates an instance of the `CppService` node.
- `rclcpp::spin(node)` keeps the node active, processing callbacks (such as those from the timer) until a shutdown signal is received.
- `node.reset()` releases the shared pointer to the node so that its resources can be deallocated and the destructor can be called before ROS 2 shuts down.
- `rclcpp::shutdown()` gracefully shuts down ROS 2, releasing all allocated resources when the node stops.
- `return 0`; indicates successful program termination.

This detailed breakdown explains each component of the ROS 2 C++ service node. For further details, refer to the [Creating a ROS 2 Service \(C++\)](#) tutorial.

3.2.3 Integrating the C++ Service Node into the Package

Modify `CMakeLists.txt` of your C++ package to add an executable for the service node:

```
# Find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(example_interfaces REQUIRED)

add_executable(server_exe src/server.cpp)
ament_target_dependencies(server_exe rclcpp example_interfaces)

install(TARGETS
  server_exe
  DESTINATION lib/${PROJECT_NAME})
```



3.2.4 Building, Sourcing, and Running the C++ Service Node

From the root of your workspace, build the package:

```
$ colcon build --packages-select s3_cpp_srvcli
```

Then, source the workspace:

```
$ source install/setup.bash
```

Finally, run the service server node:

```
$ ros2 run s3_cpp_srvcli server_exe
```

If everything is set up correctly, your node should be running and ready to receive requests.

3.2.5 Testing the C++ Service

Open a new terminal, test the service using the following command:

```
$ ros2 service call /cpp_add_two_ints_service example_interfaces/srv/AddTwoInts "{a: 2,  
→ b: 3}"
```

Alternatively, use `rqt_service_caller`:

```
$ ros2 run rqt_service_caller rqt_service_caller
```

and request the service on `/cpp_add_two_ints_service` by entering the values `a:5,b:7`.

3.2.6 Summary of C++ Service Node Key Identifiers

- `server.cpp`: Name of the source file for the C++ service node.
- `cpp_server_node`: Name of the C++ service node as defined in the constructor of the C++ class.
- `server_exe`: Name of the executable defined in `CMakeLists.txt`.
- `/cpp_add_two_ints_service`: Name of the service provided by the C++ server node.

3.3 Creating a ROS 2 Client Node in C++

3.3.1 Setting Up the C++ Client Node

Open a terminal and navigate to the `src` directory of your C++ package (e.g., `s3_cpp_srvcli`). Then, create a new C++ file for the client node, (e.g., `client.cpp`):

```
$ cd ros2_ws/src/s3_cpp_srvcli/src  
$ touch client.cpp
```

3.3.2 Editing the C++ Client Node

Open your workspace in VS Code:

```
$ cd ~/ros2_ws  
$ code .
```



In VS Code, locate and open `client.cpp` and implement your client node code (code to be provided later).

Implementing the C++ Client Node Code

Below is an example implementation of a simple ROS 2 client node in C++ that uses the asynchronous service call API (`async_send_request()`).

```
#include "rclcpp/rclcpp.hpp"
#include "example_interfaces/srv/add_two_ints.hpp"

using namespace std::chrono_literals;

class CppClientAsync : public rclcpp::Node {
public:
    CppClientAsync(int64_t a, int64_t b)
        : Node("cpp_client_async_node"), a_(a), b_(b) {
        RCLCPP_INFO(this->get_logger(), "C++ Client node has been started");
        client_ =
        ↪ this->create_client<example_interfaces::srv::AddTwoInts>("cpp_add_two_ints_service");
        // Wait for the service to become available
        while (!client_->wait_for_service(1s)) {
            if (!rclcpp::ok()) {
                RCLCPP_ERROR(this->get_logger(), "Interrupted while waiting for
        ↪ the service. Exiting.");
                return;
            }
            RCLCPP_INFO(this->get_logger(), "Waiting for service to become
        ↪ available...");
        }
        // Request is called
        send_request();
    }
private:
    void send_request() {
        std::shared_ptr<example_interfaces::srv::AddTwoInts::Request> request =
        ↪ std::make_shared<example_interfaces::srv::AddTwoInts::Request>();
        request->a = a_;
        request->b = b_;
        // Asynchronously send the request
        rclcpp::Client<example_interfaces::srv::AddTwoInts>::FutureAndRequestId
        ↪ future_and_request_id =
            client_->async_send_request(request);
        std::shared_future<example_interfaces::srv::AddTwoInts::Response::SharedPtr>
        ↪ result =
            future_and_request_id.future.share();
        // Spin until the future is complete
        if (rclcpp::spin_until_future_complete(this->get_node_base_interface(),
        ↪ result) ==
            rclcpp::FutureReturnCode::SUCCESS) {
            RCLCPP_INFO(this->get_logger(), "Received response: %ld + %ld = %ld",
        ↪ request.get()->a, request.get()->b, result.get()->sum);
        }
    }
};
```



```

        else {
            RCLCPP_ERROR(this->get_logger(), "Failed to call service add_two_ints");
        }
    }

    rclcpp::Client<example_interfaces::srv::AddTwoInts>::SharedPtr client_;
    int64_t a_;
    int64_t b_;
};

int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    if (argc != 3) {
        RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Usage: cpp_client_exe a b");
        return 1;
    }
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppClientAsync>(atoll(argv[1]),
    ↪ atoll(argv[2]));
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```

This code defines a client node that asynchronously requests the service `cpp_add_two_ints_service`, using `async_send_request()`, to add two integers.

Breaking Down the Client Node Code

We now examine each part of the code in detail to understand how the ROS 2 C++ client node works.

1. Includes and Setup

```

#include "rclcpp/rclcpp.hpp"
#include "example_interfaces/srv/add_two_ints.hpp"

```

- `rclcpp/rclcpp.hpp` provides the ROS 2 C++ API.
- `example_interfaces/srv/add_two_ints.hpp` defines the `AddTwoInts` service message type.

2. Defining the Node Class

```

class CppClientAsync : public rclcpp::Node {
public:
    CppClientAsync(int64_t a, int64_t b)
        : Node("cpp_client_async_node"), a_(a), b_(b) {
        RCLCPP_INFO(this->get_logger(), "C++ Client node has been started");
        client_ =
    ↪ this->create_client<example_interfaces::srv::AddTwoInts>("cpp_add_two_ints_service");
    }
};

```



```

        // Wait for the service to become available
        while (!client_->wait_for_service(1s)) {
            if (!rclcpp::ok()) {
                RCLCPP_ERROR(this->get_logger(), "Interrupted while waiting for
↳ the service. Exiting.");
                return;
            }
            RCLCPP_INFO(this->get_logger(), "Waiting for service to become
↳ available...");
        }
        // Request is called
        send_request();
    }

```

- The `CppClientAsync` class inherits from `rclcpp::Node`, creating a new ROS 2 node.
- `public`: This access specifier indicates that all members declared after it (until another access specifier is encountered) are publicly accessible.
- The `CppClientAsync()` constructor initializes the node with the name "`cpp_client_async_node`" and stores the input integers `a` and `b`.
- `RCLCPP_INFO(this->get_logger(), "C++Clientnodehasbeenstarted")` logs an informational message when the node starts.
- The node creates a client for the service "`cpp_add_two_ints_service`".
- A loop waits for the service to become available, logging status messages. If the ROS system is interrupted during the wait, an error is logged and the function returns.
- Once the service is available, the `send_request()` function is called.

3. Sending the Service Request

```

private:
    void send_request() {
        std::shared_ptr<example_interfaces::srv::AddTwoInts::Request> request =
↳ std::make_shared<example_interfaces::srv::AddTwoInts::Request>();
        request->a = a_;
        request->b = b_;
        // Asynchronously send the request
        rclcpp::Client<example_interfaces::srv::AddTwoInts>::FutureAndRequestId
↳ future_and_request_id =
        client_->async_send_request(request);
        std::shared_future<example_interfaces::srv::AddTwoInts::Response::SharedPtr>
↳ result =
        future_and_request_id.future.share();
        // Spin until the future is complete
        if (rclcpp::spin_until_future_complete(this->get_node_base_interface(),
↳ result) ==
        rclcpp::FutureReturnCode::SUCCESS) {
            RCLCPP_INFO(this->get_logger(), "Received response: %ld + %ld = %ld",
↳ request.get()->a, request.get()->b, result.get()->sum);
        }
    }

```



```

        else {
            RCLCPP_ERROR(this->get_logger(), "Failed to call service add_two_ints");
        }
    }

    rclcpp::Client<example_interfaces::srv::AddTwoInts>::SharedPtr client_;
    int64_t a_;
    int64_t b_;
};

```

- **private**: This access specifier indicates that all members declared after it are accessible only within the class itself.
- The **send_request()** method creates a request, assigns values to its fields, and sends it asynchronously. It waits until the future completes and then logs the received response.
- **request->a** and **request->b** store the values to be added, taken from the class member variables.
- **async_send_request(request)** sends the request asynchronously and returns a **FutureAndRequestId** object, which contains both the future result and the internal request ID.
- **future.share()** transforms the unique future into a shared future, allowing multiple accesses if needed.
- **spin_until_future_complete()** blocks the current node until the future is completed or a timeout/error occurs. Although the request is sent asynchronously, this line synchronously waits for the response.
- **result.get()->sum** accesses the result of the service call once the future has completed successfully.
- **RCLCPP_INFO()** and **RCLCPP_ERROR()** are used to print informational or error messages to the console, including the input data and the result.
- **client_**, **a_**, and **b_**: Class members used to send the request and store the request parameters.

4. The Main Function

```

int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    if (argc != 3) {
        RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Usage: cpp_client_exe a b");
        return 1;
    }
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppClientAsync>(atoll(argv[1]),
    ↪ atoll(argv[2]));
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```

- **rclcpp::init(argc, argv)** initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- **argc!=3** ensures that exactly two numeric arguments are passed from the command line. If not, a usage message is printed and the program exits.



- `node=std::make_shared<CppClientAsync>()` creates an instance of the `CppClientAsync` node, passing the two command-line arguments as integers.
- `node.reset()` releases the shared pointer to the node so that its resources can be deallocated and the destructor can be called before ROS 2 shuts down.
- `rclcpp::shutdown()` gracefully shuts down ROS 2, releasing all allocated resources when the node stops.
- `return 0;` indicates successful program termination.

This detailed breakdown explains each component of the ROS 2 C++ client node. For further details, refer to the [Creating a ROS 2 Client \(C++\)](#) tutorial.

3.3.3 Integrating the C++ Client Node into the Package

Modify `CMakeLists.txt` of your C++ package to add an executable for the client node while preserving any existing node setup (e.g., `server_exe`):

```
# Find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(example_interfaces REQUIRED)

add_executable(server_exe src/server.cpp)
ament_target_dependencies(server_exe rclcpp example_interfaces)

add_executable(client_exe src/client.cpp)
ament_target_dependencies(client_exe rclcpp example_interfaces)

install(TARGETS
  server_exe
  client_exe
  DESTINATION lib/${PROJECT_NAME}
)
```

3.3.4 Building, Sourcing, and Running the C++ Client Node

From the root of your workspace, build the package:

```
$ colcon build --packages-select s3_cpp_srvcli
```

Then, source the workspace:

```
$ source install/setup.bash
```

Finally, run the client node:

```
$ ros2 run s3_cpp_srvcli client_exe 11 13
```

If everything is set up correctly, your node should execute, send a request to the service, and wait for the response. If the service is not available, the client will wait until it becomes accessible before proceeding.



3.3.5 Summary of C++ Client Node Key Identifiers

- `client.cpp`: Name of the source file for the C++ client node.
- `cpp_client_async_node`: Name of the C++ client node as defined in the constructor of the C++ class.
- `client_exe`: Name of the executable defined in `CMakeLists.txt`.
- `/cpp_add_two_ints_service`: Name of the service that the C++ client node calls.

3.4 Creating a ROS 2 Service Node in Python

3.4.1 Setting Up the Python Service Node

Open a terminal and navigate to the `src` directory of your Python package (e.g., `s3_py_srvcli`). Then, create a new Python file for the service node (e.g., `server.py`):

```
$ cd ros2_ws/src/s3_py_srvcli/s3_py_srvcli
$ touch server.py
```

3.4.2 Editing the Python Service Node

Open your workspace in VS Code:

```
$ cd ~/ros2_ws
$ code .
```

In VS Code, locate and open `server.py` and implement your service node code (code to be provided later).

Implementing the Python Service Node Code

Below is an example implementation of a simple ROS 2 service node in Python:

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from example_interfaces.srv import AddTwoInts

class PyService(Node):
    def __init__(self):
        super().__init__('py_server_node')
        self.get_logger().info("Python Server node has been started")
        self.srv = self.create_service(AddTwoInts, 'py_add_two_ints_service',
        ↪ self.AddTwoInts_callback)
        self.get_logger().info("Service 'py_add_two_ints_service' is ready to receive
        ↪ requests")
```




```

def AddTwoInts_callback(self, request, response):
    response.sum = request.a + request.b
    self.get_logger().info('Received request:\na: %d, b: %d' % (request.a,
    ↪ request.b))
    self.get_logger().info('Sending back response: sum = %d' % (response.sum))

    return response

def main(args=None):
    rclpy.init(args=args)
    node = PyService()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.destroy_node()
    finally:
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```

This code defines a server node that offers a service to add two integers using the service name `py_add_two_ints_service`.

Breaking Down the Service Node Code

We now examine each part of the code in detail to understand how the Python service node works.

1. Imports and Initialization

```

#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from example_interfaces.srv import AddTwoInts

```

- The shebang line (`#!/usr/bin/envpython3`) ensures that the script is executed with Python 3.
- `rclpy` provides the ROS 2 Python API.
- `Node` is the base class for creating ROS 2 nodes.
- `AddTwoInts` is the service message type imported from the `example_interfaces` package.

2. Defining the Node Class

```

class PyService(Node):

    def __init__(self):
        super().__init__('py_server_node')
        self.get_logger().info("Python Server node has been started")
        self.srv = self.create_service(AddTwoInts, 'py_add_two_ints_service',
        ↪ self.AddTwoInts_callback)
        self.get_logger().info("Service 'py_add_two_ints_service' is ready to receive
        ↪ requests")

```



- The `PyService` class inherits from `Node`, creating a new ROS 2 node.
- In the `__init__` method:
 - `super().__init__()` initializes the node with the name `py_server_node`.
 - `self.get_logger().info("PythonServernodehasbeenstarted")` logs an informational message when the node starts.
 - `self.create_service()` creates a service using the `AddTwoInts` service type with the service name `"py_add_two_ints_service"`. The callback function, `AddTwoInts_callback`, is set to handle incoming requests.
 - `self.get_logger().info("Service'py_add_two_ints_service'isreadytoreceiverequests")` logs an informational message to indicate service is ready.

3. Request Callback Method

```
def AddTwoInts_callback(self, request, response):
    response.sum = request.a + request.b
    self.get_logger().info('Received request:\na: %d, b: %d' % (request.a,
    ↪ request.b))
    self.get_logger().info('Sending back response: sum = %d' % (response.sum))
    return response
```

- The `AddTwoInts_callback` method is invoked when the service receives a request.
- It calculates the sum of `a` and `b`, logs the request and response, and returns the response.

4. The Main Function

```
def main(args=None):
    rclpy.init(args=args)
    node = PyService()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.destroy_node()
    finally:
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

- `rclpy.init(args=args)` initializes ROS 2 communication by parsing command-line arguments and setting up necessary middleware.
- An instance of the `PyService` node is created.
- `rclpy.spin(node)` keeps the node active, processing callbacks until a shutdown signal is received.
- The `try-except-finally` block ensures graceful shutdown:
 - `try`: Runs the spinning loop to keep the node active.
 - `except KeyboardInterrupt`: Catches the interrupt signal (e.g., `Ctrl+C`) and explicitly destroys the node to release resources.
 - `finally`: Ensures that `rclpy.shutdown()` is called to clean up resources, regardless of how the spin loop is exited.

- `if __name__ == '__main__':` Ensures that the `main()` function is called only when the script is executed directly.

This detailed breakdown explains each component of the ROS 2 Python service node. For further details, refer to the [Creating a ROS 2 Service \(Python\)](#) tutorial.

3.4.3 Integrating the Python Service Node into the Package

Edit the `setup.py` file of your Python package to add an entry point for the service node:

```
entry_points={
    'console_scripts': [
        'server_exe = s3_py_srvcli.server:main',
    ],
}
```

3.4.4 Building, Sourcing, and Running the Python Service Node

From the root of your workspace, build the package:

```
$ colcon build --packages-select s3_py_srvcli
```

Then, source the workspace:

```
$ source install/setup.bash
```

Finally, run the service node:

```
$ ros2 run s3_py_srvcli server_exe
```

If everything is set up correctly, your node should be running and ready to receive requests.

3.4.5 Testing the Python Service

Open a new terminal, test the service using the following command:

```
$ ros2 service call /py_add_two_ints_srv example_interfaces/srv/AddTwoInts "{a: -2, b: -3}"
```

Alternatively, use `rqt_service_caller`:

```
$ ros2 run rqt_service_caller rqt_service_caller
```

and request the service on `/py_add_two_ints_service` with values `a: -5, b: -7`.

3.4.6 Summary of Python Service Node Key Identifiers

- `server.py`: Name of the source file for the Python service node.
- `py_server_node`: Name of the Python service node as defined in the constructor of the Python class.
- `server_exe`: Name of executable defined in `setup.py`.
- `/py_add_two_ints_service`: The name of the service provided by the Python server node.



3.5 Creating a ROS 2 Client Node in Python

3.5.1 Setting Up the Python Client Node

Open a terminal and navigate to the `src` directory of your Python package (e.g., `s3_py_srvcli`). Then, create a new Python file for the service node (e.g., `client.py`):

```
$ cd ros2_ws/src/s3_py_srvcli/s3_py_srvcli
$ touch client.py
```

3.5.2 Editing the Python Client Node

Open your workspace in VS Code:

```
$ cd ~/ros2_ws
$ code .
```

In VS Code, locate and open `client.py` and implement your client node code (code to be provided later).

Implementing the Python Client Node Code

Below is an example implementation of a simple ROS 2 client node in Python that uses the asynchronous service call API (`call_async()`).

```
#!/usr/bin/env python3
import sys
import rclpy
from rclpy.node import Node
from example_interfaces.srv import AddTwoInts

class PyClientAsync(Node):

    def __init__(self):
        super().__init__('py_client_async_node')
        self.get_logger().info("Python Client node has been started")
        self.client = self.create_client(AddTwoInts, 'py_add_two_ints_service')

        # Wait for the service to become available
        while not self.client.wait_for_service(timeout_sec=1.0):
            if not rclpy.ok():
                self.get_logger().error('Interrupted while waiting for the service.
                ↪ Exiting.')
            return
        self.get_logger().info('Waiting for service to become available...')
        # Prepare a persistent request object
        self.req = AddTwoInts.Request()
```



```

def send_request(self, a, b):
    self.req.a = a
    self.req.b = b
    return self.client.call_async(self.req)

def main(args=None):
    rclpy.init(args=args)
    if len(sys.argv) != 3:
        print("Usage: client_exe a b")
        sys.exit(1)
    node = PyClientAsync()
    future = node.send_request(int(sys.argv[1]), int(sys.argv[2]))
    # Spin until the future is complete
    rclpy.spin_until_future_complete(node, future)
    result = future.result()
    if result is not None:
        node.get_logger().info(
            'Received response: %d + %d = %d' %
            (int(sys.argv[1]), int(sys.argv[2]), result.sum)
        )
    else:
        node.get_logger().error('Failed to call service add_two_ints')
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

This code defines a client node that asynchronously calls the service `py_add_two_ints_service`, using `call_async()`, to add two integers.

Breaking Down the Client Node Code

We now examine each part of the code in detail to understand how the ROS 2 Python client node works.

1. Imports and Initialization

```

#!/usr/bin/env python3
import sys
import rclpy
from rclpy.node import Node
from example_interfaces.srv import AddTwoInts

```

- The shebang line (`#!/usr/bin/envpython3`) ensures that the script is executed with Python 3.
- `rclpy` provides the ROS 2 Python API.
- `Node` is the base class for creating ROS 2 nodes.
- `AddTwoInts` is the service message type imported from the `example_interfaces` package.



2. Defining the Node Class

```
class PyClientAsync(Node):

    def __init__(self):
        super().__init__('py_client_async_node')
        self.get_logger().info("Python Client node has been started")
        self.client = self.create_client(AddTwoInts, 'py_add_two_ints_service')

        # Wait for the service to become available
        while not self.client.wait_for_service(timeout_sec=1.0):
            if not rclpy.ok():
                self.get_logger().error('Interrupted while waiting for the service.
                ↪ Exiting.')
                return
            self.get_logger().info('Waiting for service to become available...')

        # Prepare a persistent request object
        self.req = AddTwoInts.Request()
```

- The `PyClientAsync` class inherits from `Node`, creating a new ROS 2 node.
- In the `__init__` method:
 - `super().__init__()` initializes the node with the name `py_client_async_node`.
 - `self.get_logger().info("PythonClientnodehasbeenstarted")` logs an informational message when the node starts.
 - `self.create_client()` creates a client to call the `AddTwoInts` service type under the name `py_add_two_ints_service`.
 - A loop waits for the service to become available, logging status messages. If the ROS system is interrupted during the wait, an error is logged and the function returns.
 - Once the service is available, a persistent request object is initialized with `AddTwoInts.Request()`.

3. Sending the Service Request

```
def send_request(self, a, b):
    self.req.a = a
    self.req.b = b
    return self.client.call_async(self.req)
```

- The `send_request` method assigns the integers `a` and `b` to the corresponding fields of the previously created request object.
- Then, it uses `call_async()` to initiate a non-blocking service call, returning a future object that will eventually contain the result.
- This approach allows the node to continue running other tasks while waiting for the service response.



4. The Main Function

```
def main(args=None):
    rclpy.init(args=args)
    if len(sys.argv) != 3:
        print("Usage: client_exe a b")
        sys.exit(1)
    node = PyClientAsync()
    future = node.send_request(int(sys.argv[1]), int(sys.argv[2]))
    # Spin until the future is complete
    rclpy.spin_until_future_complete(node, future)
    result = future.result()
    if result is not None:
        node.get_logger().info(
            'Received response: %d + %d = %d' %
            (int(sys.argv[1]), int(sys.argv[2]), result.sum)
        )
    else:
        node.get_logger().error('Failed to call service add_two_ints')
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

- `rclpy.init(args=args)` initializes ROS 2 communication by parsing command-line arguments and setting up the necessary middleware.
- `len(sys.argv) != 3` ensures that exactly two numeric arguments are passed from the command line. If not, a usage message is printed and the program exits.
- An instance of the `PyClientAsync` node is created.
- `rclpy.spin_until_future_complete()` processes callbacks until the service response is received, and the result is then logged.
- `node.destroy_node()` explicitly destroys the node to release resources.
- After execution, `rclpy.shutdown()` gracefully shuts down ROS 2, releasing all allocated resources.
- `if __name__ == '__main__':`: Ensures that the `main()` function is called only when the script is executed directly.

This detailed breakdown explains each component of the ROS 2 Python client node. For further details, refer to the [Creating a ROS 2 Client \(Python\)](#) tutorial.

3.5.3 Integrating the Python Client Node into the Package

Edit the `setup.py` file of your Python package to add an entry point for the client node while preserving any existing node setup (e.g., `server_exe`):

```
entry_points={
    'console_scripts': [
        'server_exe = s3_py_srvcli.server:main',
        'client_exe = s3_py_srvcli.client:main',
    ],
}
```



3.5.4 Building, Sourcing, and Running the Python Client Node

From the root of your workspace, build the package:

```
$ colcon build --packages-select s3_py_srvcli
```

Then, source the workspace:

```
$ source install/setup.bash
```

Finally, run the client node:

```
$ ros2 run s3_py_srvcli client_exe -11 -13
```

If everything is set up correctly, your node should execute, send a request to the service, and wait for the response. If the service is not available, the client will wait until it becomes accessible before proceeding.

3.5.5 Summary of Python Client Node Key Identifiers

- `client.py`: Name of the source file for the Python client node.
- `py_client_node`: Name of the Python client node as defined in the constructor of the Python class.
- `client_exe`: Name of executable defined in `setup.py`.
- `/py_add_two_ints_service`: The name of the service that the Python client node calls.

3.6 Practice Assignment: Running ROS 2 Services and Clients

In this assignment, you will run the ROS 2 service and client nodes implemented in both C++ and Python, as described in Sections 3.2, 3.3, 3.4, and 3.5. Capture evidence of successful execution using a screenshot and submit it in the designated assignment area on Canvas (within the ROS 2 module) as a PNG file. The filename must follow this format:

`FirstnameLastname_evidence_sX.png`

where `sX` corresponds to the session number. For example, a correct filename would be:

`EduardoDavila_evidence_s3.png`.

3.6.1 Executing ROS 2 Services and Clients

Running the Python Service and Client

After implementing and saving your Python nodes, and updating the `package.xml` and `setup.py` files, build your package and run the nodes. In three separate terminal sessions (or using the `Terminator` emulator), execute the following commands from the root of your workspace:

- In **Terminal 1**, build your package, source the setup file, and launch the Python service node:

```
$ colcon build --packages-select s3_py_srvcli
$ source install/setup.bash
$ ros2 run s3_py_srvcli server_exe
```



- In **Terminal 2**, source the setup file and run the Python client node:

```
$ source install/setup.bash
$ ros2 run s3_py_srvcli client_exe -11 -13
```

- In **Terminal 3**, send a service request directly from the command line:

```
$ ros2 service call /py_add_two_ints_service example_interfaces/srv/AddTwoInts "{a:
↪ -2, b: -3}"
```

You should observe that the `py_server` node processes service requests from both the `py_client` node and the command-line client in Terminal 3.

Running the C++ Service and Client

After implementing and saving your C++ nodes, and updating the `package.xml` and `CMakeLists.txt` files, build your package and run the nodes. In three additional terminal sessions (or three new sessions in **Terminator**), execute the following commands from the root of your workspace:

- In **Terminal 4**, build your package, source the setup file, and launch the C++ service node:

```
$ colcon build --packages-select s3_cpp_srvcli
$ source install/setup.bash
$ ros2 run s3_cpp_srvcli server_exe
```

- In **Terminal 5**, source the setup file and run the C++ client node:

```
$ source install/setup.bash
$ ros2 run s3_cpp_srvcli client_exe 11 13
```

- In **Terminal 6**, send a service request directly from the command line:

```
$ ros2 service call /cpp_add_two_ints_service example_interfaces/srv/AddTwoInts
↪ "{a: 2, b: 3}"
```

You should observe that the `cpp_server` node processes service requests from both the `cpp_client` node and the command-line client in Terminal 6.

Requesting a Service Using `rqt_service_caller`

Use the `rqt_service_caller` tool to send service requests to both the Python and C++ servers. In **Terminal 7**, execute the following command (without needing to enter the workspace directory):

```
$ ros2 run rqt_service_caller rqt_service_caller
```

Then, request a service call for:

- `/py_add_two_ints_service` by entering the values `a:-5,b:-7`.
- `/cpp_add_two_ints_service` by entering the values `a:5,b:7`.

You should observe that the `py_server` and `cpp_server` nodes successfully process the service requests sent via `rqt_service_caller`.

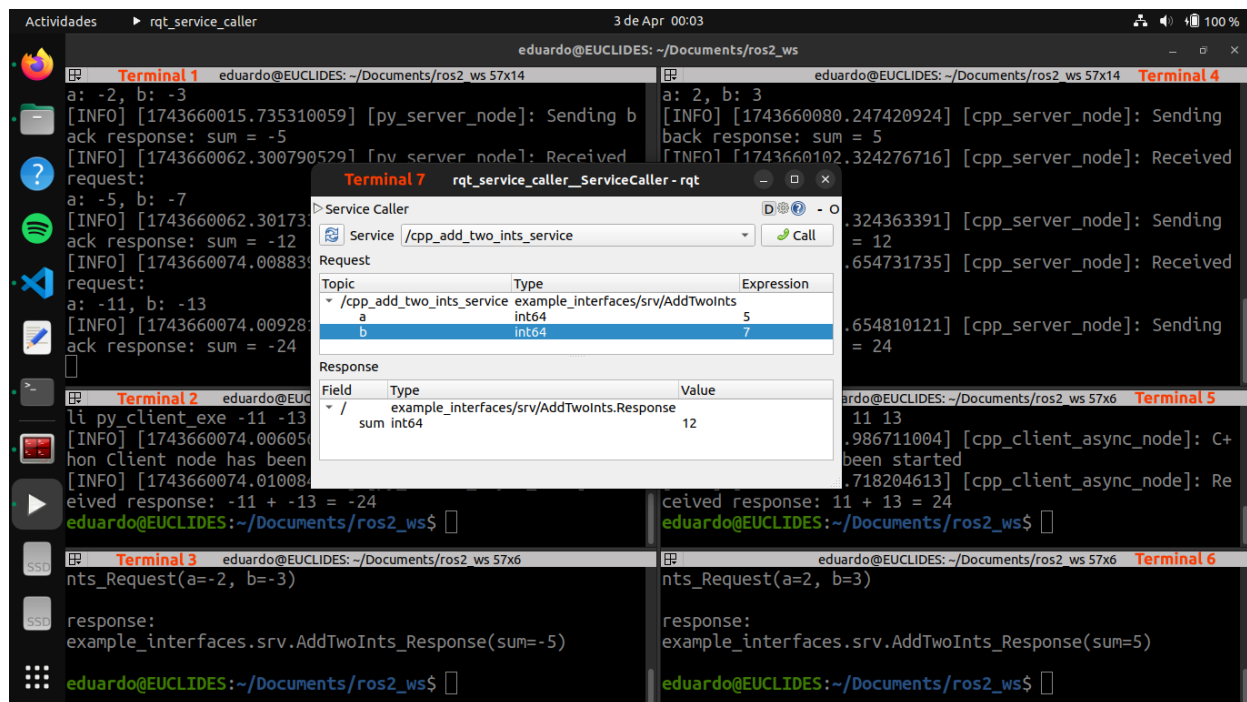


Figure 3.1: Example screenshot showing terminal outputs from the servers and clients in action.

3.6.2 Submission Instructions

1. Run the ROS 2 service and client nodes as, described above.
2. Capture a **screenshot showing the terminal outputs** from both the C++ and Python service and client nodes, along with the `rqt_service_caller` interface. See Figure 3.1 for reference.
3. Save the screenshot as a PNG file.
4. Name the file following this format: `FirstnameLastname_evidence_sX.png` (replace `X` with the week number and `Y` with the session number).
5. Submit the file as instructed by the course guidelines on Canvas.

Note: Your machine's username and hostname must be visible in the screenshot to verify the authenticity of your submission, as shown in Figure 3.1. Any submission that appears copied, unclear, or altered will be invalid and may receive a score of 0 for this assignment.

CHAPTER

4

ROS 2 Custom msg and srv Files

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✳ **Abstract** - In this chapter, we will introduce the concept of custom interfaces in the [Robot Operating System \(ROS\) 2](#), which allows you to define your own message and service types to facilitate communication between nodes. We will cover the process of creating custom message ([.msg](#)) and service ([.srv](#)) files, integrating them into a [ROS 2](#) package, and using them in C++ and Python nodes. By the end of this chapter, you will have a solid understanding of how to create and utilize custom interfaces in [ROS 2](#) to enable more complex interactions between your nodes. All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) , located in the [ros2_ws/src](#) folder, corresponding to the [ROS 2](#) packages with the [s4](#) prefix.

4.1 Introduction to ROS 2 Custom Interfaces

In previous chapters you utilized message and service interfaces to learn about topics, services, and simple publisher/subscriber (C++/Python) and service/client (C++/Python) nodes. The interfaces you used were predefined in those cases.

While it is good practice to use predefined interface definitions, you will probably need to define your own messages and services sometimes as well. This chapter will introduce you to the simplest method of creating custom interface definitions. For more details, refer to the guide on [Creating Custom msg and srv Files](#) .

4.1.1 Prerequisites

Before you begin, ensure that you are familiar with workspace and package creation, as described in Sections 2.2, 2.3, and 2.4. Additionally, this chapter also uses the source files created in the publisher/subscriber (C++ and Python) in Chapter 2 and service/client (C++ and Python) in Chapter 3 to try out the new custom messages.

Open a terminal and navigate to the `src` directory of your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.

Next, create the following packages in your workspace:

- C++ package for the custom *.msg* and *.srv* files:

```
$ ros2 pkg create s4_custom_interface --build-type ament_cmake --license Apache-2.0
→ --description "Your package description here"
```

- C++ package for the publisher/subscriber and service/client nodes:

```
$ ros2 pkg create s4_cpp_apps --build-type ament_cmake --dependencies rclcpp
→ s4_custom_interface --license Apache-2.0 --description "Your package
→ description here"
```

- Python package for the publisher/subscriber and service/client nodes:

```
$ ros2 pkg create s4_py_apps --build-type ament_python --dependencies rclpy
→ s4_custom_interface --license Apache-2.0 --description "Your package
→ description here"
```

You may replace `s4_custom_interface`, `s4_cpp_apps`, and `s4_py_apps` with package names of your preference.

The *.msg* and *.srv* files are required to be placed in directories called `msg` and `srv` respectively. Create the directories in `ros2_ws/src/s4_custom_interface`:

```
$ mkdir msg srv
```

4.2 Creating the *.msg* and *.srv* files

4.2.1 msg definition

In the `s4_custom_interface/msg` directory you just created, make a new file called `HardwareStatus.msg` with a few lines of code declaring its data structure:

```
int64 temperature
bool are_motors_up
string debug_message
builtin_interfaces/Time the_time
```

This is a custom message that transfers different message types, and uses a message from another message package (`builtin_interfaces/Time`).



Also in the `s4_custom_interface/msg` directory you just created, make a new file called `Sphere.msg` with the following content:

```
geometry_msgs/Point center
float64 radius
```

This custom message uses a message from another message package (`geometry_msgs/Point` in this case).

4.2.2 srv definition

Back in the `s4_custom_interface/srv` directory you just created, make a new file called `AddThreeInts.srv` with the following request and response structure:

```
int64 a
int64 b
int64 c
---
int64 sum
```

This is your custom service that requests a service with three integers named a, b, and c, and responds with an integer called sum.

4.3 CMakeLists.txt

To convert the interfaces you defined into language-specific code (like C++ and Python) so that they can be used in those languages, add the following lines to `CMakeLists.txt`:

```
# Added manually
find_package(builtin_interfaces REQUIRED)
find_package(geometry_msgs REQUIRED)
find_package(rosidl_default_generators REQUIRED)

# Added manually
rosidl_generate_interfaces(${PROJECT_NAME}
  "msg/HardwareStatus.msg"
  "msg/Sphere.msg"
  "srv/AddThreeInts.srv"
  DEPENDENCIES builtin_interfaces geometry_msgs # Add packages that above messages
  ↳ depend on, in this case builtin_interfaces for HardwareStatus.msg, and geometry_msgs
  ↳ for Sphere.msg
)
```

4.4 package.xml

Because the interfaces rely on `rosidl_default_generators` for generating language-specific code, you need to declare a build tool dependency on it. `rosidl_default_runtime` is a runtime or execution-stage dependency, needed to be able to use the interfaces later. The `rosidl_interface_packages` is the name of the dependency group that your package, `s4_custom_interface`, should be associated with, declared using the `<member_of_group>` tag.

Add the following lines within the `<package>` element of `package.xml`:

```
<!-- Added manually -->
<depend>builtin_interfaces</depend>
<depend>geometry_msgs</depend>

<test_depend>ament_lint_auto</test_depend>
<test_depend>ament_lint_common</test_depend>

<!-- Added manually -->
<buildtool_depend>rosidl_default_generators</buildtool_depend>
<exec_depend>rosidl_default_runtime</exec_depend>
<member_of_group>rosidl_interface_packages</member_of_group>
```

4.5 Build the *s4_custom_interface* package

Now that all the parts of your custom interfaces package are in place, you can build the package. In the root of your workspace (`/ros2_ws`), run the following command:

```
$ colcon build --packages-select s4_custom_interface
```

Now the interfaces will be discoverable by other ROS 2 packages.

4.6 Confirm msg and srv creation

In a new terminal, run the following command from within your workspace (`/ros2_ws`) to source it:

```
$ source install/setup.bash
```

Now you can confirm that your interface creation worked by using the `ros2 interface show` command:

```
$ ros2 interface show s4_custom_interface/msg/HardwareStatus
```

should return:

```
int64 temperature
bool are_motors_up
string debug_message
builtin_interfaces/Time the_time
  int32 sec
  uint32 nanosec
```

And

```
$ ros2 interface show s4_custom_interface/msg/Sphere
```

should return:

```
geometry_msgs/Point center
  float64 x
  float64 y
  float64 z
float64 radius
```



And

```
$ ros2 interface show s4_custom_interface/srv/AddThreeInts
```

should return:

```
int64 a
int64 b
int64 c
---
int64 sum
```

4.7 Test the new interfaces

For this step you can use the `s2_cpp_pubsub`, `s2_py_pubsub`, `s3_cpp_srvcli`, and `s3_py_srvcli` packages you created previously. A few simple modifications to the publisher/subscriber and service/client nodes, `CMakeLists.txt` and `package.xml` files will allow you to use your new interfaces.

4.7.1 Implementing `HardwareStatus.msg` and `Sphere.msg` in pub/sub nodes

With a few modifications to the publisher/subscriber package created in a previous tutorial (C++ or Python), you can see `Num.msg` in action. Since you will be changing the standard string msg to a numerical one, the output will be slightly different.

Publisher in C++

```
#include <chrono>
#include <memory>
#include "rclcpp/rclcpp.hpp"
#include "s4_custom_interface/msg/hardware_status.hpp"
#include "s4_custom_interface/msg/sphere.hpp"

using namespace std::chrono_literals;

class CppPublisher : public rclcpp::Node {
public:
    CppPublisher()
    : Node("cpp_publisher_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Publisher node has been started");
        publisher_ =
        ↪ this->create_publisher<s4_custom_interface::msg::HardwareStatus>("cpp_hs_topic",
        ↪ 10);
        publisher2_ =
        ↪ this->create_publisher<s4_custom_interface::msg::Sphere>("cpp_sphere_topic", 10);
        timer_ = this->create_wall_timer(500ms,
        ↪ std::bind(&CppPublisher::timer_callback, this));
    }
}
```



```

private:
    void timer_callback() {
        // HardwareStatus message
        auto message = s4_custom_interface::msg::HardwareStatus();
        message.temperature = 45;
        message.are_motors_up = true;
        message.debug_message = "Eduardo's C++ system runs smoothly";
        message.the_time = this->get_clock()->now();
        RCLCPP_INFO(this->get_logger(), "Publishing HardwareStatus: Temperature: %ld,
↪ Motors Up: %s, Debug Message: %s, Time: %d.%09d",
                    message.temperature,
                    message.are_motors_up ? "True" : "False",
                    message.debug_message.c_str(),
                    message.the_time.sec,
                    message.the_time.nanosec);

        publisher_->publish(message);

        // Sphere message
        auto message2 = s4_custom_interface::msg::Sphere();
        message2.center.x = 1.0;
        message2.center.y = 2.0;
        message2.center.z = 3.0;
        message2.radius = 4.0;

        RCLCPP_INFO(this->get_logger(), "Publishing Sphere: x: %f, y: %f, z: %f,
↪ Radius: %f",
                    message2.center.x,
                    message2.center.y,
                    message2.center.z,
                    message2.radius);

        publisher2_->publish(message2);
    }

    rclcpp::TimerBase::SharedPtr timer_;
    rclcpp::Publisher<s4_custom_interface::msg::HardwareStatus>::SharedPtr
↪ publisher_;
    rclcpp::Publisher<s4_custom_interface::msg::Sphere>::SharedPtr publisher2_;
};

int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppPublisher>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```



Publisher in Python

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s4_custom_interface.msg import HardwareStatus
from s4_custom_interface.msg import Sphere

class PyPublisher(Node):
    def __init__(self):
        super().__init__('py_publisher_node')
        self.get_logger().info("Python Publisher node has been started")
        self.publisher_ = self.create_publisher(HardwareStatus, 'py_hs_topic', 10)
        self.publisher2_ = self.create_publisher(Sphere, 'py_sphere_topic', 10)
        timer_period = 0.5 # seconds
        self.timer = self.create_timer(timer_period, self.timer_callback)

    def timer_callback(self):
        # HardwareStatus message
        msg = HardwareStatus()
        msg.temperature = 45
        msg.are_motors_up = True
        msg.debug_message = "Eduardo's Python system running smoothly"
        msg.the_time = self.get_clock().now().to_msg()
        self.get_logger().info("Publishing HardwareStatus: Temperature: %d, Motors Up:
        → %s, Debug Message: %s, Time: %d.%09d" % (
            msg.temperature,
            msg.are_motors_up,
            msg.debug_message,
            msg.the_time.sec,
            msg.the_time.nanosec))
        self.publisher_.publish(msg)

        # Sphere message
        msg2 = Sphere()
        msg2.center.x = 1.0
        msg2.center.y = 2.0
        msg2.center.z = 3.0
        msg2.radius = 4.0
        self.get_logger().info("Publishing Sphere: x: %f, y: %f, z: %f, Radius: %f" % (
            msg2.center.x,
            msg2.center.y,
            msg2.center.z,
            msg2.radius))
        self.publisher2_.publish(msg2)

def main(args=None):
    rclpy.init(args=args)
    node = PyPublisher()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
```



```

    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```

Subscriber in C++

```

#include <memory>
#include "rclcpp/rclcpp.hpp"
#include "s4_custom_interface/msg/hardware_status.hpp"
#include "s4_custom_interface/msg/sphere.hpp"

using std::placeholders::_1;
class CppSubscriber : public rclcpp::Node{
public:
    CppSubscriber()
    : Node("cpp_subscriber_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Subscriber node has been started");
        subscription_ =
        ↪ this->create_subscription<s4_custom_interface::msg::HardwareStatus>("py_hs_topic",
        ↪ 10, std::bind(&CppSubscriber::topic_callback, this, _1));
        subscription2_ =
        ↪ this->create_subscription<s4_custom_interface::msg::Sphere>("cpp_sphere_topic", 10,
        ↪ std::bind(&CppSubscriber::sphere_callback, this, _1));
    }
private:
    void topic_callback(const s4_custom_interface::msg::HardwareStatus::SharedPtr
    ↪ msg) {
        RCLCPP_INFO(this->get_logger(), "Received HardwareStatus: Temperature: %ld,
        ↪ Motors Up: %s, Debug Message: %s, Time: %d.%09d",
            msg->temperature,
            msg->are_motors_up ? "True" : "False",
            msg->debug_message.c_str(),
            msg->the_time.sec,
            msg->the_time.nanosec);
    }
    void sphere_callback(const s4_custom_interface::msg::Sphere::SharedPtr msg) {
        RCLCPP_INFO(this->get_logger(), "Received Sphere: x: %f, y: %f, z: %f, Radius:
        ↪ %f",
            msg->center.x,
            msg->center.y,
            msg->center.z,
            msg->radius);
    }
    rclcpp::Subscription<s4_custom_interface::msg::HardwareStatus>::SharedPtr
    ↪ subscription_;
    rclcpp::Subscription<s4_custom_interface::msg::Sphere>::SharedPtr
    ↪ subscription2_;
};

```



```
int main(int argc, char * argv[]) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppSubscriber>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}
```

Subscriber in Python

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s4_custom_interface.msg import HardwareStatus
from s4_custom_interface.msg import Sphere

class PySubscriber(Node):
    def __init__(self):
        super().__init__('py_subscriber_node')
        self.get_logger().info("Python Subscriber node has been started")
        self.subscription = self.create_subscription(
            HardwareStatus, 'cpp_hs_topic', self.listener_callback, 10)
        self.subscription2 = self.create_subscription(
            Sphere, 'py_sphere_topic', self.sphere_callback, 10)
        self.subscription # prevent unused variable warning
        self.subscription2 # prevent unused variable warning

    def listener_callback(self, msg):
        self.get_logger().info("Received HardwareStatus: Temperature: %d, Motors Up: %s,
        ↳ Debug Message: %s, Time: %d.%09d" % (
            msg.temperature,
            msg.are_motors_up,
            msg.debug_message,
            msg.the_time.sec,
            msg.the_time.nanosec))

    def sphere_callback(self, msg):
        self.get_logger().info("Received Sphere: x: %f, y: %f, z: %f, Radius: %f" % (
            msg.center.x,
            msg.center.y,
            msg.center.z,
            msg.radius))

def main(args=None):
    rclpy.init(args=args)
    node = PySubscriber()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
```



```

    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```

4.7.2 Implementing `AddThreeInts.srv` in service/client nodes

Service in C++

```

#include "rclcpp/rclcpp.hpp"
#include "s4_custom_interface/srv/add_three_ints.hpp"

class CppService : public rclcpp::Node{
public:
    CppService()
    : Node("cpp_server_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Server node has been started");
        service_ = this->create_service<s4_custom_interface::srv::AddThreeInts>(
            "cpp_add_three_ints_service",
            std::bind(&CppService::AddThreeInts_callback, this,
        ↪ std::placeholders::_1, std::placeholders::_2)
            );
        RCLCPP_INFO(this->get_logger(), "Service 'cpp_add_three_ints_service' is
        ↪ ready to receive requests");
    }

private:
    void AddThreeInts_callback(
        ↪ const std::shared_ptr<s4_custom_interface::srv::AddThreeInts::Request>
        ↪ request,
        ↪ std::shared_ptr<s4_custom_interface::srv::AddThreeInts::Response> response)
    ↪ {
        response->sum = request->a + request->b + request->c;
        RCLCPP_INFO(this->get_logger(), "Received request:\na: %ld, b: %ld, c:
        ↪ %ld", request->a, request->b, request->c);
        RCLCPP_INFO(this->get_logger(), "Sending back response: sum = %ld",
        ↪ response->sum);
    }

    rclcpp::Service<s4_custom_interface::srv::AddThreeInts>::SharedPtr service_;
};

int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppService>();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```



Service in Python

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s4_custom_interface.srv import AddThreeInts
class PyService(Node):
    def __init__(self):
        super().__init__('py_server_node')
        self.get_logger().info("Python Server node has been started")
        self.srv = self.create_service(AddThreeInts, 'py_add_three_ints_service',
        ↪ self.AddThreeInts_callback)
        self.get_logger().info("Service 'py_add_three_ints_service' is ready to receive
        ↪ requests")
    def AddThreeInts_callback(self, request, response):
        response.sum = request.a + request.b + request.c
        self.get_logger().info('Received request:\na: %d, b: %d, b: %d' % (request.a,
        ↪ request.b, request.c))
        self.get_logger().info('Sending back response: sum = %d' % (response.sum))
        return response
def main(args=None):
    rclpy.init(args=args)
    node = PyService()
    try:
        rclpy.spin(node)
    except:
        node.destroy_node()
    finally:
        rclpy.shutdown()
if __name__ == '__main__':
    main()
```

Client in C++

```
#include "rclcpp/rclcpp.hpp"
#include "s4_custom_interface/srv/add_three_ints.hpp"
using namespace std::chrono_literals;
class CppClientAsync : public rclcpp::Node{
public:
    CppClientAsync(int64_t a, int64_t b, int64_t c)
        : Node("cpp_client_async_node"), a_(a), b_(b), c_(c) {
        RCLCPP_INFO(this->get_logger(), "C++ Client node has been started");
        client_ = this->create_client<s4_custom_interface::srv::AddThreeInts>(
        ↪ "cpp_add_three_ints_service");
        // Wait for the service to become available
        while (!client_->wait_for_service(1s)) {
            if (!rclcpp::ok()) {
                RCLCPP_ERROR(this->get_logger(), "Interrupted while waiting for
                ↪ the service. Exiting.");
                return;
            }
        }
    }
```



```

        RCLCPP_INFO(this->get_logger(), "Waiting for service to become
↪ available...");
    }
    // Request is called
    send_request();
}

private:
    void send_request() {
        std::shared_ptr<s4_custom_interface::srv::AddThreeInts::Request> request =
↪ std::make_shared<s4_custom_interface::srv::AddThreeInts::Request>();
        request->a = a_;
        request->b = b_;
        request->c = c_;
        rclcpp::Client<s4_custom_interface::srv::AddThreeInts>::FutureAndRequestId
↪ future_and_request_id =
            client_->async_send_request(request);

↪ std::shared_future<s4_custom_interface::srv::AddThreeInts::Response::SharedPtr>
↪ result =
            future_and_request_id.future.share();
            // Spin until the future is complete
            if (rclcpp::spin_until_future_complete(this->get_node_base_interface(),
↪ result) ==
                rclcpp::FutureReturnCode::SUCCESS) {
                RCLCPP_INFO(this->get_logger(), "Received response: %ld + %ld + %ld =
↪ %ld", request.get()->a, request.get()->b, request.get()->c, result.get()->sum);
            }
            else {
                RCLCPP_ERROR(this->get_logger(), "Failed to call service
↪ add_three_ints");
            }
    }

    rclcpp::Client<s4_custom_interface::srv::AddThreeInts>::SharedPtr client_;
    int64_t a_;
    int64_t b_;
    int64_t c_;
};

int main(int argc, char **argv) {
    rclcpp::init(argc, argv);
    if (argc != 4) {
        RCLCPP_INFO(rclcpp::get_logger("rclcpp"), "Usage: client_exe a b c");
        return 1;
    }
    std::shared_ptr<rclcpp::Node> node = std::make_shared<CppClientAsync>(atoll(argv[1]),
↪ atoll(argv[2]), atoll(argv[3]));
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```



Client in Python

```
#!/usr/bin/env python3
import sys
import rclpy
from rclpy.node import Node
from s4_custom_interface.srv import AddThreeInts

class PyClientAsync(Node):
    def __init__(self):
        super().__init__('py_client_async_node')
        self.get_logger().info("Python Client node has been started")
        self.client = self.create_client(AddThreeInts, 'py_add_three_ints_service')
        # Wait for the service to become available
        while not self.client.wait_for_service(timeout_sec=1.0):
            if not rclpy.ok():
                self.get_logger().error('Interrupted while waiting for the service.
                ↪ Exiting.')
                return
            self.get_logger().info('Waiting for service to become available...')
        # Prepare a persistent request object
        self.req = AddThreeInts.Request()

    def send_request(self, a, b, c):
        self.req.a = a
        self.req.b = b
        self.req.c = c
        return self.client.call_async(self.req)

def main(args=None):
    rclpy.init(args=args)
    if len(sys.argv) != 4:
        print("Usage: client_exe a b c")
        sys.exit(1)
    node = PyClientAsync()
    future = node.send_request(int(sys.argv[1]), int(sys.argv[2]), int(sys.argv[3]))
    # Spin until the future is complete
    rclpy.spin_until_future_complete(node, future)
    result = future.result()
    if result is not None:
        node.get_logger().info(
            'Received response: %d + %d + %d = %d' %
            (int(sys.argv[1]), int(sys.argv[2]), int(sys.argv[3]), result.sum)
        )
    else:
        node.get_logger().error('Failed to call service add_three_ints')
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```



4.7.3 CMakeLists.txt

Add the following lines (C++ package only):

```
# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(s4_custom_interface REQUIRED)

add_executable(publisher_exe src/publisher.cpp)
ament_target_dependencies(publisher_exe rclcpp s4_custom_interface)

add_executable(subscriber_exe src/subscriber.cpp)
ament_target_dependencies(subscriber_exe rclcpp s4_custom_interface)

add_executable(server_exe src/server.cpp)
ament_target_dependencies(server_exe rclcpp s4_custom_interface)

add_executable(client_exe src/client.cpp)
ament_target_dependencies(client_exe rclcpp s4_custom_interface)

install(TARGETS
  publisher_exe
  subscriber_exe
  server_exe
  client_exe
  DESTINATION lib/${PROJECT_NAME})
```

4.7.4 setup.py

Add the following lines (Python package only):

```
entry_points={
    'console_scripts': [
        "publisher_exe = s4_py_apps.publisher:main",
        "subscriber_exe = s4_py_apps.subscriber:main",
        "server_exe = s4_py_apps.server:main",
        "client_exe = s4_py_apps.client:main"
    ],
},
```

4.7.5 package.xml

Add the following line (for both C++ and Python packages):

```
<depend>s4_custom_interface</depend>
```

4.7.6 Building the Packages and Running the Nodes

After making the above edits and saving all the changes, build the packages:




```
$ colcon build --packages-select s4_cpp_apps
$ colcon build --packages-select s4_py_apps
```

Then open eight new terminals, source `ros2_ws` in each, and run:

- In **terminal 1**, run the Python publisher node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps publisher_exe
```

- In **terminal 2**, run the Python subscriber node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps subscriber_exe
```

- In **terminal 3**, run the Python server node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps server_exe
```

- In **terminal 4**, run the Python client node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps client_exe -2 -3 -5
```

- In **terminal 5**, run the C++ publisher node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps publisher_exe
```

- In **terminal 6**, run the C++ subscriber node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps subscriber_exe
```

- In **terminal 7**, run the C++ server node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps server_exe
```

- In **terminal 8**, run the C++ client node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps client_exe 2 3 5
```

Additionally, you can open two new terminals, within your workspace, to run the `rqt_graph` and `rqt_service_caller` tools to visualize the current ROS 2 graph and request services to the server nodes, respectively.

4.8 Practice Assignment: ROS 2 Custom .msg and .srv Files

In this assignment, you will run the ROS 2 publisher/subscriber and service/client nodes implemented in both C++ and Python, as described in Section 4.7. Capture evidence of successful execution using two screenshots and submit them in the designated assignment area on Canvas (within the ROS 2 module) as PNG files. The filenames must follow this format:



FirstnameLastname_evidence_sX_Y.png

where **sX** corresponds to the session number, and **Y** is the evidence number. For example, correct filenames would be:

EduardoDavila_evidence_s4_1.png,
EduardoDavila_evidence_s4_2.png.

4.8.1 Executing ROS 2 Publishers, Subscribers, Services and Clients

After editing and saving your custom .msg and .srv files, as well as your C++ and Python nodes, and updating the CMakeLists.txt, setup.py, and package.xml files, build the packages:

```
$ colcon build --packages-select s4_custom_interface
$ colcon build --packages-select s4_cpp_apps
$ colcon build --packages-select s4_py_apps
```

Then, in eight separate terminal sessions (or using the Terminator emulator), source your ROS 2 workspace in each terminal and run the nodes as follows:

- In **terminal 1**, run the Python publisher node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps publisher_exe
```

- In **terminal 2**, run the Python subscriber node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps subscriber_exe
```

- In **terminal 3**, run the Python server node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps server_exe
```

- In **terminal 4**, run the Python client node:

```
$ source install/setup.bash
$ ros2 run s4_py_apps client_exe -2 -3 -5
```

- In **terminal 5**, run the C++ publisher node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps publisher_exe
```

- In **terminal 6**, run the C++ subscriber node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps subscriber_exe
```

- In **terminal 7**, run the C++ server node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps server_exe
```



- In **terminal 8**, run the C++ client node:

```
$ source install/setup.bash
$ ros2 run s4_cpp_apps client_exe 2 3 5
```

You should observe that:

1. The `py_publisher` node publishes the custom `HardwareStatus` and `Sphere` messages.
2. The `py_subscriber` node receives the custom `HardwareStatus` message from the `cpp_publisher` node, and receives the `Sphere` message from the `py_publisher` node.
3. The `py_server` node processes the service request from the `py_client` node.
4. The `cpp_publisher` node publishes the custom `HardwareStatus` and `Sphere` messages.
5. The `cpp_subscriber` node receives the custom `HardwareStatus` message from the `py_publisher` node, and receives the `Sphere` message from the `cpp_publisher` node.
6. The `cpp_server` node processes the service request from the `cpp_client` node.

4.8.2 Visualizing the ROS Graph with `rqt_graph`

To verify that the publisher, subscriber, service and client nodes are correctly connected, use the `rqt_graph` tool. Then, in **Terminal 9**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_graph rqt_graph
```

or simply:

```
$ rqt_graph
```

This will visualize the ROS 2 graph, showing the service nodes, as well as the topics connecting your publisher and subscriber nodes, as described above.

4.8.3 Submission Instructions

1. Run the ROS 2 service and client nodes as described above.
2. Capture a **screenshot showing the terminal outputs** from both the C++ and Python nodes (publishers, subscribers, services, and clients), as described in Section 4.8.1. See Figure 4.1 for reference.
3. Capture a **screenshot showing the ROS 2 graph** (from `rqt_graph`), illustrating the connections among the publisher and subscriber nodes, and the service nodes, as described in Section 4.8.2. See Figure 4.2 for reference.
4. Save the screenshots as PNG files.
5. Name the files following this format: `FirstnameLastname_evidence_sX_Y.png` (replace `X` with the session number and `Y` with the evidence number).
6. Submit the files as specified by the course guidelines on Canvas.

Note: Your machine's username and hostname must be visible in the screenshots to verify the authenticity of your submission, as shown in Figure 4.1. Any submission that appears copied, unclear, or altered will be considered invalid and may receive a score of 0.



The screenshot displays eight terminal windows arranged in a 4x2 grid, each showing the output of a different ROS 2 node. The nodes are organized into four pairs, each pair representing a specific functionality in both Python and C++:

- Terminal 1 (Python Publisher):** Outputs a bug message and a sphere description (x: 1.000000, y: 2.000000, z: 3.000000, Radius: 4.000000).
- Terminal 2 (Python Subscriber):** Receives the hardware status (Temperature: 45, Motors Up: True) and the sphere data from Terminal 1.
- Terminal 3 (Python Service):** Receives a request (a: -2, b: -3, b: -5) and sends back a response (sum = -10).
- Terminal 4 (Python Client):** Waits for the service and receives the response (sum = -10).
- Terminal 5 (C++ Publisher):** Outputs a bug message and a sphere description (x: 1.000000, y: 2.000000, z: 3.000000, Radius: 4.000000).
- Terminal 6 (C++ Subscriber):** Receives the hardware status and the sphere data from Terminal 5.
- Terminal 7 (C++ Service):** Receives a request (a: 2, b: 3, c: 5) and sends back a response (sum = 10).
- Terminal 8 (C++ Client):** Waits for the service and receives the response (sum = 10).

Figure 4.1: Example screenshot showing terminal outputs of the publishers, subscribers, services, and clients, in C++ and Python, in action.

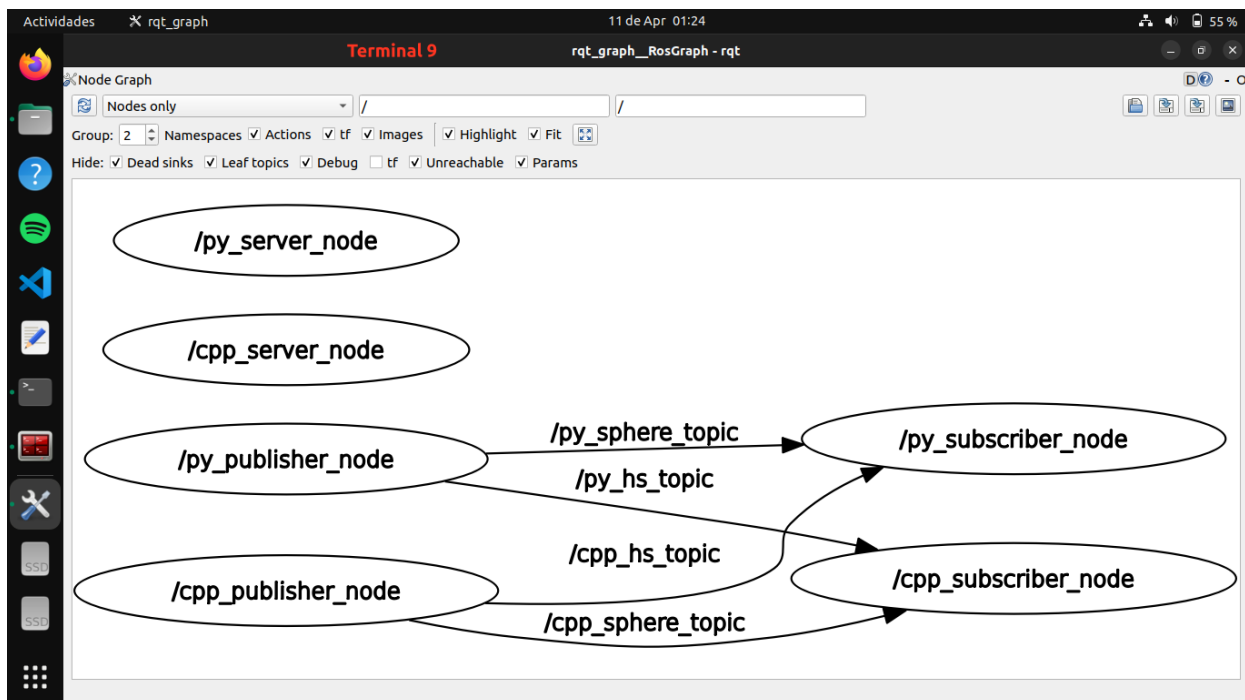


Figure 4.2: Example screenshot showing the ROS 2 graph of the publisher, subscriber, and service nodes.

CHAPTER

ROS 2 Parameters and YAML and Launch Files (Image Acquisition)

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 **Abstract** - In this module, we will explore how to create [Robot Operating System \(ROS\) 2](#) nodes that publish and subscribe to image data using both C++ and Python. We will cover the necessary setup, including defining parameters and launch files, and provide example implementations for both image publisher and subscriber nodes. By the end of this module, you will have a solid understanding of how to work with image data in [ROS 2](#) and how to integrate OpenCV for image processing tasks.

All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) , located in the `ros2_ws/src` folder, corresponding to the [ROS 2](#) packages with the `s5` prefix.

5.1 Introduction to ROS 2

A parameter is a configuration value of a node. You can think of parameters as node settings. A node can store parameters as integers, floats, booleans, strings, and lists. In [ROS 2](#), each node maintains its own parameters. Additionally, parameters can be set using YAML files, which provide a convenient way to manage and organize parameters for [ROS 2](#) nodes. YAML files allow you to define parameters in a structured format, making it easier to read and maintain configurations for your [ROS 2](#) applications. For more background on parameters, refer to the guide on [Understanding parameters](#)  and the concept document on [Parameters](#) .

On the other hand, a launch file is a script that allows you to start multiple nodes and set parameters in a single command. ROS 2 launch files can be written in XML, YAML, and Python. However, the recommended format for ROS 2 launch files is Python, as it provides greater flexibility and functionality, and it uses the `launch` library to define the nodes to be launched and their configurations. Each launch file can perform the following actions:

- Start a node, setting its namespace, and setting parameters in that node (using the args).
- Start a node and setting its namespace.
- Create a node to remap messages from one topic to another.
- Include another launch file.
- Include another launch file in another namespace.
- Setup command line arguments with defaults.

For more background on launch files, refer to the tutorial on [Creating a launch file](#) and the guide on [Using XML, YAML, and Python for ROS 2 Launch Files](#).

5.2 Prerequisites

Before you begin, ensure that you are familiar with workspace and package creation, as described in Sections 2.2, 2.3, and 2.4.

Open a terminal and navigate to the `src` directory of your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.

Next, create the following packages in your workspace:

- C++ package for the image publisher node:

```
$ ros2 pkg create s5_cpp_camera --build-type ament_cmake --dependencies rclcpp
→ sensor_msgs image_transport cv_bridge --license Apache-2.0 --description "Your
→ package description here"
```

- C++ package for the image subscriber node:

```
$ ros2 pkg create s5_cpp_camera_usr --build-type ament_cmake --dependencies rclcpp
→ sensor_msgs image_transport cv_bridge --license Apache-2.0 --description -"Your
→ package description here"
```

- Python package for the image publisher node:

```
$ ros2 pkg create s5_py_camera --build-type ament_python --dependencies rclpy
→ sensor_msgs image_transport cv_bridge --license Apache-2.0 --description "Your
→ package description here"
```

- Python package for the image subscriber node:

```
$ ros2 pkg create s5_py_camera_usr --build-type ament_python --dependencies rclpy
→ sensor_msgs image_transport cv_bridge --license Apache-2.0 --description -"Your
→ package description here"
```



You may replace `s5_cpp_camera`, `s5_cpp_camera_usr`, `s5_py_camera`, and `s5_py_camera_usr` with package names of your preference.

The parameter (`.yaml`) and launch (`.py`) files are required to be placed in directories called `config` and `launch`, respectively. Create the directories in `ros2_ws/src/s5_cpp_camera` for the C++ package, and in `ros2_ws/src/s5_py_camera` for the Python package:

```
$ mkdir config launch
```

5.3 Creating the parameter (`.yaml`) and launch (`.py`) Files

5.3.1 Parameter definition

In the `s5_cpp_camera/config` and `s5_py_camera/config` directories you just created, make a new file called `camera_params.yaml` with a few lines of code declaring its data structure. For the C++ package:

```
py_camera_node:
  ros__parameters:
    camera:
      deviceID: 0
      width: 640
      height: 480
```

And for the Python package:

```
cpp_camera_node:
  ros__parameters:
    camera:
      deviceID: 0
      width: 640
      height: 480
```

These are `.yaml` files that assign different parameters for the image publisher nodes.

5.3.2 Launch file definition

Back in the `s5_cpp_camera/launch` and `s5_py_camera/launch` directories you just created, make a new file called `camera_launch.py` with the following structure. For the C++ package:

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to the YAML file
    params_file = os.path.join(
        get_package_share_directory('s5_cpp_camera'),
        'config',
        'camera_params.yaml'
    )
```



```

return LaunchDescription([
    Node(
        package='s5_cpp_camera',
        name='cpp_camera_node',
        executable='camera_exe',
        parameters=[params_file],
        output = 'screen'
    )
])

```

And for the Python package:

```

import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to the YAML file
    params_file = os.path.join(
        get_package_share_directory('s5_py_camera'),
        'config',
        'camera_params.yaml'
    )

    return LaunchDescription([
        Node(
            package='s5_py_camera',
            name='py_camera_node',
            executable='camera_exe',
            parameters=[params_file],
            output = 'screen'
        )
    ])

```

These are your launch files that run the image publisher nodes and setting the `deviceId`, `width`, and `height` parameters.

5.4 CMakeLists.txt

To make accessible the parameter (`.yaml`) and launch (`.py`) files you defined in the C++ package, add the following lines to `CMakeLists.txt`:

```

# Added manually
install(DIRECTORY
    config
    launch
    DESTINATION share/${PROJECT_NAME}
)

```



5.5 setup.py

To make accessible the parameter (.yaml) and launch (.py) files you defined in the Python package, add the data files while preserving any existing file setup, with the following lines to `setup.py`:

```
data_files=[
    ('share/ament_index/resource_index/packages',
     ['resource/' + package_name]),
    ('share/' + package_name, ['package.xml']),
    ('share/' + package_name + '/config', ['config/camera_params.yaml']),
    ('share/' + package_name + '/launch', ['launch/camera_launch.py'])
],
```

5.6 package.xml

Make sure the required dependencies are defined within the `package.xml` files, for all the C++ packages and the Python packages:

```
<depend>sensor_msgs</depend>
<depend>image_transport</depend>
<depend>cv_bridge</depend>
```

Note:

For packages with launch files, it is a good idea to add an `exec_depend` dependency on the `ros2launch` package in the `package.xml` file from each package that contains launch files:

```
<exec_depend>ros2launch</exec_depend>
```

This helps make sure that the `ros2 launch` command is available after building the package(s). It also ensures that all launch file formats (XML, YAML, or Python) are recognized.

5.7 Creating a ROS 2 Image Publisher Node with C++

5.7.1 Setting Up the C++ Node

Navigating to the Package Directory

Open a terminal and navigate to your C++ package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s5_cpp_camera/src
```

Replace `s5_cpp_camera` with your actual package name.



Creating the C++ Node File

Create a new C++ file for your image publisher node, for example, `camera.cpp`:

```
$ touch camera.cpp
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `camera.cpp` file.

5.7.2 Editing the C++ Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `camera.cpp` file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 image publisher node in C++:

Image Publisher in C++

```
#include <rclcpp/rclcpp.hpp>
#include <sensor_msgs/msg/image.hpp>
#include <image_transport/image_transport.hpp>
#include <cv_bridge/cv_bridge.h>
#include <opencv2/opencv.hpp>
#include <chrono>

class CameraNode : public rclcpp::Node {
public:
    CameraNode()
        : Node("cpp_camera_node"), frame_count_(0), fps_(0.0) {
        RCLCPP_INFO(this->get_logger(), "C++ Camera node has been started");

        // Declare parameters with default values
        this->declare_parameter<int>("camera.deviceID", 0);
        this->declare_parameter<int>("camera.width", 960);
        this->declare_parameter<int>("camera.height", 540);

        // Get parameters
        this->get_parameter("camera.deviceID", deviceID_);
        this->get_parameter("camera.width", width_);
        this->get_parameter("camera.height", height_);
```

```

    // Open the camera
    cap_.open(deviceID_);
    if (!cap_.isOpened()) {
        RCLCPP_ERROR(this->get_logger(), "Failed to open camera");
        rclcpp::shutdown();
    }

    // Set camera properties
    cap_.set(cv::CAP_PROP_FRAME_WIDTH, width_);
    cap_.set(cv::CAP_PROP_FRAME_HEIGHT, height_);
    std::cout << "Set Frame Width: " << width_ << std::endl;
    std::cout << "Set Frame Height: " << height_ << std::endl;

    cv::Mat frame;
    cap_ >> frame;

    // Current cam roperties
    std::cout << "Current camera properties:" << std::endl;
    std::cout << "Device ID: " << deviceID_ << std::endl;
    std::cout << "Frame Width: " << frame.cols << std::endl;
    std::cout << "Frame Height: " << frame.rows << std::endl;

    // Initialize start time
    start_time_ = std::chrono::steady_clock::now();
}

~CameraNode() {
    if (cap_.isOpened()) {
        cap_.release();
        RCLCPP_INFO(this->get_logger(), "Camera resource released.");
    }
}

void init() {
    // Initialize ImageTransport and create a publisher
    image_transport::ImageTransport it(shared_from_this());
    pub_ = it.advertise("camera/image_raw", 1);

    // Start the timer for capturing and publishing images
    timer_ = this->create_wall_timer(
        std::chrono::milliseconds(16),
        std::bind(&CameraNode::CaptureAndPublish, this));
}

private:
    void CaptureAndPublish() {
        cv::Mat frame;
        cap_ >> frame;
        if (!frame.empty()) {
            // Increment frame count
            frame_count_++;
        }
    }
}

```



```

        // Calculate FPS every second
        std::chrono::time_point<std::chrono::steady_clock> now =
→ std::chrono::steady_clock::now();
        std::chrono::duration<double> elapsed = now - start_time_;
        if (elapsed.count() >= 1.0) {
            fps_ = frame_count_ / elapsed.count();
            frame_count_ = 0;
            start_time_ = now;
        }

        // Convert FPS to string
        std::ostringstream fps_text;
        fps_text << "FPS: " << std::fixed << std::setprecision(2) << fps_;

        // Put FPS text on the frame
        cv::putText(frame, fps_text.str(), cv::Point(10,
→ 30), cv::FONT_HERSHEY_SIMPLEX, 1.0, cv::Scalar(0, 255, 0), 2);

        // Convert OpenCV image to ROS message
        std_msgs::msg::Header header;
        header.stamp = this->now();
        sensor_msgs::Image::SharedPtr msg = cv_bridge::CvImage(header, "bgr8",
→ frame).toImageMsg();

        // Publish the image
        pub_.publish(msg);
    }
}

int deviceID_;
int width_;
int height_;
cv::VideoCapture cap_;
image_transport::Publisher pub_;
rclcpp::TimerBase::SharedPtr timer_;

// FPS calculation variables
std::chrono::steady_clock::time_point start_time_;
int frame_count_;
double fps_;
};

int main(int argc, char** argv) {
    rclcpp::init(argc, argv);
    std::shared_ptr<CameraNode> node = std::make_shared<CameraNode>();
    node->init(); // Call the initialization function after construction
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```



5.7.3 Integrating the C++ Image Publisher Node into the Package

Modify `CMakeLists.txt` of your C++ package (`s5_cpp_camera`) to add an executable for the image publisher node:

```
# Added manually
add_executable(camera_exe src/camera.cpp)
ament_target_dependencies(camera_exe rclcpp sensor_msgs image_transport cv_bridge)
target_link_libraries(camera_exe
    ${OpenCV_LIBRARIES}
)

# Added manually
install(TARGETS
    camera_exe
    DESTINATION lib/${PROJECT_NAME}
)

# Added manually
install(DIRECTORY
    config
    launch
    DESTINATION share/${PROJECT_NAME}
)
```

5.7.4 Summary of C++ Image Publisher Node Key Identifiers

- `camera.cpp`: Name of the source file for the C++ image publisher node.
- `cpp_camera_node`: Name of the C++ image publisher node as defined in the constructor of the C++ class.
- `camera_exe`: Name of the executable defined in `CMakeLists.txt`.
- `camera/image_raw`: Name of the topic provided by the C++ image publisher node.

5.8 Creating a ROS 2 Image Subscriber Node with C++

5.8.1 Setting Up the C++ Node

Navigating to the Package Directory

Open a terminal and navigate to your C++ package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s5_cpp_camera_usr/src
```

Replace `s5_cpp_camera_usr` with your actual package name.

Creating the C++ Node File

Create a new C++ file for your image subscriber node, for example, `camera_usr.cpp`:

```
$ touch camera_usr.cpp
```



Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `camera_usr.cpp` file.

5.8.2 Editing the C++ Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `camera_usr.cpp` file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 image subscriber node in C++:

Image Subscriber in C++

```
#include <rclcpp/rclcpp.hpp>
#include <sensor_msgs/msg/image.hpp>
#include <image_transport/image_transport.hpp>
#include <cv_bridge/cv_bridge.h>
#include <opencv2/opencv.hpp>

class ImageSubscriber : public rclcpp::Node {
public:
    ImageSubscriber()
        : Node("cpp_camera_user_node") {
        RCLCPP_INFO(this->get_logger(), "C++ Camera User node has been started");
    }
    ~ImageSubscriber() {
        cv::destroyAllWindows();
        RCLCPP_INFO(this->get_logger(), "All OpenCV windows destroyed.");
    }

    void init() {
        // Initialize ImageTransport and create a subscriber using image_transport
        image_transport::ImageTransport it(shared_from_this());
        subscription_ = it.subscribe("camera/image_raw", 1,
        ↪ std::bind(&ImageSubscriber::image_callback, this, std::placeholders::_1));
    }
}
```



```

private:
    void image_callback(const sensor_msgs::msg::Image::ConstSharedPtr & msg) {
        try {
            // Convert ROS Image message to OpenCV image
            cv::Mat cv_image = cv_bridge::toCvShare(msg, "bgr8")->image;
            // Display the image
            cv::imshow("C++ Camera User", cv_image);
            cv::waitKey(1);
            // ImageSubscriber::main_process(cv_image);
        }
        catch (cv_bridge::Exception & e) {
            RCLCPP_ERROR(this->get_logger(), "cv_bridge exception: %s", e.what());
        }
    }

    void main_process(const cv::Mat image) {
        // Resize the image to half its original size
        cv::Mat resized_image;
        cv::resize(image, resized_image, cv::Size(), 0.5, 0.5);

        // Convert the resized image to grayscale
        cv::Mat gray_image;
        cv::cvtColor(resized_image, gray_image, cv::COLOR_BGR2GRAY);

        // Apply standard histogram equalization
        cv::Mat equalized_image;
        cv::equalizeHist(gray_image, equalized_image);

        // Apply CLAHE
        cv::Mat clahe_image;
        cv::Ptr<cv::CLAHE> clahe = cv::createCLAHE(0.5, cv::Size(8, 8));
        clahe->apply(gray_image, clahe_image);

        // Concatenate images horizontally
        cv::Mat concatenated_image;
        std::vector<cv::Mat> images = {gray_image, equalized_image, clahe_image};
        cv::hconcat(images, concatenated_image);

        // Display the concatenated images
        cv::imshow("Grayscale - HE - CLAHE (CPP)", concatenated_image);
        cv::waitKey(1);
    }

    image_transport::Subscriber subscription_;
};

int main(int argc, char** argv) {
    rclcpp::init(argc, argv);
    std::shared_ptr<ImageSubscriber> node = std::make_shared<ImageSubscriber>();
    node->init();
    rclcpp::spin(node);
    node.reset();
    rclcpp::shutdown();
    return 0;
}

```



5.8.3 Integrating the C++ Image Subscriber Node into the Package

Modify `CMakeLists.txt` of your C++ package (`s5_cpp_camera_usr`) to add an executable for the image subscriber node:

```
# Added manually
add_executable(camera_usr_exe src/camera_usr.cpp)
ament_target_dependencies(camera_usr_exe rclcpp sensor_msgs image_transport cv_bridge)
target_link_libraries(camera_usr_exe
    ${OpenCV_LIBRARIES}
)

# Added manually
install(TARGETS
    camera_usr_exe
    DESTINATION lib/${PROJECT_NAME}
)
```

5.8.4 Summary of C++ Image Subscriber Node Key Identifiers

- `camera_usr.cpp`: Name of the source file for the C++ image subscriber node.
- `cpp_camera_user_node`: Name of the C++ image publisher node as defined in the constructor of the C++ class.
- `camera_usr_exe`: Name of the executable defined in `CMakeLists.txt`.
- `camera/image_raw`: Name of the topic that the C++ image subscriber node is subscribed.

5.9 Creating a ROS 2 Image Publisher Node with Python

5.9.1 Setting Up the Python Node

Navigating to the Package Directory

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s5_py_camera/s5_py_camera
```

Replace `s5_py_camera` with your actual package name.

Creating the Python Node File

Create a new Python file for your image publisher node, for example, `camera.py`:

```
$ touch camera.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `camera.py` file.



5.9.2 Editing the Python Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `camera.py` file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 image publisher node in Python:

Image Publisher in Python

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge
import cv2
from std_msgs.msg import Header
import time

class CameraNode(Node):
    def __init__(self):
        super().__init__('py_camera_node')
        self.get_logger().info("Python Camera node has been started")

        # Declare parameters with default values
        self.declare_parameter('camera.deviceID', 0)
        self.declare_parameter('camera.width', 960)
        self.declare_parameter('camera.height', 540)

        # Get parameters
        self.deviceID =
            self.get_parameter('camera.deviceID').get_parameter_value().integer_value
        self.width =
            self.get_parameter('camera.width').get_parameter_value().integer_value
        self.height =
            self.get_parameter('camera.height').get_parameter_value().integer_value

        # Open the camera
        self.cap = cv2.VideoCapture(self.deviceID)
        if not self.cap.isOpened():
            self.get_logger().error('Failed to open camera')
            rclpy.shutdown()
```



```

# Set camera properties
self.cap.set(cv2.CAP_PROP_FRAME_WIDTH, self.width)
self.cap.set(cv2.CAP_PROP_FRAME_HEIGHT, self.height)
print(f"Set Frame Width: {self.width}")
print(f"Set Frame Height: {self.height}")

ret, frame = self.cap.read()

# Current cam roperties
print(f"Current camera properties:")
print(f"Device ID: {self.deviceID}")
print(f"Frame Width: {frame.shape[1]}")
print(f"Frame Height: {frame.shape[0]}")

# Initialize start time
self.start_time = time.time()
self.fps = 0.0

# Initialize CvBridge and create a publisher
self.bridge = CvBridge()
self.publisher = self.create_publisher(Image, 'camera/image_raw', 1)

# Start the timer for capturing and publishing images
self.timer = self.create_timer(0.016, self.CaptureAndPublish)
self.frame_count = 0

def CaptureAndPublish(self):
    ret, frame = self.cap.read()
    if ret:
        # Increment frame count
        self.frame_count += 1

        # Calculate FPS every second
        elapsed_time = time.time() - self.start_time
        if elapsed_time >= 1.0:
            self.fps = self.frame_count / elapsed_time
            self.frame_count = 0
            self.start_time = time.time()

        # Convert FPS to string
        fps_text = f"FPS: {self.fps:.2f}"

        # Put FPS text on the frame
        cv2.putText(frame, fps_text, (10, 30), cv2.FONT_HERSHEY_SIMPLEX, 1.0, (0,
        ↪ 255, 0), 2)

        # Convert OpenCV image to ROS message
        header = Header()
        header.stamp = self.get_clock().now().to_msg()
        image_message = self.bridge.cv2_to_imgmsg(frame, encoding='bgr8')
        image_message.header = header

        # Publish the image
        self.publisher.publish(image_message)

```



```
def main(args=None):
    rclpy.init(args=args)
    node = CameraNode()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        node.cap.release()
        node.get_logger().info("Camera resource released.")
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

5.9.3 Integrating the Python Image Publisher Node into the Package

Modify `setup.py` of your Python package (`s5_py_camera`) to add an executable for the image publisher node:

```
entry_points={
    'console_scripts': [
        'camera_exe = s5_py_camera.camera:main',
    ],
}
```

5.9.4 Summary of Python Image Publisher Node Key Identifiers

- `camera.py`: Name of the source file for the Python image publisher node.
- `py_camera_node`: Name of the Python image publisher node as defined in the constructor of the Python class.
- `camera_exe`: Name of the executable defined in `setup.py`.
- `camera/image_raw`: Name of the topic provided by the Python image publisher node.

5.10 Creating a ROS 2 Image Subscriber Node with Python

5.10.1 Setting Up the Python Node

Navigating to the Package Directory

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s5_py_camera_usr/s5_py_camera_usr
```

Replace `s5_py_camera_usr` with your actual package name.



Creating the Python Node File

Create a new Python file for your image subscriber node, for example, `camera_usr.py`:

```
$ touch camera_usr.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `camera_usr.py` file.

5.10.2 Editing the Python Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `camera_usr.py` file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 image subscriber node in Python:

Image Subscriber in Python

```
import rclpy
from rclpy.node import Node
from sensor_msgs.msg import Image
from cv_bridge import CvBridge, CvBridgeError
import cv2
import numpy as np

class ImageSubscriber(Node):
    def __init__(self):
        super().__init__('py_camera_user_node')
        self.get_logger().info("Python Camera User node has been started")

        # Initialize CvBridge and create a subscriber
        self.bridge = CvBridge()
        self.subscription = self.create_subscription(Image, 'camera/image_raw',
            ↪ self.image_callback, 1)

    def image_callback(self, msg):
        try:
            # Convert ROS Image message to OpenCV image
            cv_image = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
```



```
        # Display the image
        cv2.imshow('Python Camera User', cv_image)
        cv2.waitKey(1)
        # self.main_process(cv_image)
    except CvBridgeError as e:
        self.get_logger().error(f'CvBridge Error: {e}')

def main_process(self, image):
    # Get the height and width
    height, width = image.shape[:2]

    # Resize the image
    resized_image = cv2.resize(image, (int(0.5*width), int(0.5*height)))

    # Convert the image to grayscale
    gray_image = cv2.cvtColor(resized_image, cv2.COLOR_BGR2GRAY)

    # Apply histogram equalization
    equalized_image = cv2.equalizeHist(gray_image)

    # create a CLAHE object (Arguments are optional).
    clahe = cv2.createCLAHE(clipLimit=0.5, tileGridSize=(8,8))
    clahe_image = clahe.apply(gray_image)

    # Display the images, stacking them side-by-side
    stacked_img = np.hstack((gray_image, equalized_image, clahe_image))
    cv2.imshow("Grayscale - HE - CLAHE (PY)", stacked_img)
    cv2.waitKey(1)

def main(args=None):
    rclpy.init(args=args)
    node = ImageSubscriber()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        cv2.destroyAllWindows()
        node.get_logger().info("All OpenCV windows destroyed.")
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```



5.10.3 Integrating the Python Image Subscriber Node into the Package

Modify `setup.py` of your Python package (`s5_py_camera_usr`) to add an executable for the image subscriber node:

```
entry_points={
    'console_scripts': [
        'camera_usr_exe = s5_py_camera_usr.camera_usr:main',
    ],
},
```

5.10.4 Summary of Python Image Subscriber Node Key Identifiers

- `camera_usr.py`: Name of the source file for the Python image subscriber node.
- `py_camera_user_node`: Name of the Python image publisher node as defined in the constructor of the Python class.
- `camera_usr_exe`: Name of the executable defined in `setup.py`.
- `camera/image_raw`: Name of the topic that the Python image subscriber node is subscribed.

5.11 Building the Packages and Running the Nodes

As a good practice, you can run `rosdep` in the root of the workspace (`ros2_ws`) to check for missing dependencies before building:

```
$ rosdep install -i --from-path src --rosdistro humble -y
```

After making the above edits and saving all the changes, in the root of your workspace (e.g., `/ros2_ws`), build the packages:

```
$ colcon build --packages-select s5_cpp_camera
$ colcon build --packages-select s5_cpp_camera_usr
$ colcon build --packages-select s5_py_camera
$ colcon build --packages-select s5_py_camera_usr
```

Then, in three terminals, source `ros2_ws` in each, and run:

- In **terminal 1**, run either the C++ image publisher node or the Python image publisher node:

```
$ source install/setup.bash
$ ros2 run s5_cpp_camera camera_exe
```

- In **terminal 2**, run the C++ image subscriber node:

```
$ source install/setup.bash
$ ros2 run s5_cpp_camera_usr camera_usr_exe
```

- In **terminal 3**, run the Python image subscriber node:

```
$ source install/setup.bash
$ ros2 run s5_py_camera_usr camera_usr_exe
```

Additionally, you can open two additional terminals to run the `rqt_graph` and `rqt_image_view` tools to visualize the current ROS 2 graph and visualize the enabled image topic, respectively.



5.12 Practice Assignment: ROS 2 Parameters and YAML and Launch files (Image Acquisition)

In this assignment, you will run the ROS 2 image publisher and subscriber nodes implemented in both C++ and Python, as described in Sections 5.7, 5.8, 5.9, and 5.10. Capture evidence of successful execution using two screenshots and submit them in the designated assignment area on Canvas (within the ROS 2 module) as PNG files. The filenames must follow this format:

`FirstnameLastname_evidence_sX_Y.png`

where `sX` corresponds to the session number, and `Y` is the evidence number. For example, correct filenames would be:

`EduardoDavila_evidence_s5_1.png`,
`EduardoDavila_evidence_s5_2.png`.

5.12.1 Executing ROS 2 Image Publishers and Subscribers

After editing and saving your C++ and Python nodes, and updating the `CMakeLists.txt`, `setup.py`, and `package.xml` files, build the packages:

```
$ colcon build --packages-select s5_cpp_camera
$ colcon build --packages-select s5_cpp_camera_usr
$ colcon build --packages-select s5_py_camera
$ colcon build --packages-select s5_py_camera_usr
```

Then, in three separate terminal sessions (or using the Terminator emulator), source your ROS 2 workspace in each terminal and run the nodes as follows:

- In **terminal 1**, run either the C++ image publisher node or the Python image publisher node:

```
$ source install/setup.bash
$ ros2 run s5_cpp_camera camera_exe
```

- In **terminal 2**, run the C++ image subscriber node:

```
$ source install/setup.bash
$ ros2 run s5_cpp_camera_usr camera_usr_exe
```

- In **terminal 3**, run the Python image subscriber node:

```
$ source install/setup.bash
$ ros2 run s5_py_camera_usr camera_usr_exe
```

You should observe that:

1. The `cpp_camera` `py_camera` node publishes the image message.
2. The `cpp_camera_usr` and `py_camera_usr` nodes receive the image message.



5.12.2 Visualizing the image topic with `rqt_image_view`

To verify that the publisher node is correctly publishing the image topic, use the `rqt_image_view` tool. Then, in **Terminal 4**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_image_view rqt_image_view
```

This will visualize the image acquisition provided by either the C++ publisher node or the Python publisher node.

5.12.3 Visualizing the ROS Graph with `rqt_graph`

To verify that the publisher and subscriber nodes are correctly connected, use the `rqt_graph` tool. Then, in **Terminal 5**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_graph rqt_graph
```

or simply:

```
$ rqt_graph
```

This will visualize the [ROS 2](#) graph, showing the topic connecting your publisher and subscriber nodes, as described above.

5.12.4 Submission Instructions

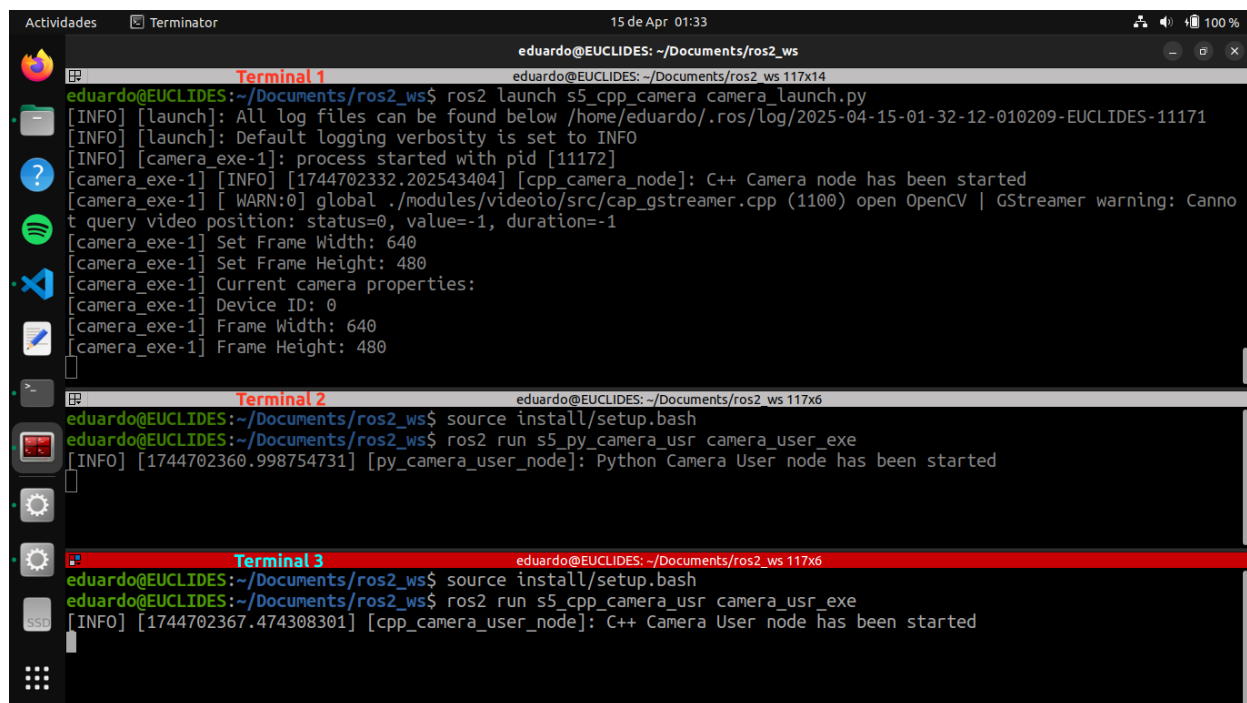


Figure 5.1: Example screenshot showing terminal outputs of the publisher and subscribers, in C++ and Python, in action.



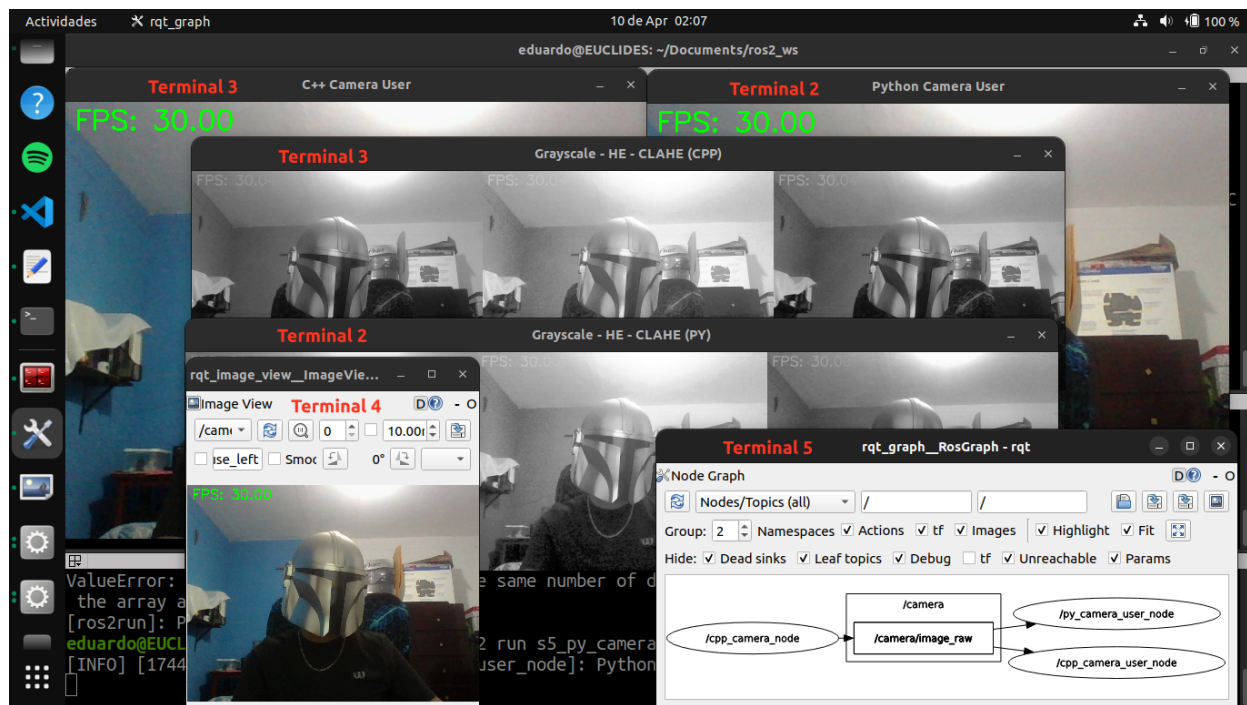


Figure 5.2: Example screenshot showing the image windows and the ROS 2 graph of the publisher and subscriber nodes.

1. Run the ROS 2 image publisher and subscriber nodes as described above.
2. Capture a **screenshot showing the terminal outputs** from both the C++ and Python nodes (publisher and subscribers), as described in Section 5.12.1. See Figure 5.1 for reference.
3. Capture a **screenshot showing the image windows and the ROS 2 graph** (from OpenCV, rqt_image_view and rqt_graph), illustrating the image publication and subscription, and the connections among the publisher and subscriber nodes, as described in Sections 5.12.1, 5.12.2, and 5.12.3. See Figure 5.2 for reference.
4. Save the screenshots as PNG files.
5. Name the files following this format: `FirstnameLastname_evidence_sX_Y.png` (replace X with the session number and Y with the evidence number).
6. Submit the files as specified by the course guidelines on Canvas.

Note: Your machine's username and hostname must be visible in the screenshots to verify the authenticity of your submission, as shown in Figure 5.1. Any submission that appears copied, unclear, or altered will be considered invalid and may receive a score of 0.

CHAPTER

6

URDF Modeling and Launch Files (Robot States)

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✳ **Abstract** - This chapter provides an introduction to the [Unified Robot Description Format \(URDF\)](#) and its application in modeling and visualizing robots within the [Robot Operating System \(ROS\) 2](#) ecosystem. We will explore various [URDF](#) models, ranging from simple geometric shapes to more complex articulated robots, some of them including physical properties such as collision and inertia, and even, the use of meshes for creating detailed robot models. Additionally, we will cover the necessary prerequisites for working with [URDF](#) files, including package creation, installation of required packages, and recommended [VS Code](#) extensions for an enhanced development experience. Moreover, we will discuss the necessary setup, including defining the joint state publisher nodes and launch files, to visualize the [URDF](#) models in [ROS Visualization \(RViz\)](#). All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) , located in the `ros2_ws/src` folder, corresponding to the [ROS 2](#) packages with the `s6` prefix.

6.1 Introduction

[Unified Robot Description Format \(URDF\)](#) is an XML-based format used to describe the physical configuration, geometry, and properties of robots in [ROS 2](#). It provides a standardized way to represent the structure of a robot, including its links, joints, and visual and collision properties.

URDF files are essential for simulating robots in ROS 2 environments, as they allow for accurate visualization and interaction with the robot's model in tools like **ROS Visualization (RViz)**.

6.2 Prerequisites

6.2.1 Workspace and Package Creation

Before getting started, make sure you are familiar with workspace and package creation, as described in Sections 2.2, 2.3, and 2.4.

Open a terminal and navigate to the **src** directory of your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.

Next, create the following packages in your workspace:

- A C++ package to contain the **URDF** models and state publisher nodes:

```
$ ros2 pkg create s6_cpp_urdf --build-type ament_cmake --dependencies rclcpp  
→ geometry_msgs sensor_msgs tf2_ros tf2_geometry_msgs --license Apache-2.0  
→ --description "Your package description here"
```

- A Python package to contain the **URDF** models and state publisher nodes:

```
$ ros2 pkg create s6_py_urdf --build-type ament_python --dependencies rclpy  
→ geometry_msgs sensor_msgs tf2_ros tf_transformations --license Apache-2.0  
→ --description "Your package description here"
```

Feel free to replace `s6_cpp_urdf` and `s6_py_urdf` with names of your preference.

The launch (`.py`), mesh (`.STL` and `.dae`), **RViz** (`.rviz`), and **URDF** (`.urdf`) files should be organized into directories named **launch**, **meshes**, **rviz**, and **urdf**, respectively. Create these directories in both the C++ and Python packages:

```
$ mkdir launch meshes rviz urdf
```

6.2.2 Installing Required Packages

Before proceeding, ensure that the following library and packages are installed. Update your package list (apt repository) and install them with:

```
$ sudo apt update  
$ sudo apt install ros-humble-joint-state-publisher  
$ sudo apt install ros-humble-urdf-tutorial  
$ sudo apt install ros-humble-tf-transformations  
$ sudo apt install ros-humble-tf2-tools # or ros-humble-tf2*  
$ pip3 install transforms3d
```

Also, from the root of your workspace, use **rosdep** to check for any other missing dependencies before building:

```
$ rosdep install -i --from-path src --rosdistro humble -y
```



If this is your first time using **rosdep**, you will most likely encounter the following error related to its initialization:

```
ERROR: your rosdep installation has not been initialized yet. Please run:

sudo rosdep init
rosdep update
```

Given this, run the commands as suggested by the error message to initialize **rosdep**:

```
$ sudo rosdep init
Wrote /etc/ros/rosdep/sources.list.d/20-default.list
Recommended: please run

rosdep update
$ rosdep update
```

Finally, run **rosdep** again and build your workspace to ensure that all dependencies are installed correctly and packages are properly configured.

6.2.3 Installing Recommended VS Code Extensions

For an improved development experience, we recommend installing the following **VS Code** extensions, already listed in Section 1.5.2:

- **URDF** by *smilerobotics* - Adds syntax highlighting and support for **URDF** files.
- **URDF Visualizer** by *morningfrog* - Allows to preview **URDF** files within the editor.

6.3 Exploring and Visualizing URDF (.urdf) Files

In this section, we will explore and analyze various visual **URDF** models. Some models are simple and consist of just a few lines, whereas others are more complex, exceeding 200 lines of **XML**.

To help you understand and experiment with these models, you may choose one of the following options:

- **Copy and paste** the contents of the provided **URDF** files directly into new files within the **meshes** directory of your own C++ and Python packages.
- **Download or clone** the entire **ROS 2** course repository and explore the models directly.

Each **URDF** model will be briefly described, and where appropriate, a preview of its structure will be provided. For larger files, only the most relevant sections will be highlighted, with explanations of components such as **link**, **joint**, **visual**, **inertial**, and **collision** elements.

At this stage, our main objective is to understand the geometric structure of the robot and how its parts are connected. In some models, we will incorporate articulation, as well as visual and physical properties.

We recommend becoming familiar with the **XML** specifications that define **URDF** modeling. For reference, consult the official documentation on the [<link>](#) element, which describes the kinematic and dynamic properties of a link, and the [<joint>](#) element, which defines the corresponding kinematic and dynamic properties of a joint, as illustrated in Figure 6.1.

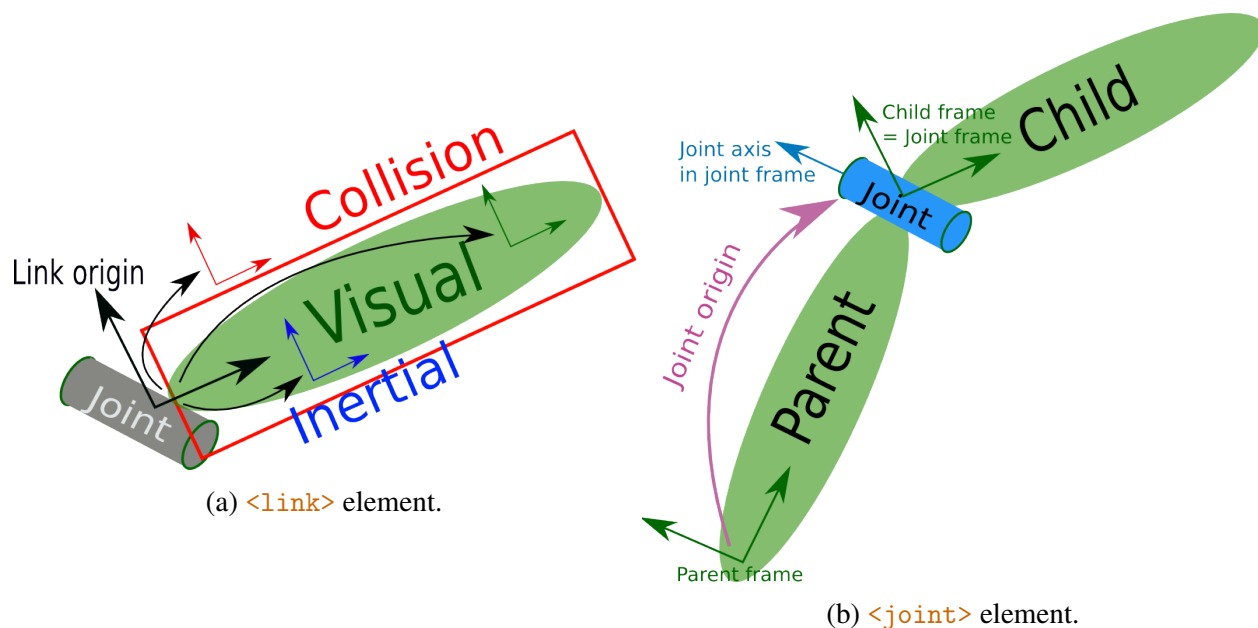


Figure 6.1: XML specifications for URDF modeling.

6.3.1 One Shape

We will begin by exploring some basic shapes using simple URDF models. These serve as straightforward examples to introduce key URDF concepts.

`0_sphere.urdf`

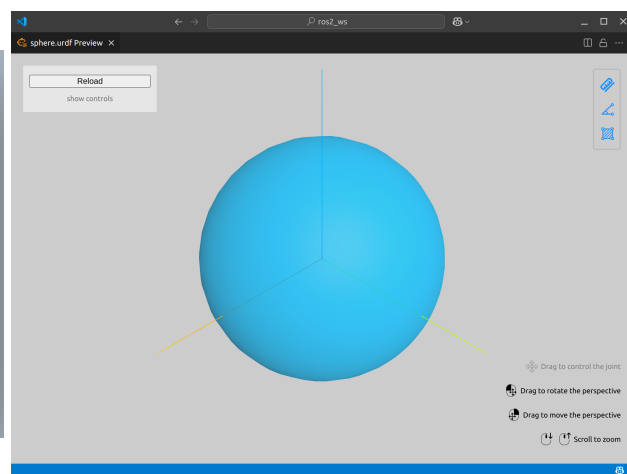
Source: [src/s6_cpp_urdf/urdf/0_sphere.urdf](#) or [src/s6_py_urdf/urdf/0_sphere.urdf](#)

```

1  <?xml version="1.0"?>
2  <robot name="sphere_robot">
3    <link name="base_link">
4      <visual>
5        <!-- <origin rpy="0 0 0" xyz="0 0 0.2"/> -->
6        <geometry>
7          <sphere radius="0.2"/>
8        </geometry>
9        <material name="blue">
10         <color rgba="0.0 0.5 1.0 0.75" />
11        </material>
12      </visual>
13    </link>
14  </robot>

```

(a) URDF file.



(b) URDF visualization.

Figure 6.2: URDF modeling of a sphere using the URDF development and visualization extensions.

In plain terms, the XML code defines a robot named `sphere_robot` with a single link called `base_link`. The link includes one visual element: a sphere with a radius of 0.2 m, as defined and displayed in Figure 6.2. Whereas the structure may seem verbose for such a simple model, it lays the foundation for more complex and articulated URDF models.

Key observations:

- The *fixed frame* refers to the coordinate frame used as the reference for the simulation environment. In this case, it is defined by the only link, `base_link`.
- By default, the origin of the sphere's geometry is at its center. As a result, half of the sphere is correctly placed below the grid plane in the visualization.

`1_box.urdf`

Source: [src/s6_cpp_urdf/urdf/1_box.urdf](#) or [src/s6_py_urdf/urdf/1_box.urdf](#)

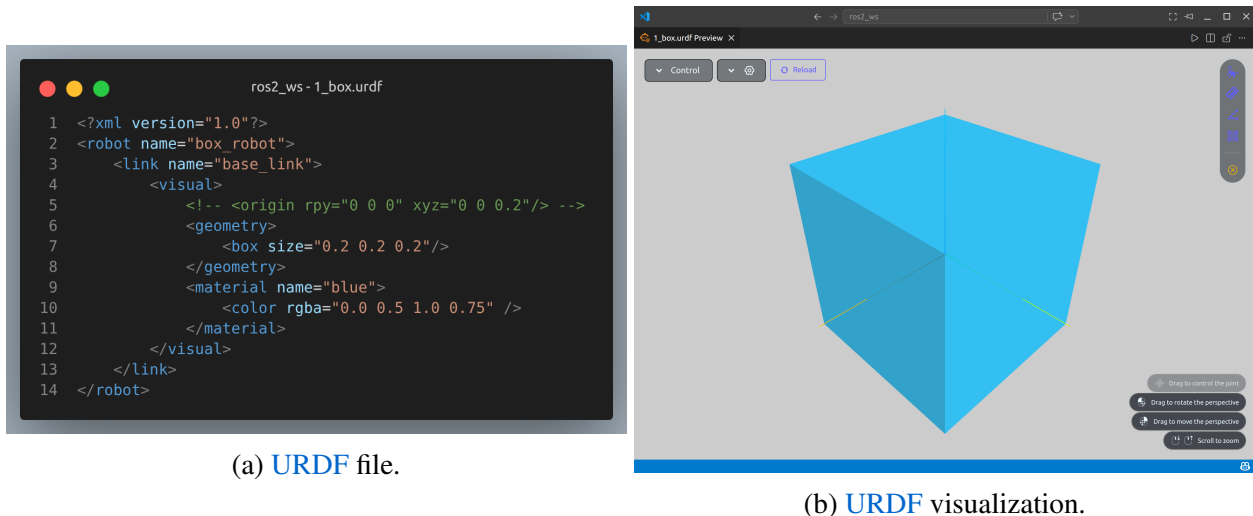


Figure 6.3: URDF modeling of a sphere using the URDF development and visualization extensions.

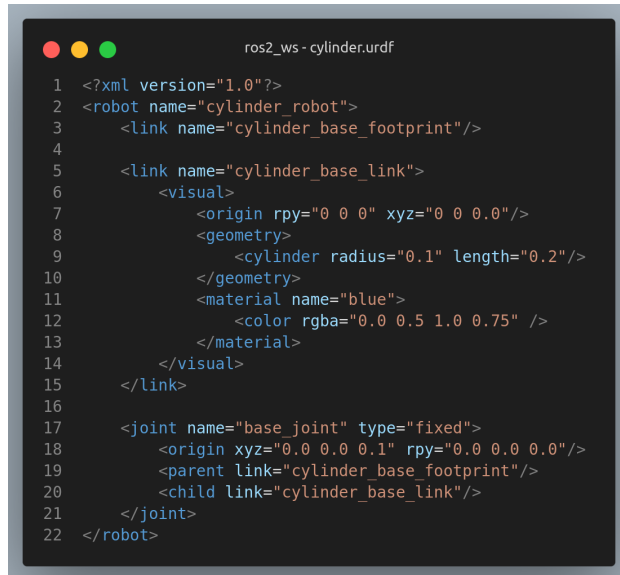
In plain terms, the XML code defines a robot named `box_robot` with a single link called `base_link`. The link includes one visual element: a box with dimensions of 0.2 m x 0.2 m x 0.2 m, as defined and displayed in Figure 6.3. Whereas the structure may seem verbose for such a simple model, it lays the foundation for more complex and articulated URDF models.

Key observations:

- The *fixed frame* refers to the coordinate frame used as the reference for the simulation environment. In this case, it is defined by the only link, `base_link`.
- By default, the origin of the box's geometry is at its center. As a result, the box is correctly placed on the grid plane in the visualization.

2_cylinder.urdf

Source: [src/s6_cpp_urdf/urdf/2_cylinder.urdf](#) or [src/s6_py_urdf/urdf/2_cylinder.urdf](#)

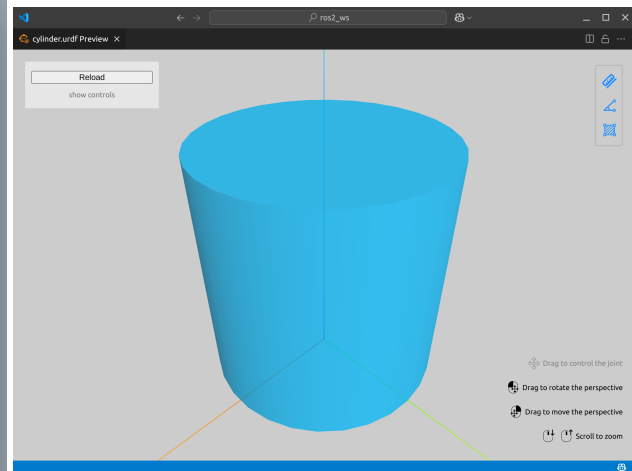


```

1 <?xml version="1.0"?>
2 <robot name="cylinder_robot">
3   <link name="cylinder_base_footprint"/>
4
5   <link name="cylinder_base_link">
6     <visual>
7       <origin rpy="0 0 0" xyz="0 0 0.0"/>
8       <geometry>
9         <cylinder radius="0.1" length="0.2"/>
10      </geometry>
11      <material name="blue">
12        <color rgba="0.0 0.5 1.0 0.75" />
13      </material>
14    </visual>
15  </link>
16
17  <joint name="base_joint" type="fixed">
18    <origin xyz="0.0 0.0 0.1" rpy="0.0 0.0 0.0"/>
19    <parent link="cylinder_base_footprint"/>
20    <child link="cylinder_base_link"/>
21  </joint>
22 </robot>

```

(a) URDF file.



(b) URDF visualization.

Figure 6.4: URDF modeling of a cylinder using the URDF development and visualization extensions.

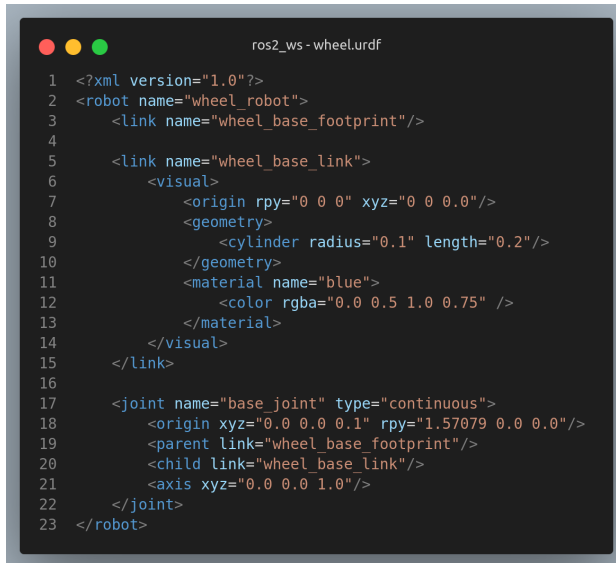
The XML code defines a robot named `cylinder_robot`, composed of a parent link called `cylinder_base_footprint` and a child link called `cylinder_base_link`. These are connected via a fixed joint named `base_joint`, which positions the child link's coordinate frame 0.1 m above that of the parent link. The robot contains a single visual element, a cylinder with a radius of 0.1 m and a height of 0.2 m, as defined and displayed in Figure 6.4.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `cylinder_base_footprint` link.
- By default, the origin of the cylinder's geometry is at its base. As a result, the cylinder is correctly placed on the grid plane in the visualization.

3_wheel.urdf

Source: [src/s6_cpp_urdf/urdf/3_wheel.urdf](#) or [src/s6_py_urdf/urdf/3_wheel.urdf](#)

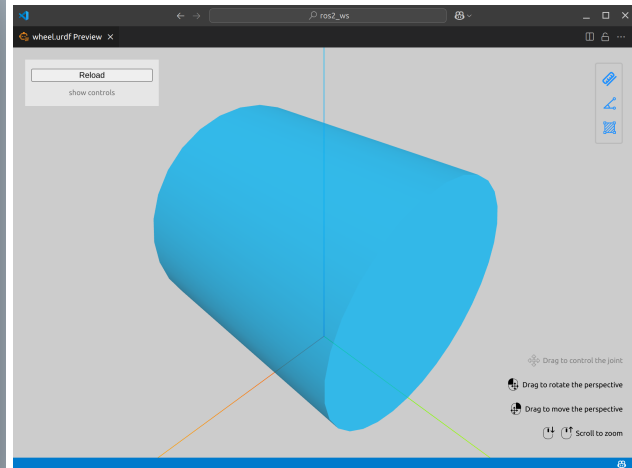


```

1 <?xml version="1.0"?>
2 <robot name="wheel_robot">
3   <link name="wheel_base_footprint"/>
4
5   <link name="wheel_base_link">
6     <visual>
7       <origin rpy="0 0 0" xyz="0 0 0.0"/>
8       <geometry>
9         <cylinder radius="0.1" length="0.2"/>
10      </geometry>
11      <material name="blue">
12        <color rgba="0.0 0.5 1.0 0.75" />
13      </material>
14    </visual>
15  </link>
16
17  <joint name="base_joint" type="continuous">
18    <origin xyz="0.0 0.0 0.1" rpy="1.57079 0.0 0.0"/>
19    <parent link="wheel_base_footprint"/>
20    <child link="wheel_base_link"/>
21    <axis xyz="0.0 0.0 1.0"/>
22  </joint>
23 </robot>

```

(a) URDF file.



(b) URDF visualization.

Figure 6.5: URDF modeling of a wheel using the URDF development and visualization extensions.

The XML code defines a robot named `wheel_robot`, composed of a parent link called `wheel_base_footprint` and a child link called `wheel_base_link`. These are connected via a continuous joint named `base_joint`, which positions the child link's coordinate frame 0.1 m above that of the parent link. The child link is also rotated by $\frac{\pi}{2}$ radians around the x-axis (roll), aligning the cylinder's axis of symmetry with the x-axis. The joint allows continuous rotation about the z-axis of the child link, simulating the spinning behavior of a wheel. The visual element is a cylinder (representing the wheel) with a radius of 0.1 m and a height of 0.2 m, as defined and displayed in Figure 6.5.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `wheel_base_footprint` link.
- By default, the wheel is visualized as a rotated cylinder, aligned horizontally due to the roll transformation. Its origin is positioned at the base of its geometry, so it appears correctly placed on the grid.

6.3.2 Multiple Shapes

Now let's explore how to add multiple shapes and links to form a more complex robot model.

4_robot.urdf

Source: [src/s6_cpp_urdf/urdf/4_robot.urdf](#) or [src/s6_py_urdf/urdf/4_robot.urdf](#)

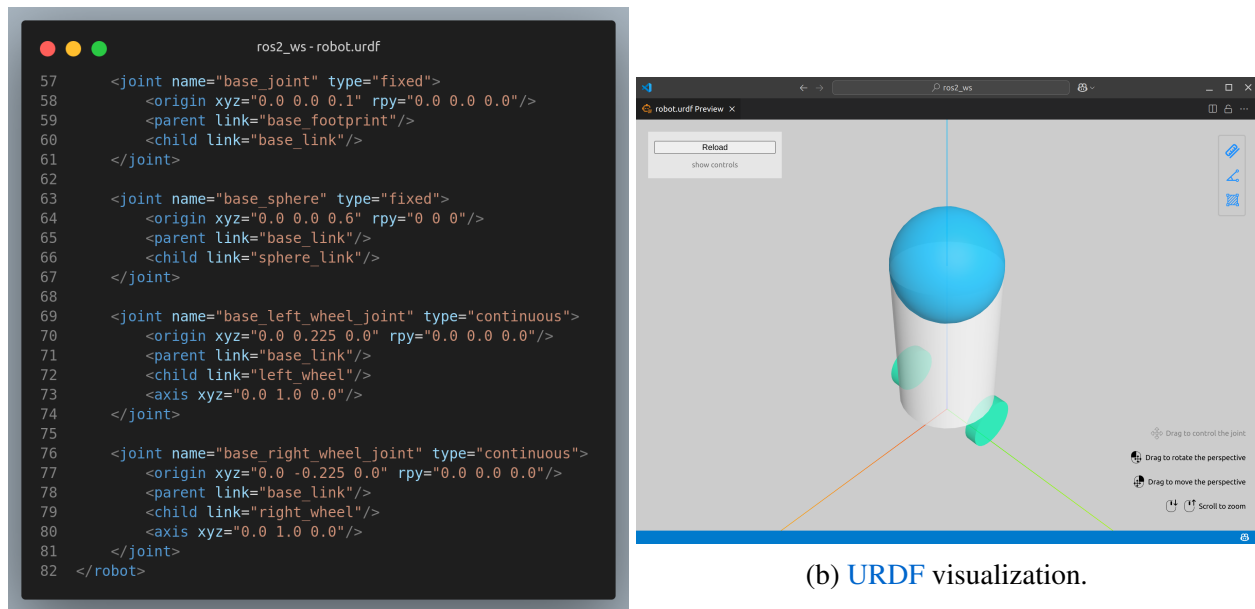


Figure 6.6: URDF modeling of a robot using the URDF development and visualization extensions.

The XML code defines a robot named `my_robot`, composed of multiple links connected by both fixed and continuous joints. The main fixed joint, `base_joint`, positions the `base_link` 0.1 m above the `base_footprint` link. Additional joints connect the robot's body and its wheels, allowing continuous rotation of the wheels about their respective y-axis, thus simulating spinning behavior. The robot includes several visual elements: a blue sphere, a white cylinder, and two additional green cylinders representing the wheels, as defined and displayed in Figure 6.6.

Key observations:

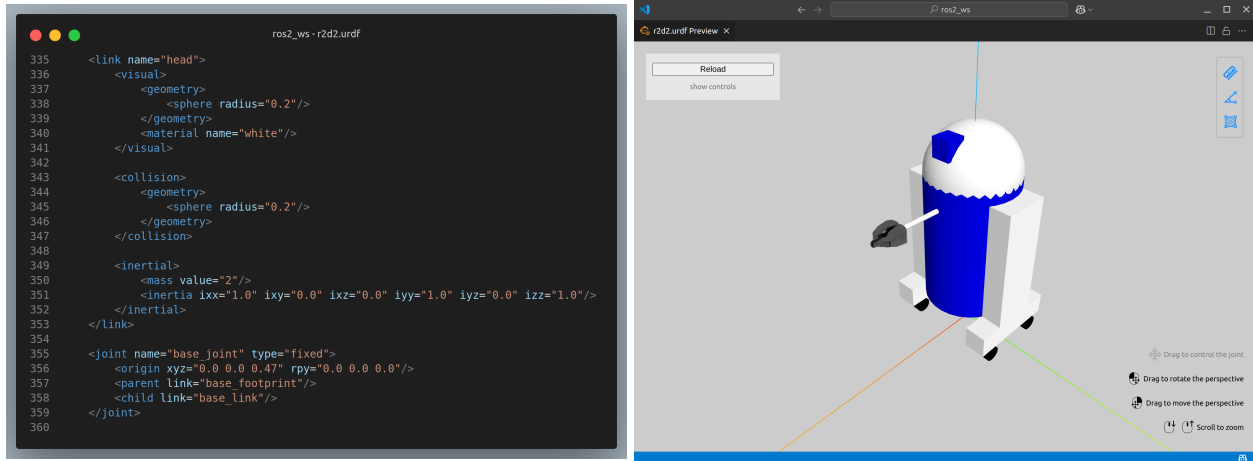
- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `base_footprint` link.
- By default, the origin of the robot's geometry is placed at its base, ensuring that the entire robot model is correctly aligned with the grid plane in the visualization.

6.3.3 Physical Properties

Now, we will look at how to add some basic physical properties to your URDF model and how to specify its collision and inertial properties.

`r2d2.urdf`

Source: [src/s6_cpp_urdf/urdf/r2d2.urdf](#) or [src/s6_py_urdf/urdf/r2d2.urdf](#)



(a) URDF file.

(b) URDF visualization.

Figure 6.7: URDF modeling of a R2D2 using the URDF development and visualization extensions.

The XML code defines a robot named `r2d2_physics`, composed of multiple links connected by fixed, continuous, prismatic, and revolute joints. The main fixed joint, `base_joint`, positions the `base_link` 0.47 m above the `base_footprint` link. Additional joints connect the robot's body, wheels, and gripper, allowing continuous rotation of the wheels about their respective y-axes to simulate spinning behavior, and prismatic motion of the gripper along its respective x-axis. The R2D2 model includes several visual elements: a white sphere, blue and white boxes, and black cylinders representing the wheels, as defined and displayed in Figure 6.7.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `base_footprint` link.
- By default, the origin of the R2D2's geometry is placed at its base, ensuring that the entire robot model is correctly aligned with the grid plane in the visualization.

r2d2_2.urdf

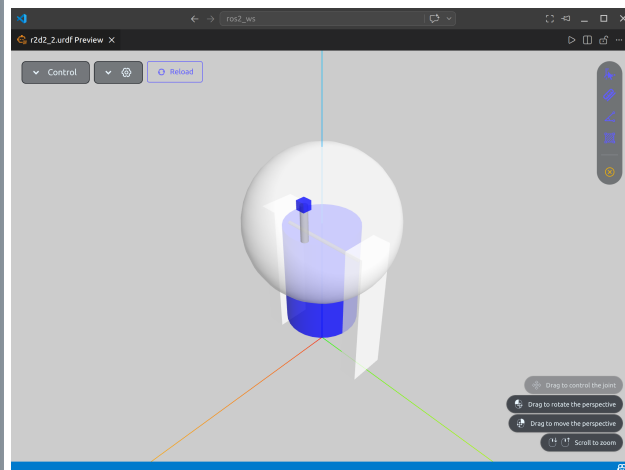
Source: [src/s6_cpp_urdf/urdf/r2d2_2.urdf](#) or [src/s6_py_urdf/urdf/r2d2_2.urdf](#)

```

1 <?xml version="1.0"?>
2 <robot name="r2d2_2">
3   <link name="base_footprint"/>
4
5   <link name="axis">
6     <visual>
7       <origin xyz="0 0 0" rpy="1.57 0 0"/>
8       <geometry>
9         <cylinder radius="0.01" length=".5"/>
10      </geometry>
11      <material name="gray">
12        <color rgba=".2 .2 .2 1"/>
13      </material>
14    </visual>
15
16    <collision>
17      <origin xyz="0 0 0" rpy="1.57 0 0"/>
18      <geometry>
19        <cylinder radius="0.01" length=".5"/>
20      </geometry>
21      <contact_coefficients mu="0" kp="1000.0" kd="1.0"/>
22    </collision>
23  </link>
24
25  <joint name="base_joint" type="fixed">
26    <origin xyz="0.0 0.0 0.7" rpy="0.0 0.0 0.0"/>
27    <parent link="base_footprint"/>
28    <child link="axis"/>
29  </joint>

```

(a) URDF file.



(b) URDF visualization.

Figure 6.8: URDF modeling of a R2D2 using the URDF development and visualization extensions.

The XML code defines a robot named `r2d2_2`, composed of multiple links connected by fixed, continuous, prismatic, and revolute joints. The main fixed joint, `base_joint`, positions the `base_link` 0.7 m above the `base_footprint` link. Additional joints connect the robot's body and periscope, allowing prismatic motion of the periscope along its respective z-axis. The R2D2 model includes several visual elements: a sphere, boxes, and cylinders, as defined and displayed in Figure 6.8.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `base_footprint` link.
- By default, the origin of the R2D2's geometry is placed at its base, ensuring that the entire robot model is correctly aligned with the grid plane in the visualization.

tello_uav.urdf

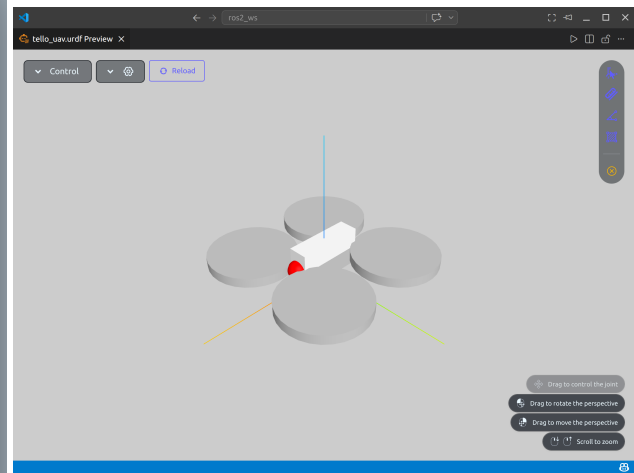
Source: [src/s6_cpp_urdf/urdf/tello_uav.urdf](#) or [src/s6_py_urdf/urdf/tello_uav.urdf](#)

```

79 <!-- Camera link -->
80 <link name="camera_link">
81   <visual>
82     <origin xyz="0 0 0"/>
83     <geometry>
84       <sphere radius="0.01"/>
85     </geometry>
86     <material name="red"/>
87   </visual>
88   <collision>
89     <geometry>
90       <sphere radius="0.01"/>
91     </geometry>
92   </collision>
93   <inertial>
94     <mass value="1e-5"/>
95     <inertia ixx="1e-6" ixy="0" ixz="0" iyy="1e-6" iyz="0" izz="1e-6"/>
96   </inertial>
97 </link>
98
99 <!-- Base joint -->
100 <joint name="base_joint" type="fixed">
101   <parent link="base_footprint"/>
102   <child link="base_link"/>
103   <origin xyz="0.0 0.0 0.02" rpy="0 0 0"/>
104 </joint>
105
106 <!-- Fixed joint -->
107 <joint name="camera_joint" type="fixed">
108   <origin xyz="0.035 0 0" rpy="0 0 0"/> <!-- TODO fix origin -->
109   <parent link="base_link"/>
110   <child link="camera_link"/>
111 </joint>

```

(a) URDF file.



(b) URDF visualization.

Figure 6.9: URDF modeling of a Tello AMR using the URDF development and visualization extensions.

The XML code defines a Unmanned Aerial Vehicle (UAV) named `tello_uav`, composed of some links connected by fixed joints. The main fixed joint, `base_joint`, positions the `base_link` 0.02 m above the `base_footprint` link. The Tello UAV model includes some visual elements: boxes and cylinders, as defined and displayed in Figure 6.9.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `base_footprint` link.
- By default, the origin of the Tello UAV's geometry is placed at its base, ensuring that the entire robot model is correctly aligned with the grid plane in the visualization.

6.3.4 Meshes

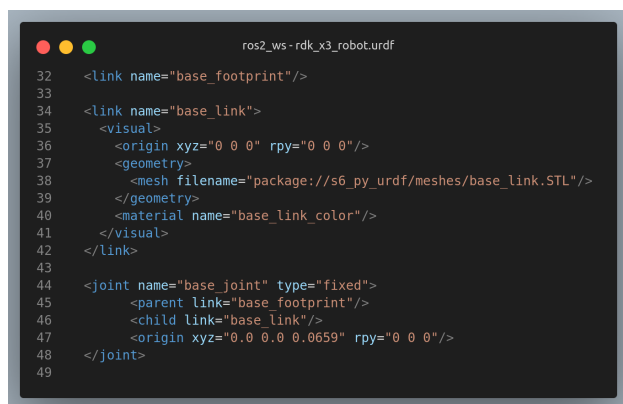
Meshes are separate files whose paths must be explicitly specified using the `package://NAME_OF_PACKAGE/path` notation. The meshes referenced in the previous `r2d2_2.urdf` file are located within the `urdf_tutorial` package, in a folder named `meshes`. Refer to the Appendix C to load the mesh files to correctly visualize the URDF models.

Meshes can be imported in various formats. STL is quite common, but the engine also supports DAE files, which can include embedded color data, eliminating the need to specify colors or materials separately. Often, these are provided as separate files. Meshes can be resized using relative scaling parameters or a bounding box size. Additionally, it is possible to reference meshes from entirely different packages.

Next, we will explore the URDF model for our challenge project robot: the `RDkX3Robot`. Refer to the Appendix C to load the mesh files to correctly visualize the URDF model.

`rdk_x3_amr.urdf`

Source: `src/s6_cpp_urdf/urdf/rdk_x3_amr.urdf` or `src/s6_py_urdf/urdf/rdk_x3_amr.urdf`

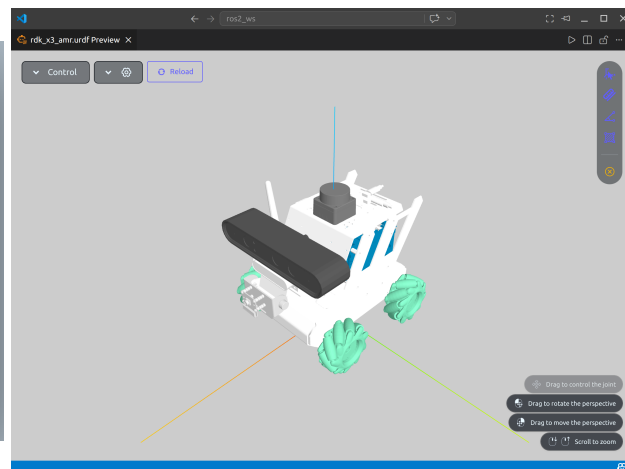


```

32 <link name="base_footprint"/>
33 <link name="base_link">
34 <visual>
35 <origin xyz="0 0 0" rpy="0 0 0"/>
36 <geometry>
37 <mesh filename="package://s6_py_urdf/meshes/base_link.STL"/>
38 </geometry>
39 <material name="base_link_color"/>
40 </visual>
41 </link>
42 <joint name="base_joint" type="fixed">
43 <parent link="base_footprint"/>
44 <child link="base_link"/>
45 <origin xyz="0 0 0.0659" rpy="0 0 0"/>
46 </joint>
47
48
49

```

(a) URDF file.



(b) URDF visualization.

Figure 6.10: URDF modeling of the RDK X3 AMR using URDF development and visualization extensions.

The XML code defines a **Autonomous Mobile Robot (AMR)** named `rdk_x3_amr`, composed of multiple links connected by fixed, continuous, and revolute joints. The main fixed joint, `base_joint`, positions the `base_link` 0.0659 m above the `base_footprint` link. Additional joints connect the robot's body and wheels, allowing continuous rotation of the wheels about their respective y-axes to simulate spinning behavior. The RDK X3 model includes several visual elements, all of them corresponding to STL files, as defined and displayed in Figure 6.10.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In this case, it is defined by the `base_footprint` link.

- By default, the origin of the RDK X3 AMR's geometry is placed at its base, ensuring that the entire robot model is correctly aligned with the grid plane in the visualization.

6.4 CMakeLists.txt

To make accessible the launch (.py), mesh (.STL and .dae), RViz (.rviz), and URDF (.urdf) files we have defined in the C++ package, add the following lines to CMakeLists.txt:

```
# Added manually
install(DIRECTORY
  launch
  meshes
  rviz
  urdf
  DESTINATION share/${PROJECT_NAME}
)
```

6.5 setup.py

To make accessible the launch (.py), mesh (.STL and .dae), RViz (.rviz), and URDF (.urdf) files we have defined in the Python package, add the data files while preserving any existing file setup, with the following lines to setup.py:

```
data_files=[
    ('share/ament_index/resource_index/packages',
     ['resource/' + package_name]),
    ('share/' + package_name, ['package.xml']),
    (os.path.join('share', package_name, 'meshes'), glob('meshes/*')),
    (os.path.join('share', package_name, 'urdf'), glob('urdf/*')),
    (os.path.join('share', package_name, 'rviz'), glob('rviz/*')),
    (os.path.join('share', package_name, 'launch'), glob('launch/*'))
],
```

6.6 package.xml

Make sure the required dependencies are defined within the package.xml files. For the C++ package:

```
<depend>rclcpp</depend>
<depend>geometry_msgs</depend>
<depend>sensor_msgs</depend>
<depend>tf2_ros</depend>
<depend>tf2_geometry_msgs</depend>
```

And for the Python package:

```
<depend>rclpy</depend>
<depend>geometry_msgs</depend>
<depend>sensor_msgs</depend>
<depend>tf2_ros</depend>
<depend>tf_transformations</depend>
```

Note:

As suggested in Chapter 5, for packages with launch files, it is a good idea to add an `exec_depend` dependency on the `ros2launch` package in the `package.xml` file from each package that contains launch files, as well as other dependencies that were not previously defined, such as `robot_state_publisher` and `rviz2`:

```
<exec_depend>ros2launch</exec_depend>
<exec_depend>robot_state_publisher</exec_depend>
<exec_depend>rviz2</exec_depend>
```

This is not strictly necessary, but it is a good practice to ensure that all required dependencies are properly declared.

6.7 Visualizing URDF (.urdf) Files with RViz

In addition to the `URDF` visualizer extension introduced in the Section 6.3, we can also use the `ROS 2` environment to observe `URDF` models through `ROS Visualization (RViz)`. To visualize a `URDF` model, launch the `display.launch.py` file from the `urdf_tutorial` package using the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/<urdf_file>.urdf
```

This command performs the following actions:

- Loads the specified model and sets it as a parameter for the `robot_state_publisher` node.
- Launches nodes that publish `sensor_msgs/msg/JointState` and coordinate transforms.
- Starts `RViz` with a predefined configuration file.

After executing `display.launch.py`, `RViz` will display a grid, a fixed (main) coordinate frame, by default set to `base_link`, and the loaded `URDF` model.

Key observations:

- The *fixed frame* is the reference coordinate frame for the simulation environment. In most cases, this should be updated to `base_footprint`.
- When using this fixed frame, the origin of most `URDF` models aligns with their base, ensuring the robot model is correctly positioned on the grid plane in `RViz`.



6.7.1 Visualizing URDF Models with the `urdf_tutorial` Package

After saving your changes, navigate to the root of your workspace (e.g., `/ros2_ws`), build the relevant packages, and source the workspace:

```
$ colcon build --packages-select s6_cpp_urdf
$ colcon build --packages-select s6_py_urdf
$ source install/setup.bash
```

Then, launch `display.launch.py` from the `urdf_tutorial` package to load the URDF models from either the C++ or Python package.

Note: Ensure that your workspace is properly sourced, as `display.launch.py` requires access to the mesh (`.dae` and `.STL`) files used by the `r2d2.urdf` and `rdk_x3_amr.urdf` models, respectively.

`0_sphere.urdf`

To visualize the `0_sphere.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/0_sphere.urdf
```

Replace `<file_path>` with the absolute path to the `0_sphere.urdf` file. Figure 6.11 shows the simulation of this model in RViz.

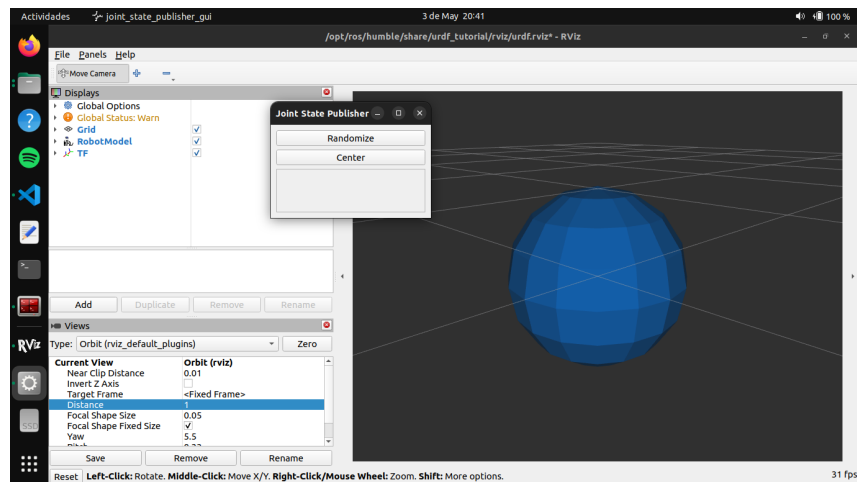


Figure 6.11: Visualization of the sphere using RViz.

`1_box.urdf`

To visualize the `1_box.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/1_box.urdf
```

Replace `<file_path>` with the absolute path to the `1_box.urdf` file. Figure 6.12 shows the simulation of this model in RViz.



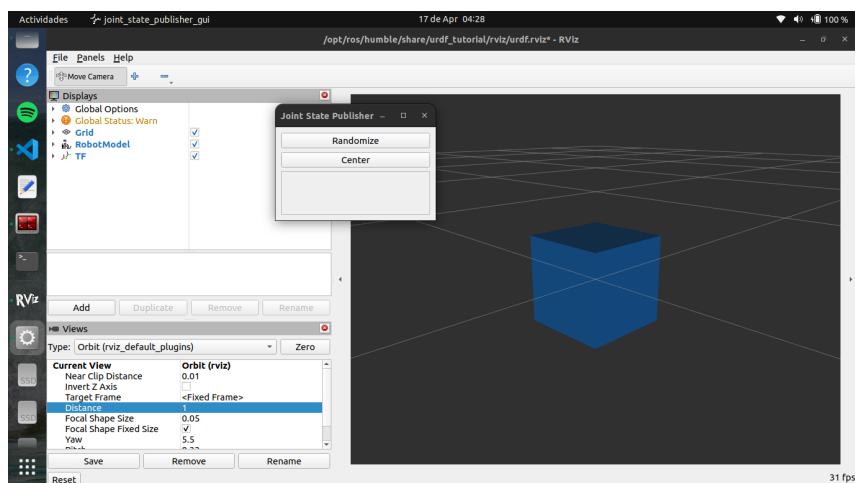


Figure 6.12: Visualization of the box using RViz.

2_cylinder.urdf

To visualize the `2_cylinder.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/2_cylinder.urdf
```

Replace `<file_path>` with the absolute path to the `2_cylinder.urdf` file. Figure 6.13 shows the simulation of this model in RViz.

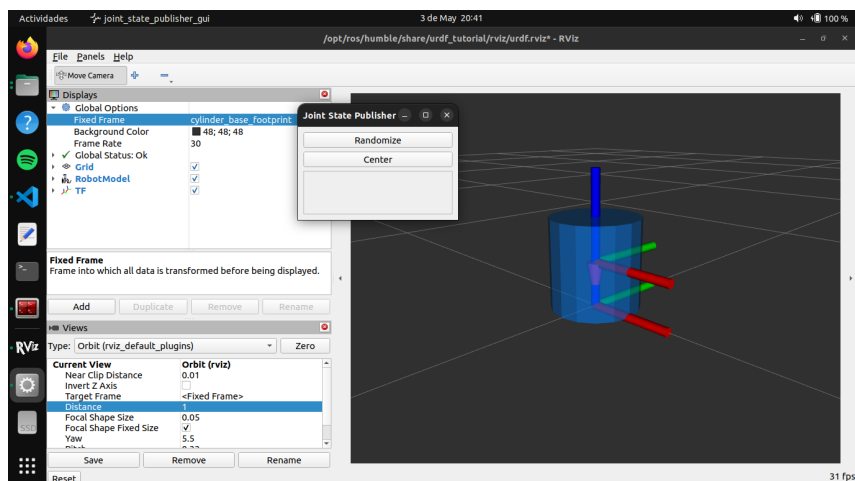


Figure 6.13: Visualization of the cylinder using RViz.

3_wheel.urdf

To visualize the `3_wheel.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/3_wheels.urdf
```



Replace `<file_path>` with the absolute path to the `3_wheel.urdf` file. Figure 6.14 shows the simulation of this model in `RViz`.

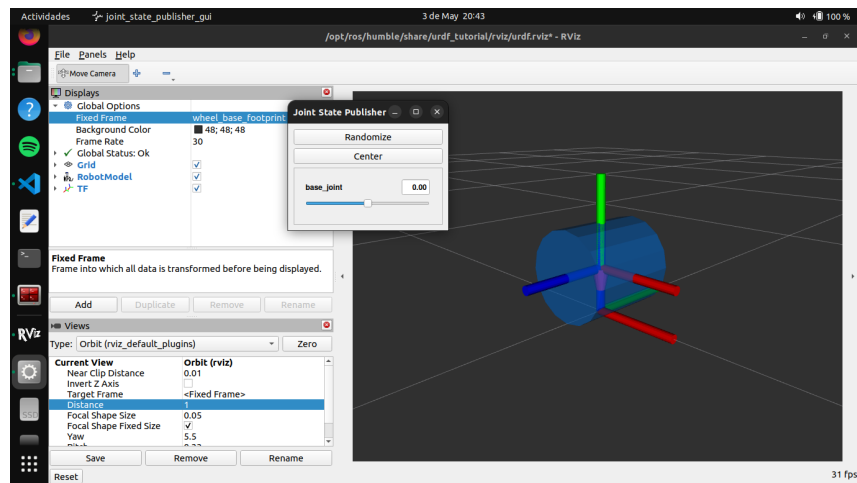


Figure 6.14: Visualization of the wheel using `RViz`.

`4_robot.urdf`

To visualize the `4_robot.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/4_robot.urdf
```

Replace `<file_path>` with the absolute path to the `4_robot.urdf` file. Figure 6.15 shows the simulation of this model in `RViz`.

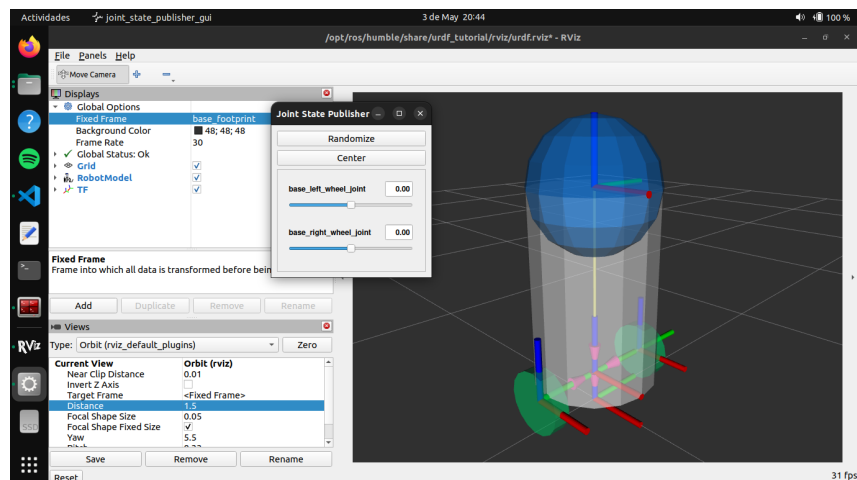


Figure 6.15: Visualization of the robot using `RViz`.

`r2d2.urdf`

To visualize the `r2d2.urdf` model from either the C++ or Python package, use the following command:



```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/r2d2.urdf
```

Replace `<file_path>` with the absolute path to the `r2d2.urdf` file. Figure 6.16 shows the simulation of this model in `RViz`.

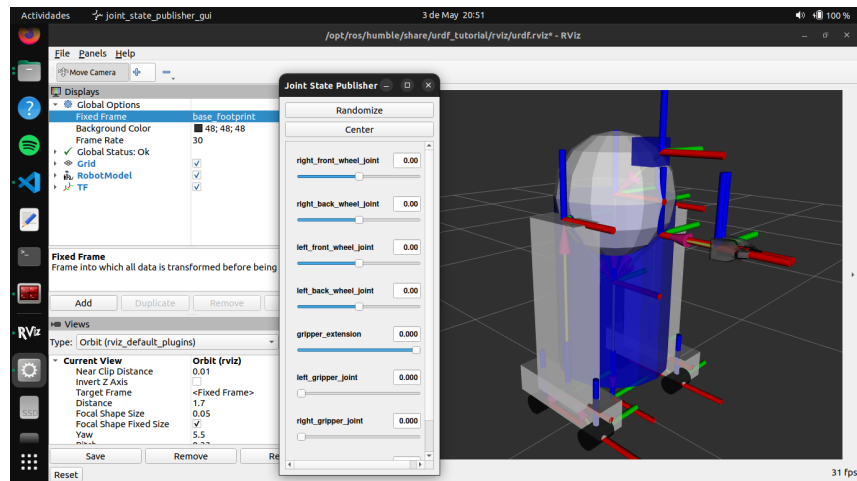


Figure 6.16: Visualization of the R2D2 using `RViz`.

`r2d2_2.urdf`

To visualize the `r2d2_2.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/r2d2\_2.urdf
```

Replace `<file_path>` with the absolute path to the `r2d2_2.urdf` file. Figure 6.17 shows the simulation of this model in `RViz`.

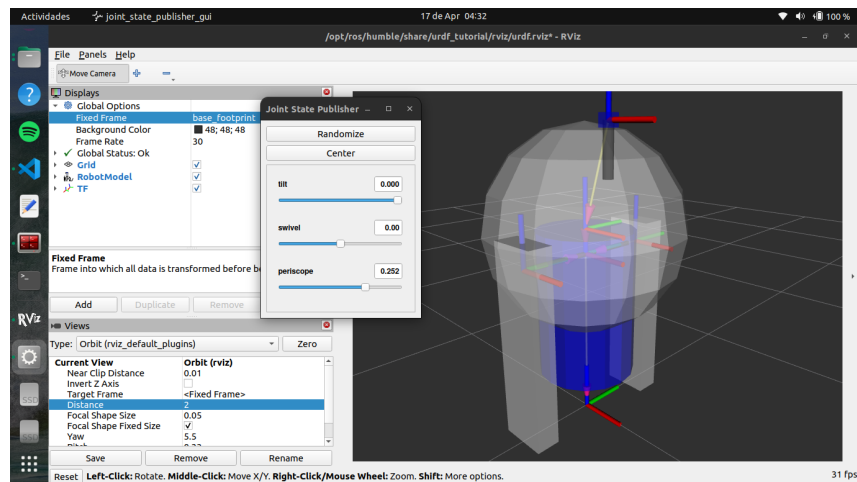


Figure 6.17: Visualization of the R2D2_2 using `RViz`.

tello_uav.urdf

To visualize the `tello_uav.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/tello\_uav.urdf
```

Replace `<file_path>` with the absolute path to the `tello_uav.urdf` file. Figure 6.18 shows the simulation of this model in RViz.

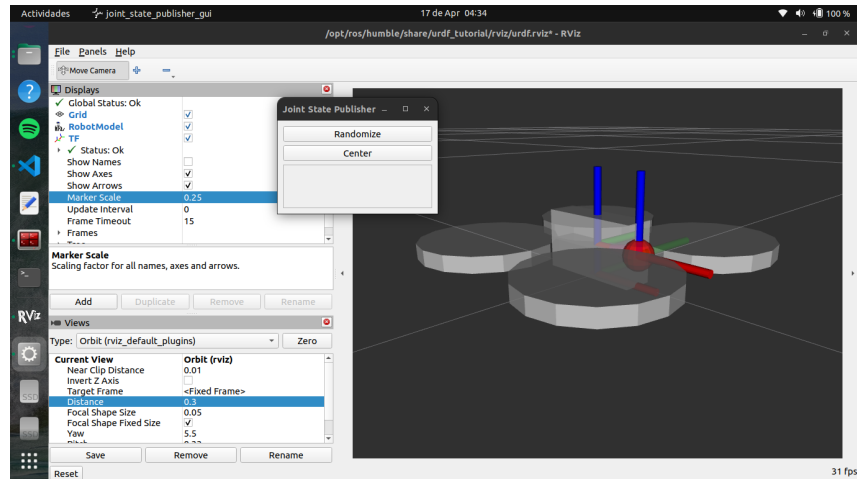


Figure 6.18: Visualization of the Tello UAV using RViz.

rdk_x3_amr.urdf

To visualize the `rdk_x3_amr.urdf` model from either the C++ or Python package, use the following command:

```
$ ros2 launch urdf_tutorial display.launch.py model:=<file_path>/rdk\_x3\_amr.urdf
```

Replace `<file_path>` with the absolute path to the `rdk_x3_amr.urdf` file. Figure 6.19 shows the simulation of this model in RViz.

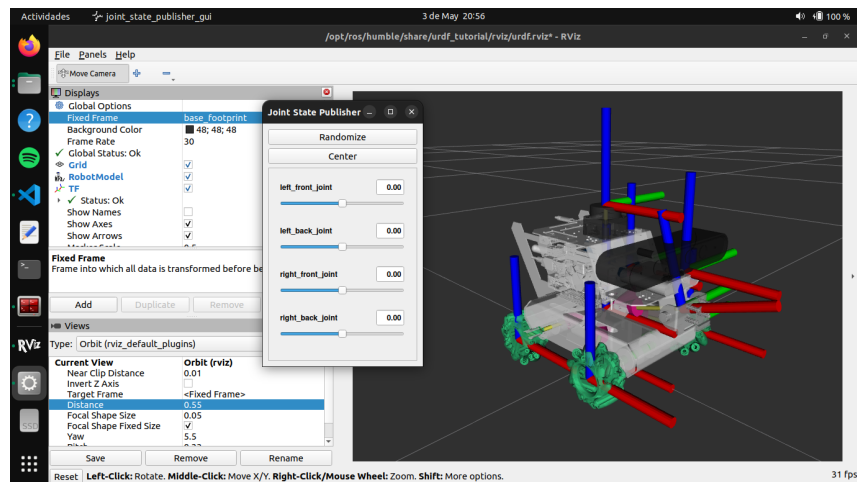


Figure 6.19: Visualization of the RDK X3 AMR using RViz.

6.8 Using URDF with *robot_state_publisher* in C++

In this section, we cover how to setting up the *robot_state_publisher* node in C++ by working with the *rdk_x3_amr.cpp* file.

The corresponding C++ nodes for the cylinder, wheel, and R2D2 models are also available in our [ROS 2 course GitHub repository](#).

6.8.1 Setting Up the C++ Node

Navigating to the Package Directory

Open a terminal and navigate to your C++ package directory within your ROS 2 workspace (e.g., *~/ros2_ws*):

```
$ cd ros2_ws/src/s6_cpp_urdf/src
```

Replace *s6_cpp_urdf* with your actual package name.

Creating the C++ Node File

Create a new C++ file for your state publisher node:

```
$ touch rdk_x3_amr.cpp
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the *rdk_x3_amr.cpp* file.

6.8.2 Editing the C++ Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In [VS Code](#), locate and open the *rdk_x3_amr.cpp* file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 state publisher node in C++:



State Publisher in C++

```
#include <rclcpp/rclcpp.hpp>
#include <geometry_msgs/msg/quaternion.hpp>
#include <sensor_msgs/msg/joint_state.hpp>
#include <tf2_ros/transform_broadcaster.h>
#include <tf2_geometry_msgs/tf2_geometry_msgs.hpp>
#include <cmath>
#include <thread>
#include <chrono>

using namespace std::chrono;

class StatePublisher : public rclcpp::Node{
public:
    StatePublisher(rclcpp::NodeOptions options=rclcpp::NodeOptions()):
        Node("state_publisher_node",options){
        RCLCPP_INFO(this->get_logger(),"C++ state publisher node has been
→ started");

        // Create a publisher to tell robot_state_publisher the JointState
→ information.
        // robot_state_publisher will deal with this transformation
        joint_pub_ =
→ this->create_publisher<sensor_msgs::msg::JointState>("joint_states",10);

        // Create a broadcaster to publish the transform between coordinate
→ frames that
        // will determine the position of coordinate system 'base_footprint' in
→ coordinate system 'odom'
        broadcaster = std::make_shared<tf2_ros::TransformBroadcaster>(this);

        // Timer for publishing at ~30Hz
→ timer_ = this->create_wall_timer(33ms, std::bind(&StatePublisher::publish, this));
        }

    void publish();

private:
    rclcpp::Publisher<sensor_msgs::msg::JointState>::SharedPtr joint_pub_;
    std::shared_ptr<tf2_ros::TransformBroadcaster> broadcaster;
    rclcpp::TimerBase::SharedPtr timer_;

    // State variables
    // degree means one degree
    const double degree = M_PI/180.0;
    double wheel_angle = 0.0;
    double wheel_step = degree;
    double angle = 0.0;
};
```



```

void StatePublisher::publish(){
    // Create JointState message
    sensor_msgs::msg::JointState joint_state;
    // Add time stamp
    joint_state.header.stamp=this->get_clock()->now();
    // Specify joints' name which are defined in the URDF file and their content
    joint_state.name={"left_front_joint","left_back_joint",
                     "right_front_joint","right_back_joint"};
    joint_state.position={wheel_angle,wheel_angle,-wheel_angle,-wheel_angle};

    // Create TransformStamped message
    geometry_msgs::msg::TransformStamped t;
    // Add time stamp
    t.header.stamp=this->get_clock()->now();
    // Specify the father and child frames
    // odom is the base coordinate system of tf2
    t.header.frame_id="odom";
    // base_footprint is defined in URDF file and it is the base coordinate of model
    t.child_frame_id="base_footprint";

    // Add translation change
    t.transform.translation.x = cos(angle) * 1.0;
    t.transform.translation.y = sin(angle) * 1.0;
    t.transform.translation.z = 0.0;

    // Euler angle into Quaternion and add rotation change
    tf2::Quaternion q;
    q.setRPY(0,0,angle + M_PI / 2);
    t.transform.rotation.x = q.x();
    t.transform.rotation.y = q.y();
    t.transform.rotation.z = q.z();
    t.transform.rotation.w = q.w();

    // Update state for next cycle
    wheel_angle += wheel_step * 4;
    if (wheel_angle<-M_PI || wheel_angle>M_PI){
        wheel_angle *= -1;
    }

    angle += degree; // Change the angle at a slow pace

    // Publish messages
    joint_pub->publish(joint_state);
    broadcaster->sendTransform(t);

    RCLCPP_INFO(this->get_logger(),"Publishing joint state and transform");
}

int main(int argc, char * argv[]){
    rclcpp::init(argc,argv);
    std::shared_ptr<StatePublisher> node = std::make_shared<StatePublisher>();
    rclcpp::spin(node);
    rclcpp::shutdown();
    return 0;
}

```



6.8.3 Integrating the C++ State Publisher Node into the Package

Modify `CMakeLists.txt` of your C++ package (`s6_cpp_urdf`) to add an executable for the image publisher node(s):

```
# Added manually
add_executable(cylinder_exe src/cylinder.cpp)
ament_target_dependencies(cylinder_exe rclcpp geometry_msgs sensor_msgs tf2_ros
  ↪ tf2_geometry_msgs)

add_executable(wheel_exe src/wheel.cpp)
ament_target_dependencies(wheel_exe rclcpp geometry_msgs sensor_msgs tf2_ros
  ↪ tf2_geometry_msgs)

add_executable(r2d2_exe src/r2d2.cpp)
ament_target_dependencies(r2d2_exe rclcpp geometry_msgs sensor_msgs tf2_ros
  ↪ tf2_geometry_msgs)

add_executable(rdk_x3_amr_exe src/rdk_x3_amr.cpp)
ament_target_dependencies(rdk_x3_amr_exe rclcpp geometry_msgs sensor_msgs tf2_ros
  ↪ tf2_geometry_msgs)

add_executable(tello_uav_exe src/tello_uav.cpp)
ament_target_dependencies(tello_uav_exe rclcpp geometry_msgs sensor_msgs tf2_ros
  ↪ tf2_geometry_msgs)

# Added manually
install(TARGETS
  cylinder_exe
  wheel_exe
  r2d2_exe
  rdk_x3_amr_exe
  tello_uav_exe
  DESTINATION lib/${PROJECT_NAME}
)

# Added manually
install(DIRECTORY
  launch
  meshes
  rviz
  urdf
  DESTINATION share/${PROJECT_NAME}/
)
```

6.8.4 Summary of C++ State Publisher Node Key Identifiers

- `rdk_x3_amr.cpp`: Name of the source file for the C++ state publisher node.
- `cpp_state_publisher_node`: Name of the C++ state publisher node as defined in the constructor of the C++ class.
- `rdk_x3_amr_exe`: Name of the executable defined in `CMakeLists.txt`.
- `joint_states`: Name of the topic provided by the C++ state publisher node.



6.9 Using URDF with *robot_state_publisher* in Python

In this section, we cover how to setting up the *robot_state_publisher* node in Python by working with the *rdk_x3_amr.py* file.

The corresponding Python nodes for the cylinder, wheel, and R2D2 models are also available in our [ROS 2 course GitHub repository](#).

6.9.1 Setting Up the Python Node

Navigating to the Package Directory

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., *~/ros2_ws*):

```
$ cd ros2_ws/src/s6_py_urdf/s6_py_urdf
```

Replace *s6_py_urdf* with your actual package name.

Creating the Python Node File

Create a new Python file for your state publisher node:

```
$ touch rdk_x3_amr.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the *rdk_x3_amr.py* file.

6.9.2 Editing the Python Node

Opening the Workspace in VS Code

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In [VS Code](#), locate and open the *rdk_x3_amr.py* file for editing.

Implementing the Node Code

Below is an example implementation of a ROS 2 state publisher node in Python:



State Publisher in Python

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import TransformStamped
from sensor_msgs.msg import JointState
from tf2_ros import TransformBroadcaster
import tf_transformations
import math

class StatePublisher(Node):
    def __init__(self):
        super().__init__('py_state_publisher_node')
        self.get_logger().info("Python state publisher node has been started")

        # Create a publisher to tell robot_state_publisher the JointState information.
        # robot_state_publisher will deal with this transformation
        self.joint_pub = self.create_publisher(JointState, 'joint_states', 10)

        # Create a broadcaster to publish the transform between coordinate frames that
        # will determine the position of coordinate system 'base_footprint' in
        # → coordinate system 'odom'
        self.broadcaster = TransformBroadcaster(self)

        # Timer for publishing at ~30Hz
        self.timer = self.create_timer(0.033, self.publish)

        # State variables
        self.degree = math.pi / 180.0 # degree means one degree
        self.wheel_angle = 0.0
        self.wheel_step = self.degree
        self.angle = 0.0

    def publish(self):
        # Create JointState message
        joint_state = JointState()
        # Add time stamp
        joint_state.header.stamp = self.get_clock().now().to_msg()
        # Specify joints' name which are defined in the URDF file and their content
        joint_state.name =
            ['left_front_joint', 'left_back_joint', 'right_front_joint', 'right_back_joint']
        joint_state.position =
            [self.wheel_angle, self.wheel_angle, -self.wheel_angle, -self.wheel_angle]

        # Create TransformStamped message
        t = TransformStamped()
        # Add time stamp
        t.header.stamp = self.get_clock().now().to_msg()
        # Specify the father and child frames
        # odom is the base coordinate system of tf2
        t.header.frame_id = 'odom'
        # base_footprint is defined in URDF file and it is the base coordinate of model
        t.child_frame_id = 'base_footprint'
```



```

    # Add translation change
    t.transform.translation.x = math.cos(self.angle) * 1.0
    t.transform.translation.y = math.sin(self.angle) * 1.0
    t.transform.translation.z = 0.0

    # Euler angle into Quaternion and add rotation change
    q = tf_transformations.quaternion_from_euler(0, 0, self.angle + math.pi / 2)
    t.transform.rotation.x = q[0]
    t.transform.rotation.y = q[1]
    t.transform.rotation.z = q[2]
    t.transform.rotation.w = q[3]

    # Update state for next cycle
    self.wheel_angle += self.wheel_step * 4
    if self.wheel_angle < -math.pi or self.wheel_angle > math.pi:
        self.wheel_angle *= -1

    self.angle += self.degree # Change the angle at a slow pace

    # Publish messages
    self.joint_pub.publish(joint_state)
    self.broadcaster.sendTransform(t)

    self.get_logger().info("Publishing joint state and transform")

def main(args=None):
    rclpy.init(args=args)
    node = StatePublisher()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```

6.9.3 Integrating the Python State Publisher Node into the Package

Modify `setup.py` of your Python package (`s6_py_urdf`) to add an executable for the image publisher node(s):

```

entry_points={
    'console_scripts': [
        "cylinder_exe = s6_py_urdf.cylinder:main",
        "wheel_exe = s6_py_urdf.wheel:main",
        "r2d2_exe = s6_py_urdf.r2d2:main",
        "rdk_x3_amr_exe = s6_py_urdf.rdk_x3_amr:main",
        "tello_uav_exe = s6_py_urdf.tello_uav:main"
    ],
},

```



6.9.4 Summary of Python State Publisher Node Key Identifiers

- `rdk_x3_amr.py`: Name of the source file for the Python state publisher node.
- `py_state_publisher_node`: Name of the Python state publisher node as defined in the constructor of the Python class.
- `rdk_x3_amr_exe`: Name of the executable defined in `setup.py`.
- `joint_states`: Name of the topic provided by the Python state publisher node.

6.10 Launch File Definition

Within the `s6_cpp_urdf/launch` and `s6_py_urdf/launch` directories, create (or open) a file named `rdk_x3_amr_launch.py` with the following structure. For the C++ package:

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to the URDF file
    urdf_file = os.path.join(
        get_package_share_directory('s6_cpp_urdf'),
        'urdf',
        'rdk_x3_amr.urdf'
    )

    # open the whole urdf_file_name file and read its content to robot_desc
    with open(urdf_file, 'r') as infp:
        robot_desc = infp.read()

    return LaunchDescription([
        Node(
            package='robot_state_publisher',
            executable='robot_state_publisher',
            name='robot_state_publisher',
            output='screen',
            parameters=[{'robot_description': robot_desc}],
            arguments=[urdf_file]
        ),
        Node(
            package='s6_cpp_urdf',
            executable='rdk_x3_amr_exe',
            name='rdk_x3_amr',
            output='screen'
        ),
    ])

```



And for the Python package:

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to the URDF file
    urdf_file = os.path.join(
        get_package_share_directory('s6_py_urdf'),
        'urdf',
        'rdk_x3_amr.urdf'
    )

    # open the whole urdf_file_name file and read it content to robot_desc
    with open(urdf_file, 'r') as infp:
        robot_desc = infp.read()

    return LaunchDescription([
        Node(
            package='robot_state_publisher',
            executable='robot_state_publisher',
            name='robot_state_publisher',
            output='screen',
            parameters=[{'robot_description': robot_desc}],
            arguments=[urdf_file]
        ),
        Node(
            package='s6_py_urdf',
            executable='rdk_x3_amr_exe',
            name='rdk_x3_amr',
            output='screen'
        ),
    ])

```

These launch files start two nodes:

- The `robot_state_publisher` node, which publishes the [URDF](#) model and its transformations.
- The `rdk_x3_amr` node, responsible for publishing the joint state values and the overall robot geometry.

6.11 Building the Packages and Running the Nodes

After making the above edits and saving all the changes, go to the root of your workspace (e.g., `/ros2_ws`) and build the packages:

```
$ colcon build --packages-select s6_cpp_urdf
$ colcon build --packages-select s6_py_urdf

```

Then, in two terminals, source the workspace environment in each, and proceed as follows:



- In **Terminal 1**, launch either the C++ or Python robot state publisher nodes:

```
$ source install/setup.bash
$ ros2 launch s6_py_urdf rdk_x3_amr_launch.py
```

- In **Terminal 2**, start **RViz** with the configuration file stored in the respective package:

```
$ source install/setup.bash
$ rviz2 -d src/s6_py_urdf/rviz/rdk_x3_amr.rviz
```

Additionally, open two terminals for diagnostics and introspection tools:

- **Terminal 3:** Use the **view_frames** tool from the **tf2_tools** package to visualize the current coordinate frame hierarchy (tree):

```
$ ros2 run tf2_tools view_frames
```

- **Terminal 4:** Launch the **rqt_graph** tool to view the current **ROS 2** node and topic graph:

```
$ rqt_graph
```

6.12 Practice Assignment: URDF Modeling and Launch Files (Robot States)

In this assignment, you will run the ROS 2 robot state publisher node implemented in both C++ and Python, as described in Sections 6.8 and 6.9. Capture evidence of successful execution using two screenshots as PNG files and one PDF file, and submit them in the designated assignment area on Canvas (within the ROS 2 module). The filenames must follow this format:

`FirstnameLastname_evidence_sX_Y.png`

where `sX` corresponds to the session number, and `Y` is the evidence number. For example, correct filenames would be:

`EduardoDavila_evidence_s6_1.png`,
`EduardoDavila_evidence_s6_2.png`,
`EduardoDavila_evidence_s6_3.pdf`.

6.12.1 Executing ROS 2 Robot State Publishers

After editing and saving your C++ and Python nodes, and updating the `CMakeLists.txt`, `setup.py`, and `package.xml` files, build the packages:

```
$ colcon build --packages-select s6_cpp_urdf
$ colcon build --packages-select s6_py_urdf
```

Then, in two terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, launch either the C++ or Python robot state publisher nodes:

```
$ source install/setup.bash
$ ros2 launch s6_py_urdf rdk_x3_amr_launch.py
```

- In **Terminal 2**, start **RViz** with the configuration file stored in the respective package:

```
$ source install/setup.bash
$ rviz2 -d src/s6_py_urdf/rviz/rdk_x3_amr.rviz
```

You should observe that:

1. The `robot_state_publisher` node publishes the **URDF** model and its coordinate transforms for visualization in **RViz**.
2. The `rdk_x3_amr` node publishes the joint state values, enabling dynamic updates of the model in **RViz**.

6.12.2 Visualizing the frames with `view_frames`

To observe the current coordinate frame hierarchy, use the `image_frames` tool from the `tf2_tools` package. Then, in **Terminal 3**, execute the following command (without the need to enter the workspace directory):




```
$ ros2 run tf2_tools view_frames
```

This tool will generate a PDF file in the current directory of the terminal. The file provides a graphical representation of the current coordinate frame tree published by the `robot_state_publisher` node from the robot URDF model.

6.12.3 Visualizing the ROS Graph with `rqt_graph`

To verify that the robot state nodes are correctly connected, use the `rqt_graph` tool. Then, in **Terminal 4**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_graph rqt_graph
```

or simply:

```
$ rqt_graph
```

This will visualize the ROS 2 graph, showing the `joint_states` topic connecting the robot state publisher nodes, as described above.

6.12.4 Submission Instructions

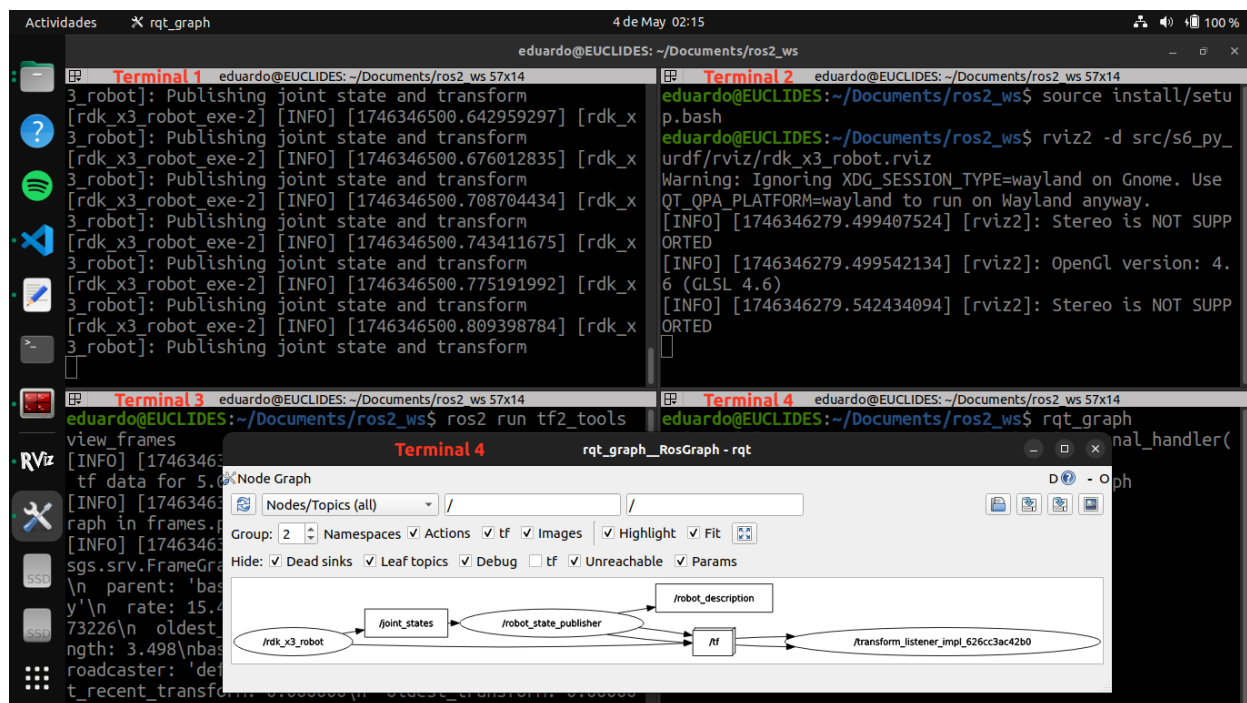


Figure 6.20: Example screenshot showing terminal outputs and the ROS 2 graph of the robot state publishers in action.

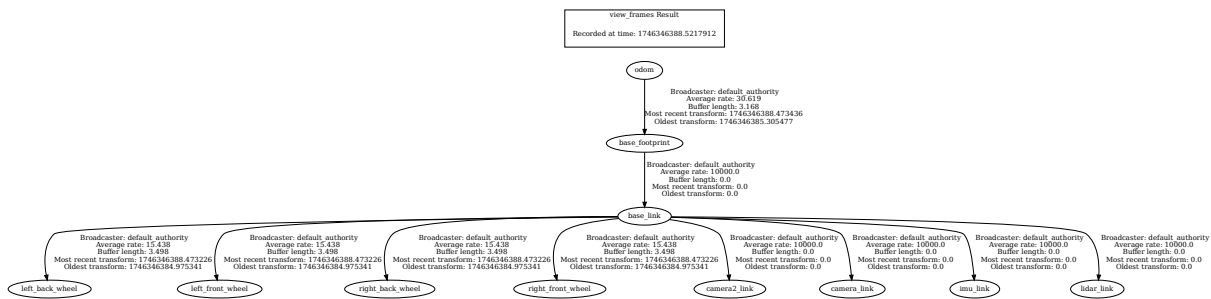


Figure 6.21: PDF showing the graphical representation of the current coordinate frame tree.

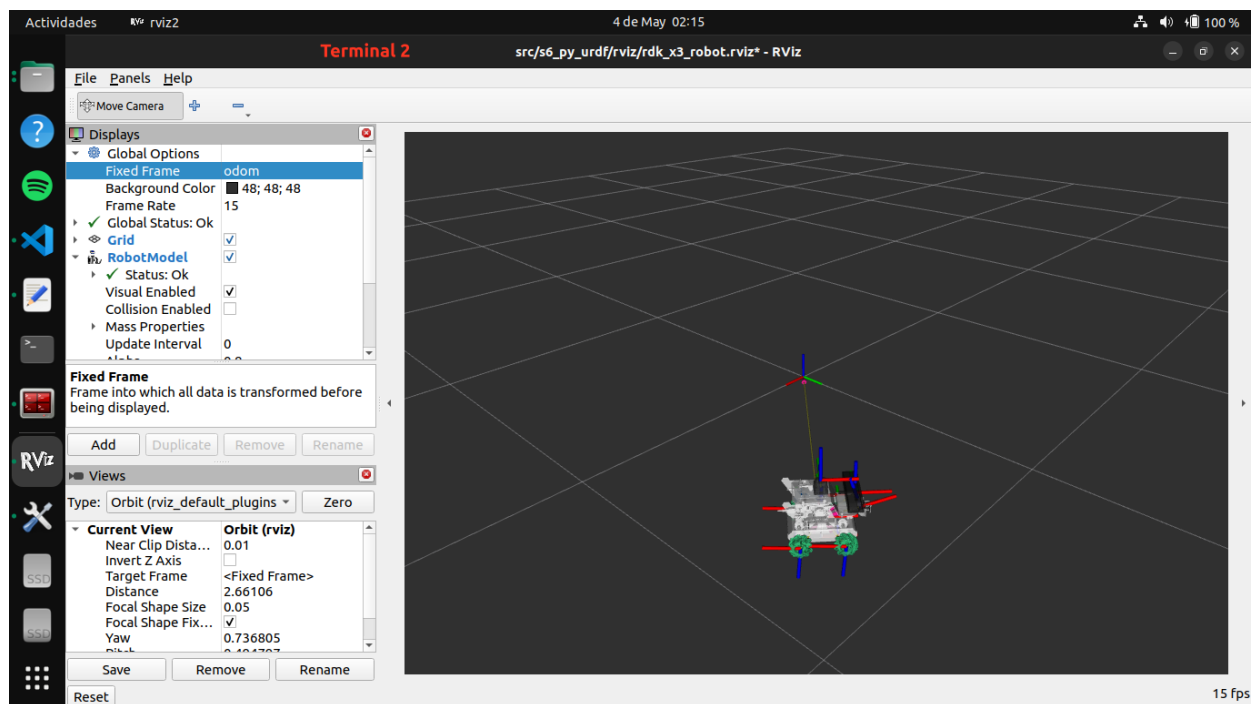


Figure 6.22: Example screenshot showing the simulation of the RDK X3 Robot in RViz.

1. Run the ROS 2 robot state publisher nodes as described above.
2. As first evidence, capture a screenshot showing the terminal outputs and the ROS 2 graph from either the C++ or Python nodes, as described in Sections 6.12.1 and 6.12.3. See Figure 6.20 for reference.
3. As second evidence, locate the PDF file generated by the view_frames tool from the tf2_tools package, as described in Section 6.12.2. See Figure 6.21 for reference.
4. As third evidence, capture a screenshot showing the simulation of the RDK X3 Robot in RViz. See Figure 6.22 for reference.
5. Save the screenshots as PNG files.
6. Name the files following this format: `FirstnameLastname_evidence_sX_Y.png` (replace X with the session number, and Y with the evidence number).
7. Submit the files as specified by the course guidelines on Canvas.



Note: Your machine's username and hostname must be visible in the screenshots to verify the authenticity of your submission, as shown in Figure [6.20](#). Any submission that appears copied, unclear, or altered will be considered invalid and may receive a score of 0.



CHAPTER

7

tf2 Library – Robot Network

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✳ **Abstract** - This chapter focuses on the implementation of a multi-robot network system using [Robot Operating System \(ROS\) 2](#), where a central task manager node provides pickup and delivery poses to robot clients. The chapter covers the creation of custom message and service interfaces, the development of a Python-based task manager service node, the implementation of robot client nodes that interact with the task manager to execute the task of moving from one location to another, and the integration of subscriber-publisher nodes to visualize and log the robot states in both terminal and [ROS Visualization \(RViz\)](#) environments. The setup includes package creation, dependency management, and integration of the nodes into a cohesive system.

All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) [↗], located in the `ros2_ws/src` folder, corresponding to the [ROS 2](#) packages with the `s7` prefix.

7.1 Introduction

In ROS 2, [Transform \(TF\)](#) (specifically `tf2`, the second-generation transform library) is the package and system used to **keep track of multiple coordinate frames and their relationships over time**. `tf2` maintains the relationship between coordinate frames in a tree structure, as shown in Figure 7.1, buffered in time and lets the user transform points, vectors, etc., between any two coordinate frames at any desired point in time.

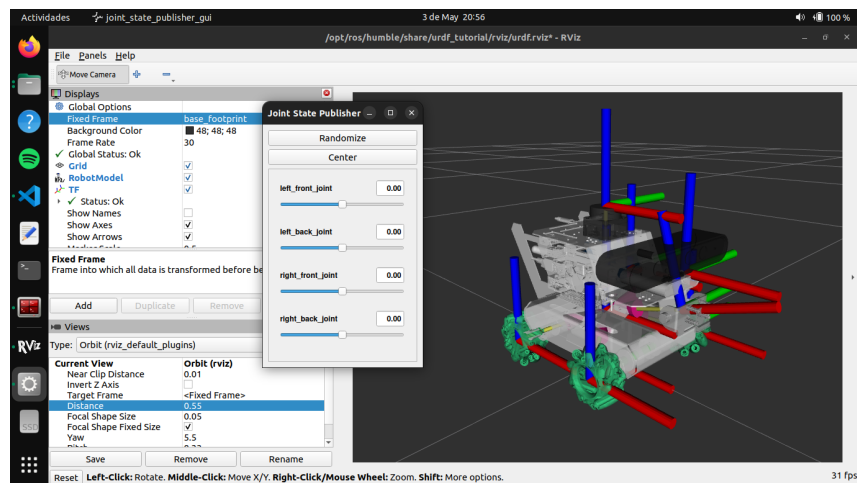


Figure 7.1: Visualization of joint states and coordinate frames of the RDK X3 Robot in **RViz** using `display.launch.py` from the `urdf_tutorial` package.

7.1.1 Properties of `tf2`

A robotic system typically has many 3D coordinate frames that change over time, such as a world frame, base frame, gripper frame, head frame, etc. `tf2` keeps track of all these frames over time, and allows you to ask questions like:

- Where was the head frame relative to the world frame 5 seconds ago?
- What is the pose of the object in my gripper relative to my base?
- What is the current pose of the base frame in the map frame?

7.2 Prerequisites

7.2.1 Workspace and Package Creation

Before getting started, make sure you are familiar with workspace and package creation, as described in Sections 2.2, 2.3, and 2.4.

Open a terminal and navigate to the `src` directory of your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.

Next, create the following packages in your workspace:

- C++ package for the custom `.msg` and `.srv` files:

```
$ ros2 pkg create s7_robot_network_interface --build-type ament_cmake --license
→ Apache-2.0 --description "Task Manager Interface: .msg and .srv files"
```

- Python package to contain the task manager node (server):

```
$ ros2 pkg create s7_py_task_manager --build-type ament_python --dependencies rclpy
→ s7_robot_network_interface geometry_msgs --license Apache-2.0 --description
→ "Task Manager - Server"
```

- Python package to contain the robot state machine nodes (clients):

```
$ ros2 pkg create s7_py_client_robot --build-type ament_python --dependencies rclpy
→ s7_robot_network_interface geometry_msgs sensor_msgs tf2_ros tf_transformations
→ --license Apache-2.0 --description "Robot State Machine - Clients"
```

- Python package to contain the nodes for monitoring the robot network tasks:

```
$ ros2 pkg create s7_py_robot_task_monitoring --build-type ament_python
→ --dependencies rclpy s7_robot_network_interface geometry_msgs sensor_msgs
→ tf2_ros tf_transformations --license Apache-2.0 --description "Robot Network
→ Task Monitoring"
```

Feel free to replace `s7_robot_network_interface`, `s7_py_task_manager`, `s7_py_client_robot`, and `s7_py_robot_task_monitoring` with names of your preference.

7.2.2 Installing Required Packages

Before proceeding, ensure that the following library and packages are installed. Update your package list (apt repository) and install them with:

```
$ sudo apt update
$ sudo apt install ros-humble-joint-state-publisher
$ sudo apt install ros-humble-tf-transformations
$ sudo apt install ros-humble-tf2-tools # or ros-humble-tf2*
$ pip3 install transforms3d
```

Also, from the root of your workspace, use `rosdep` to check for any other missing dependencies before building:

```
$ rosdep install -i --from-path src --rosdistro humble -y
```

7.3 Package Setup: `s7_robot_network_interface`

The `.msg` and `.srv` files are required to be placed in directories called `msg` and `srv` respectively. Create the directories in `ros2_ws/src/s7_robot_network_interface`:

```
$ mkdir msg srv
```

7.3.1 Creating the `.msg` and `.srv` files

msg definition

In the `s7_robot_network_interface/msg` directory you just created, make a new file called `AMR_Status.msg` with a few lines of code declaring its data structure:

```
builtin_interfaces/Time timestamp
int64 robot_id
int64 task_stage
geometry_msgs/Pose2D pose
bool align_yaw
```



This is a custom message that transfers the timestamp from the `builtin_interfaces` interface package, an integer-typed robot ID, an integer-typed task stage, a 2D pose message from the `geometry_msgs` interface package, and a boolean value to indicate whether the robot will align its yaw.

Additionally, create another message file called `UAVStatus.msg` with the following data structure:

```
builtin_interfaces/Time timestamp
int64 robot_id
geometry_msgs/Pose pose
```

This is another custom message that transfers the timestamp from the `builtin_interfaces` interface package, a robot ID, and a 3D pose message from the `geometry_msgs` interface package.

srv definition

Back in the `s7_robot_network_interface/srv` directory you just created, make a new file called `GetTwoPoses.srv` with the following request and response structure:

```
int64 robot_id
---
geometry_msgs/Pose2D pickup_pose
geometry_msgs/Pose2D delivery_pose
```

This is your custom service that requests a service with one integer named `robot_id`, and responds with two 2D poses called `pickup_pose` and `delivery_pose`.

7.3.2 Configuring `CMakeLists.txt`

To convert the interfaces you defined into language-specific code (like C++ and Python) so that they can be used in those languages, add the following lines to `CMakeLists.txt`:

```
# Added manually
find_package(builtin_interfaces REQUIRED)
find_package(geometry_msgs REQUIRED)
find_package(rosidl_default_generators REQUIRED)

# Added manually
rosidl_generate_interfaces(${PROJECT_NAME}
  "msg/AMRStatus.msg"
  "msg/UAVStatus.msg"
  "srv/GetTwoPoses.srv"
  DEPENDENCIES builtin_interfaces geometry_msgs
)
```

7.3.3 Configuring `package.xml`

Because the interfaces rely on `rosidl_default_generators` for generating language-specific code, you need to declare a build tool dependency on it. `rosidl_default_runtime` is a runtime or execution-stage dependency, needed to be able to use the interfaces later. The `rosidl_interface_packages` is the name of the dependency group that your package, `s7_robot_network_interface`, should be associated with, declared using the `<member_of_group>` tag.



Add the following lines within the `<package>` element of `package.xml`:

```
<!-- Added manually -->
<depend>builtin_interfaces</depend>
<depend>geometry_msgs</depend>

<test_depend>ament_lint_auto</test_depend>
<test_depend>ament_lint_common</test_depend>

<!-- Added manually -->
<buildtool_depend>rosidl_default_generators</buildtool_depend>
<exec_depend>rosidl_default_runtime</exec_depend>
<member_of_group>rosidl_interface_packages</member_of_group>
```

7.3.4 Build the `s7_robot_network_interface` package

Now that all the parts of your custom interfaces package are in place, you can build the package. In the root of your workspace (`/ros2_ws`), run the following command:

```
$ colcon build --packages-select s7_robot_network_interface
```

Now the interfaces will be discoverable by other ROS 2 packages.

7.3.5 Confirm msg and srv creation

In a new terminal, run the following command from within your workspace (`/ros2_ws`) to source it:

```
$ source install/setup.bash
```

Now you can confirm that your interface creation worked by using the `ros2 interface show` command. For the `AMRStatus.msg` file, run:

```
$ ros2 interface show s7_robot_network_interface/msg/AMRStatus
```

that should return:

```
builtin_interfaces/Time timestamp
  int32 sec
  uint32 nanosec
int64 robot_id
int64 task_stage
geometry_msgs/Pose2D pose
  float64 x
  float64 y
  float64 theta
bool align_yaw
```

For the `UAVStatus.msg` file, run:

```
$ ros2 interface show s7_robot_network_interface/msg/UAVStatus
```



that should return:

```
builtin_interfaces/Time timestamp
  int32 sec
  uint32 nanosec
int64 robot_id
geometry_msgs/Pose pose
  Point position
    float64 x
    float64 y
    float64 z
  Quaternion orientation
    float64 x 0
    float64 y 0
    float64 z 0
    float64 w 1
```

And similarly, for the *GetTwoPoses.srv* file, run:

```
$ ros2 interface show s7_robot_network_interface/srv/GetTwoPoses
```

that should return:

```
int64 robot_id
---
geometry_msgs/Pose2D pickup_pose
  float64 x
  float64 y
  float64 theta
geometry_msgs/Pose2D delivery_pose
  float64 x
  float64 y
  float64 theta
```

7.4 Package Setup: *s7_py_task_manager*

7.4.1 Creating a ROS 2 Pose Server Node with Python

Setting Up the Python Node

Navigating to the Package Directory:

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., *~/ros2_ws*):

```
$ cd ros2_ws/src/s7_py_task_manager/s7_py_task_manager
```

Replace *s7_py_task_manager* with your actual package name.

Creating the Python Node File:

Create a new Python file for your pose server node, for example, *task_manager.py*:

```
$ touch task_manager.py
```



Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `task_manager.py` file.

Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `task_manager.py` file for editing.

Implementing the Node Code:

Below is an example implementation of a ROS 2 pose server node in Python:

Pose Server in Python (`task_manager.py`):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s7_robot_network_interface.srv import GetTwoPoses
from geometry_msgs.msg import Pose2D
import random

class TaskManagerService(Node):
    def __init__(self):
        super().__init__('task_manager')
        self.get_logger().info("Task Manager node has been started")

        # Create service
        self.srv = self.create_service(GetTwoPoses, 'get_two_poses',
            ↪ self.GetTwoPoses_callback)
        self.get_logger().info("Service 'get_two_poses' is ready to receive requests")

        # --- Number of robots ---
        self.robot_num = 1

        # Create publishers for each robot ID
        self.pickup_publishers = {}
        self.delivery_publishers = {}
        for robot_id in range(1, self.robot_num + 1):
            ns = '/rdk_x3_amr' # f'/robot{robot_id}'
            self.pickup_publishers[robot_id] = self.create_publisher(Pose2D,
                ↪ f'{ns}/pickup_pose', 10)
            self.delivery_publishers[robot_id] = self.create_publisher(Pose2D,
                ↪ f'{ns}/delivery_pose', 10)
```

```

def GetTwoPoses_callback(self, request, response):
    lim = 2.0

    # Generate random pickup pose
    pickup_pose = Pose2D()
    pickup_pose.x = round(random.uniform(-lim, lim), 2)
    pickup_pose.y = round(random.uniform(-lim, lim), 2)
    pickup_pose.theta = round(random.uniform(-3.14, 3.14), 2)

    # Generate random delivery pose
    delivery_pose = Pose2D()
    delivery_pose.x = round(random.uniform(-lim, lim), 2)
    delivery_pose.y = round(random.uniform(-lim, lim), 2)
    delivery_pose.theta = round(random.uniform(-3.14, 3.14), 2)

    # Fill response
    response.pickup_pose = pickup_pose
    response.delivery_pose = delivery_pose

    self.get_logger().info(f'Received request from ID: {request.robot_id}')
    self.get_logger().info(f'Providing pickup pose: {pickup_pose}')
    self.get_logger().info(f'Providing delivery pose: {delivery_pose}')

    # Publish to the robot-specific topics
    robot_id = request.robot_id
    if robot_id in self.pickup_publishers:
        self.pickup_publishers[robot_id].publish(pickup_pose)
        self.delivery_publishers[robot_id].publish(delivery_pose)
    else:
        self.get_logger().warn(f"No publishers set up for robot ID {robot_id}")

    return response

def main(args=None):
    rclpy.init(args=args)
    node = TaskManagerService()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```



7.4.2 Integrating the Python Pose Server Node into the Package

Modify `setup.py` of your Python package (`s7_py_task_manager`) to add an executable for the pose server node:

```
entry_points={
    'console_scripts': [
        "task_manager_exe = s7_py_task_manager.task_manager:main"
    ],
}
```

7.4.3 Summary of Python Pose Server Node Key Identifiers

- `task_manager.py`: Name of the source file for the Python pose server node.
- `py_task_manager_node`: Name of the Python pose server node as defined in the constructor of the Python class.
- `task_manager_exe`: Name of the executable defined in `setup.py`.
- `get_two_poses`: Name of the service provided by the Python server node.

7.5 Package Setup: `s7_py_client_robot`

In this Section, we present the setup of the `s7_py_client_robot` package for one robot client named, labeled, and namespaced as `rdk_x3_amr`, including the corresponding node in Python, and configuration in the `client_robots_launch.py` and `setup.py` files.

On the other hand, in our [ROS 2 course GitHub repository](#), you can find the corresponding source files and configuration for the Tello [Unmanned Aerial Vehicle \(UAV\)](#), which has been named, labeled, and namespaced as `tello_uav`.

7.5.1 Creating a ROS 2 Robot Client Node with Python

Setting Up the Python Node

Navigating to the Package Directory:

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s7_py_client_robot/s7_py_client_robot
```

Replace `s7_py_client_robot` with your actual package name.

Creating the Python Node File:

Create a new Python file for your robot client node, for example, `client_amr.py`:

```
$ touch client_amr.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `client_amr.py` file.



Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `client_amr.py` file for editing.

Implementing the Node Code:

Below is an example implementation of a ROS 2 robot client node in Python:

Robot Client in Python (`client_amr.py`):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s7_robot_network_interface.srv import GetTwoPoses
from s7_robot_network_interface.msg import AMRStatus
from sensor_msgs.msg import JointState
from geometry_msgs.msg import TransformStamped
from tf2_ros import TransformBroadcaster
import tf_transformations
import math

class PosePublisher(Node):
    def __init__(self):
        super().__init__('pose_publisher_node')
        self.get_logger().info('Pose Publisher node has been started')

        # --- Service client setup ---
        self.client = self.create_client(GetTwoPoses, 'get_two_poses')
        while not self.client.wait_for_service(timeout_sec=1.0):
            if not rclpy.ok():
                self.get_logger().error('Interrupted while waiting for service.
                ↪ Exiting.')
            return
        self.get_logger().info('Waiting for service to become available...')
        self.req = GetTwoPoses.Request()

        # --- Robot ID ---
        self.robot_id = 1

        # --- Robot state ---
        self.x = 0.0
        self.y = 0.0
        self.yaw = 0.0
        # self.wheel_angle = 0.0
```



```
# --- Movement parameters ---
self.step_size = 0.01          # meters per update
self.yaw_step = math.pi / 180.0 # radians per update
self.wheel_step = math.pi / 180.0 # radians per update

# --- State machine stage ---
# 0: rotate to pickup
# 1: move to pickup
# 2: rotate to pickup final yaw
# 3: rotate to delivery
# 4: move to delivery
# 5: rotate to delivery final yaw
# 6: request new poses
self.stage = 0

# --- Fetch initial poses ---
self.get_new_poses()

# --- Publishers ---
self.robot_pub = self.create_publisher(AMRStatus, 'robot_status', 10)
# self.joint_pub = self.create_publisher(JointState, 'joint_states', 10)
self.broadcaster = TransformBroadcaster(self)

# --- Timer for periodic updates (~30 Hz) ---
self.timer = self.create_timer(0.033, self.publish)

def get_new_poses(self):
    """Call the get_two_poses service."""
    self.req.robot_id = self.robot_id
    future = self.client.call_async(self.req)
    future.add_done_callback(self.handle_service_response)

def handle_service_response(self, future):
    """Store pickup & delivery poses."""
    resp = future.result()
    if resp is None:
        self.get_logger().error('Failed to call service get_two_poses')
        return

    # Store poses
    self.pickup_pose = resp.pickup_pose
    self.delivery_pose = resp.delivery_pose

    self.pickup_yaw = self.pickup_pose.theta
    self.delivery_yaw = self.delivery_pose.theta

    # Pre-compute yaw needed to face each goal
    self.pickup_to_yaw = math.atan2(
        self.pickup_pose.y - self.y,
        self.pickup_pose.x - self.x
    )
```



```

self.get_logger().info(f'New pickup_pose: ({self.pickup_pose.x:.2f},
→ {self.pickup_pose.y:.2f}, {self.pickup_pose.theta:.2f} rad)')

self.get_logger().info(f'New delivery_pose: ({self.delivery_pose.x:.2f},
→ {self.delivery_pose.y:.2f}, {self.delivery_pose.theta:.2f} rad)')

def normalize_angle(self, angle: float) -> float:
    """Normalize an angle to [-pi, pi]."""
    while angle > math.pi:
        angle -= 2 * math.pi
    while angle < -math.pi:
        angle += 2 * math.pi
    return angle

def publish(self):
    # --- State machine logic ---
    if self.stage == 0:
        # 0. Rotate toward pickup
        yaw_diff = self.normalize_angle(self.pickup_to_yaw - self.yaw)
        if abs(yaw_diff) > self.yaw_step:
            self.yaw += math.copysign(self.yaw_step, yaw_diff)
            self.yaw = self.normalize_angle(self.yaw)
        else:
            self.yaw = self.pickup_to_yaw
            self.stage = 1

    elif self.stage == 1:
        # 1. Move toward pickup position
        dx = self.pickup_pose.x - self.x
        dy = self.pickup_pose.y - self.y
        dist = math.hypot(dx, dy)
        if dist > self.step_size:
            direction = math.atan2(dy, dx)
            self.x += math.cos(direction) * self.step_size
            self.y += math.sin(direction) * self.step_size
        else:
            self.x = self.pickup_pose.x
            self.y = self.pickup_pose.y
            self.stage = 2

    elif self.stage == 2:
        # 2. Rotate to final pickup yaw
        yaw_diff = self.normalize_angle(self.pickup_yaw - self.yaw)
        if abs(yaw_diff) > self.yaw_step:
            self.yaw += math.copysign(self.yaw_step, yaw_diff)
            self.yaw = self.normalize_angle(self.yaw)
        else:
            self.yaw = self.pickup_yaw
            # Prepare rotation to delivery
            self.delivery_to_yaw = math.atan2(
                self.delivery_pose.y - self.y,
                self.delivery_pose.x - self.x
            )
            self.stage = 3

```




```

elif self.stage == 3:
    # 3. Rotate toward delivery
    yaw_diff = self.normalize_angle(self.delivery_to_yaw - self.yaw)
    if abs(yaw_diff) > self.yaw_step:
        self.yaw += math.copysign(self.yaw_step, yaw_diff)
        self.yaw = self.normalize_angle(self.yaw)
    else:
        self.yaw = self.delivery_to_yaw
        self.stage = 4

elif self.stage == 4:
    # 4. Move toward delivery position
    dx = self.delivery_pose.x - self.x
    dy = self.delivery_pose.y - self.y
    dist = math.hypot(dx, dy)
    if dist > self.step_size:
        direction = math.atan2(dy, dx)
        self.x += math.cos(direction) * self.step_size
        self.y += math.sin(direction) * self.step_size
    else:
        self.x = self.delivery_pose.x
        self.y = self.delivery_pose.y
        self.stage = 5

elif self.stage == 5:
    # 5. Rotate to final delivery yaw
    yaw_diff = self.normalize_angle(self.delivery_yaw - self.yaw)
    if abs(yaw_diff) > self.yaw_step:
        self.yaw += math.copysign(self.yaw_step, yaw_diff)
        self.yaw = self.normalize_angle(self.yaw)
    else:
        self.yaw = self.delivery_yaw
        self.stage = 6

elif self.stage == 6:
    # 6. Completed delivery, request new goals
    self.get_logger().info('Completed delivery. Requesting new goals...')
    self.get_new_poses()
    self.stage = 0

# --- Publish the current Robot Status ---
status_msg = AMRStatus()
status_msg.timestamp = self.get_clock().now().to_msg()
status_msg.robot_id = self.robot_id
status_msg.task_stage = self.stage
status_msg.pose.x = self.x
status_msg.pose.y = self.y
status_msg.pose.theta = self.yaw
status_msg.align_yaw = True
self.robot_pub.publish(status_msg)

```



```

# --- Publish joint states ---
# self.wheel_angle += 4 * self.wheel_step
# if abs(self.wheel_angle) > math.pi:
#     self.wheel_angle *= -1
# js = JointState()
# js.header.stamp = self.get_clock().now().to_msg()
# js.name = [
#     'left_front_joint',
#     'left_back_joint',
#     'right_front_joint',
#     'right_back_joint'
# ]
# js.position = [
#     self.wheel_angle,
#     self.wheel_angle,
#     -self.wheel_angle,
#     -self.wheel_angle
# ]
# self.joint_pub.publish(js)

# --- Publish transform for RViz ---
# t = TransformStamped()
# t.header.stamp = self.get_clock().now().to_msg()
# t.header.frame_id = 'odom'
# namespace = self.get_namespace().strip('/')
# t.child_frame_id = f'{namespace}/base_footprint'
# t.transform.translation.x = self.x
# t.transform.translation.y = self.y
# t.transform.translation.z = 0.0
# q_tf = tf_transformations.quaternion_from_euler(0, 0, self.yaw)
# t.transform.rotation.x = q_tf[0]
# t.transform.rotation.y = q_tf[1]
# t.transform.rotation.z = q_tf[2]
# t.transform.rotation.w = q_tf[3]
# self.broadcaster.sendTransform(t)

def main(args=None):
    rclpy.init(args=args)
    node = PosePublisher()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```



7.5.2 Integrating the Python Robot Client Node into the Package

Modify `setup.py` of your Python package (`s7_py_client_robot`) to add an executable for the robot client node:

```
entry_points={
    'console_scripts': [
        "client_amr_exe = s7_py_client_robot.client_amr:main"
    ],
}
```

7.5.3 Summary of Python Robot Client Node Key Identifiers

- `client_amr.py`: Name of the source file for the Python robot client node.
- `pose_publisher_node`: Name of the Python robot client node as defined in the constructor of the Python class. In the next Subsection, we will remap this name to `client_amr` in the launch file.
- `client_amr_exe`: Name of the executable defined in `setup.py`.
- `get_two_poses`: Name of the service that the Python client node calls.

7.5.4 Launch File Definition

Within the `s7_py_robot_client/launch` directory, create (or open) a file named `client_robot_launch.py` with the following structure.

```
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # AMR: RobotStatus publisher node
    amr_status = Node(
        package='s7_py_client_robot',
        executable='client_amr_exe',
        name='client_amr',
        namespace='rdk_x3_amr',
        remappings=[ # <<< this makes /rdk_x3_amr/get_two_poses → /get_two_poses
                     ('get_two_poses', '/get_two_poses'),
        ],
        output='screen'
    )

    return LaunchDescription([
        amr_status
    ])
```

This launch file starts the robot client node with the following features:

- Adds the `rdk_x3_amr` namespace to the robot client node.
- Adds the `rdk_x3_amr` name to the robot client node instead of `pose_publisher_node`, as previously defined in the constructor of the Python class.
- Remaps the `/rdk_x3_amr/get_two_poses` service name to `/get_two_poses` to correctly call the pose service.



7.6 Package Setup: `s7_py_robot_task_monitoring`

In this Section, we present the setup of the `s7_py_robot_task_monitoring` package for the robot state publishers of one robot and the two cylinders that represent the pickup and delivery poses, respectively. The source files of these publishers are named, labeled, and namespaced as `rdk_x3_amr`, `tello_uav`, `cylinder1`, and `cylinder2`, respectively, including the corresponding node in Python, [Unified Robot Description Format \(URDF\)](#) model, and configuration in the `robot_network_launch.py` and `setup.py` files.

On the other hand, in our [ROS 2 course GitHub repository](#), you can find the corresponding source files and configuration of the robot state publishers for the Tello UAV, which has been named, labeled, and namespaced as `tello_uav`.

7.6.1 Creating a Pose Sub-Pub Node with Python: Pickup and Delivery Poses

Setting Up the Python Node

Navigating to the Package Directory:

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s7_py_robot_task_monitoring/s7_py_robot_task_monitoring
```

Replace `s7_py_robot_task_monitoring` with your actual package name.

Creating the Python Node File:

Create a new Python file for your pose sub-pub node, for example, `cylinders.py`:

```
$ touch cylinders.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `cylinders.py` file.

Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `cylinders.py` file for editing.



Implementing the Node Code:

Below is an example implementation of a ROS 2 pose sub-pub node in Python:

Pose Sub-Pub in Python (*cylinders.py*):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Pose2D
from geometry_msgs.msg import TransformStamped
from tf2_ros import TransformBroadcaster
import tf_transformations

class CylinderVisualizer(Node):
    def __init__(self):
        super().__init__('cylinder_visualizer')
        self.get_logger().info("Cylinder Visualizer node has been started")

        # TF broadcaster for all cylinders
        self.broadcaster = TransformBroadcaster(self)

        # --- Number of robots ---
        self.robot_num = 1

        # Create subscribers for pickup & delivery for robots: 1 to robot_num
        self.subs = [] # keep subscriptions alive
        for rid in range(1, self.robot_num + 1):
            ns = '/rdk_x3_amr' # f'/robot{rid}'
            sub_pick = self.create_subscription(
                Pose2D, f'{ns}/pickup_pose',
                lambda msg, r=rid: self.pose_callback(msg, r, 'pickup'),
                10)
            sub_del = self.create_subscription(
                Pose2D, f'{ns}/delivery_pose',
                lambda msg, r=rid: self.pose_callback(msg, r, 'delivery'),
                10)
            self.subs += [sub_pick, sub_del]

    def pose_callback(self, msg: Pose2D, robot_id: int, kind: str):
        """
        Broadcast a TF frame for a cylinder based on:
        - robot_id: 1 - robot_num
        - kind: 'pickup' or 'delivery'
        """
        # Determine cylinder index: pickup1+1, delivery1+2, pickup2+3, etc.
        base = (robot_id - 1) * 2
        cyl_idx = base + (1 if kind == 'pickup' else 2)

        t = TransformStamped()
        t.header.stamp = self.get_clock().now().to_msg()
        t.header.frame_id = 'odom'
        t.child_frame_id = f'cylinder{cyl_idx}/base_footprint'
```



```

    # Translation from Pose2D
    t.transform.translation.x = msg.x
    t.transform.translation.y = msg.y
    t.transform.translation.z = 0.0

    # Rotation from yaw (theta)
    q = tf_transformations.quaternion_from_euler(0.0, 0.0, msg.theta)
    t.transform.rotation.x = q[0]
    t.transform.rotation.y = q[1]
    t.transform.rotation.z = q[2]
    t.transform.rotation.w = q[3]

    # Publish to tf2
    self.broadcaster.sendTransform(t)
    self.get_logger().info(f"Broadcasted cylinder{cyl_idx}: ({msg.x:.2f},
    ↪ {msg.y:.2f}, {msg.theta:.2f})")

def main(args=None):
    rclpy.init(args=args)
    node = CylinderVisualizer()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```

7.6.2 Integrating the Python Pose Sub-Pub Node into the Package

Modify `setup.py` of your Python package (*s7_py_robot_task_monitoring*) to add an executable for the pose sub-pub node:

```

entry_points={
    'console_scripts': [
        "cylinders_exe = s7_py_robot_task_monitoring.cylinders:main"
    ],
},

```

7.6.3 Summary of Python Pose Sub-Pub Node Key Identifiers

- `cylinders.py`: Name of the source file for the Python pose sub-pub node.
- `cylinder_visualizer`: Name of the Python pose sub-pub node as defined in the constructor of the Python class.
- `cylinders_exe`: Name of the executable defined in `setup.py`.
- `/rdk_x3_amr/pickup_pose`: Name of the topic that the Python node subscribes to get the pickup pose for the `rdk_x3_amr`.



- `/rdk_x3_amr/delivery_pose`: Name of the topic that the Python node subscribes to get the delivery pose for the `rdk_x3_amr`.

7.6.4 Creating a Status Sub-Pub Node with Python: Robot Status

Setting Up the Python Node

Navigating to the Package Directory:

Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s7_py_robot_task_monitoring/s7_py_robot_task_monitoring
```

Replace `s7_py_robot_task_monitoring` with your actual package name.

Creating the Python Node File:

Create a new Python file for your status sub-pub node, for example, `robot.py`:

```
$ touch robot.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `robot.py` file.

Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `robot.py` file for editing.

Implementing the Node Code:

Below is an example implementation of a ROS 2 status sub-pub node in Python:

Status Sub-Pub in Python (`robot.py`):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s7_robot_network_interface.msg import AMRStatus
from sensor_msgs.msg import JointState
from geometry_msgs.msg import TransformStamped
from tf2_ros import TransformBroadcaster
import tf_transformations
import math
```



```

class RobotVisualizer(Node):
    def __init__(self):
        super().__init__('robot_visualizer')
        self.get_logger().info("Robot Visualizer node has been started")

        # --- Robot wheel state ---
        self.wheel_angle = 0.0

        # --- Movement parameters ---
        self.wheel_step = math.pi / 180.0 # radians per update

        # --- Number of robots ---
        self.robot_num = 1

        # Create subscribers to robot_status topic for each robot
        self.subs = [] # keep subscriptions alive
        for rid in range(1, self.robot_num + 1):
            topic = 'rdk_x3_amr/robot_status' # f'/robot{rid}/robot_status'
            # Subscribe to '/robotN/robot_status'
            sub = self.create_subscription(
                AMRStatus, topic,
                lambda msg, r=rid: self.RobotPose_callback(msg, r),
                10)
            self.subs.append(sub)

        # Create publishers to joint_states topic for each robot
        self.pubs = [] # keep publications alive
        for rid in range(1, self.robot_num + 1):
            topic = 'rdk_x3_amr/joint_states' # f'/robot{rid}/joint_states'
            # Publish to '/robotN/joint_states'
            pub = self.create_publisher(JointState, topic, 10)
            self.pubs.append(pub)

        # --- Timer for periodic updates (~30 Hz) ---
        self.timer = self.create_timer(0.033, self.publish)

        # TF broadcaster for all robots
        self.broadcaster = TransformBroadcaster(self)

    def RobotPose_callback(self, msg: AMRStatus, robot_id: int):
        """
        Broadcast 'odom'→'rdk_x3_amr(robotN)/base_footprint' transform
        based on Pose2D field from AMRStatus message.
        """
        t = TransformStamped()
        t.header.stamp = msg.timestamp # self.get_clock().now().to_msg()
        t.header.frame_id = 'odom'
        t.child_frame_id = 'rdk_x3_amr/base_footprint' #
        ↪ f'/robot{robot_id}/base_footprint'

        # Translation from Pose2D
        t.transform.translation.x = msg.pose.x
        t.transform.translation.y = msg.pose.y
        t.transform.translation.z = 0.0

```




```

    # Rotation from yaw (theta)
    q = tf_transformations.quaternion_from_euler(0.0, 0.0, msg.pose.theta)
    t.transform.rotation.x = q[0]
    t.transform.rotation.y = q[1]
    t.transform.rotation.z = q[2]
    t.transform.rotation.w = q[3]

    # Publish transform to tf2
    self.broadcaster.sendTransform(t)
    self.get_logger().debug(
        f"Broadcasted robot{robot_id} at ({msg.pose.x:.2f}, {msg.pose.y:.2f},
        ↪ {msg.pose.theta:.2f})"
    )

def publish(self):
    # --- Publish joint states ---
    for robot_id in range(1, self.robot_num + 1):
        self.wheel_angle += 4 * self.wheel_step
        if abs(self.wheel_angle) > math.pi:
            self.wheel_angle *= -1

        js = JointState()
        js.header.stamp = self.get_clock().now().to_msg()
        js.name = [
            'left_front_joint',
            'left_back_joint',
            'right_front_joint',
            'right_back_joint'
        ]
        js.position = [
            self.wheel_angle,
            self.wheel_angle,
            -self.wheel_angle,
            -self.wheel_angle
        ]
        self.pubs[robot_id-1].publish(js)

def main(args=None):
    rclpy.init(args=args)
    node = RobotVisualizer()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()

```



7.6.5 Integrating the Python Status Sub-Pub Node into the Package

Modify `setup.py` of your Python package (`s7_py_robot_task_monitoring`) to add an executable for the status sub-pub node after any existing ROS 2 node entries (e.g., `cylinders_exe`):

```
entry_points={
    'console_scripts': [
        "cylinders_exe = s7_py_robot_task_monitoring.cylinders:main",
        "robot_exe = s7_py_robot_task_monitoring.robot:main"
    ],
}
```

7.6.6 Summary of Python Status Sub-Pub Node Key Identifiers

- `robot.py`: Name of the source file for the Python status sub-pub node.
- `robot_visualizer`: Name of the Python status sub-pub node as defined in the constructor of the Python class.
- `robot_exe`: Name of the executable defined in `setup.py`.
- `/rdk_x3_amr/robot_status`: Name of the topic (of `AMRStatus.msg` type) that the Python node subscribes to get the status from the `rdk_x3_amr` client.
- `/rdk_x3_amr/joint_states`: Name of the topic in which the Python node publishes wheel joint stats from the `rdk_x3_amr` client.

7.6.7 Launch File Definition

Within the `s7_py_task_monitoring/launch` directory, create (or open) a file named `robot_network_launch.py` with the following structure.

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to package
    pkg_share = get_package_share_directory('s7_py_robot_task_monitoring')

    # Paths to URDF files
    cylinder1_urdf = os.path.join(pkg_share, 'urdf', 'cylinder1.urdf')
    cylinder2_urdf = os.path.join(pkg_share, 'urdf', 'cylinder2.urdf')
    amr_urdf = os.path.join(pkg_share, 'urdf', 'rdk_x3_amr.urdf')
```



```

# Read URDF files
with open(cylinder1_urdf, 'r') as infp:
    cylinder1_description = infp.read()

with open(cylinder2_urdf, 'r') as infp:
    cylinder2_description = infp.read()

with open(amr_urdf, 'r') as infp:
    amr_description = infp.read()

# cylinder1 (pickup pose for the AMR) robot_state_publisher
cylinder1_amr_desc = Node(
    package='robot_state_publisher',
    executable='robot_state_publisher',
    # name='cylinder1_state_publisher',
    namespace='cylinder1',
    parameters=[{'robot_description': cylinder1_description}],
    output='screen'
)

# cylinder2 (delivery pose for the AMR) robot_state_publisher
cylinder2_amr_desc = Node(
    package='robot_state_publisher',
    executable='robot_state_publisher',
    # name='cylinder2_state_publisher',
    namespace='cylinder2',
    parameters=[{'robot_description': cylinder2_description}],
    output='screen'
)

# cylinders pose subscriber-publisher node
cylinders_tf = Node(
    package='s7_py_robot_task_monitoring',
    executable='cylinders_exe',
    # name='cylinders_exe',
    output='screen'
)

# amr robot_state_publisher
amr_desc = Node(
    package='robot_state_publisher',
    executable='robot_state_publisher',
    # name='amr_state_publisher',
    namespace='rdk_x3_amr',
    parameters=[{'robot_description': amr_description}],
    output='screen'
)

# robot (amr) pose subscriber-publisher node
robot_tf = Node(
    package='s7_py_robot_task_monitoring',
    executable='robot_exe',
    name='robot',
    output='screen'
)

```



```
return LaunchDescription([
    cylinder1_amr_desc,
    cylinder2_amr_desc,
    amr_desc,
    cylinders_tf,
    robot_tf
])
```

This launch file starts five nodes:

- The `cylinder1/robot_state_publisher` node (`cylinder1_amr_desc`), which publishes the `URDF` model and the transformations of `cylinder1` that represents the pickup pose for the `rdk_x3_amr` client.
- The `cylinder2/robot_state_publisher` node (`cylinder2_amr_desc`), which publishes the `URDF` model and the transformations of `cylinder2` that represents the delivery pose for the `rdk_x3_amr` client.
- The `rdk_x3_amr/robot_state_publisher` node (`amr_desc`), which publishes the `URDF` model and the transformations of `rdk_x3_amr` client.
- The `cylinders` node (`cylinders_tf`), responsible for subscribing and publishing the pose values from the pickup and delivery cylinders for the `rdk_x3_amr` client.
- The `robot` node (`robot_tf`), responsible for subscribing and publishing the status from `rdk_x3_amr` client, its joint state values, and the overall robot geometry.

7.6.8 Creating a Status Subscriber Node with Python: Robot Status Logger

Setting Up the Python Node

Navigating to the Package Directory:

Open a terminal and navigate to your Python package directory within your `ROS 2` workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s7_py_robot_task_monitoring/s7_py_robot_task_monitoring
```

Replace `s7_py_robot_task_monitoring` with your actual package name.

Creating the Python Node File:

Create a new Python file for your status subscriber node, for example, `robot_status_logger.py`:

```
$ touch robot_status_logger.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `robot_status_logger.py` file.

Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```



Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `robot_status_logger.py` file for editing.

Implementing the Node Code:

Below is an example implementation of a ROS 2 status subscriber node in Python:

Status Subscriber in Python (`robot_status_logger.py`):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from s7_robot_network_interface.msg import UAVStatus
from s7_robot_network_interface.msg import AMRStatus
from datetime import datetime, timezone
from zoneinfo import ZoneInfo # Available in Python 3.9+ (For Humble/Ubuntu 22.04 with
    ↪ Python 3.10)
import tf_transformations
import math

class RobotStatusLogger(Node):
    def __init__(self):
        super().__init__('robot_status_logger')
        self.get_logger().info("RobotStatusLogger node has been started")

        # --- Number of robots to be monitored ---
        self.robot_num = 2 # 0: UAV, 1: AMR

        # Robot ID → ID & color label map
        self.rid_label = {
            1: "1 - AMR",
            2: "2 - UAV",
        }

        # Stage → human-readable description map
        self.stage_desc = {
            0: "Rotating to pickup point",
            1: "Moving to pickup point",
            2: "Rotating to pickup final yaw",
            3: "Rotating to delivery point",
            4: "Moving to delivery point",
            5: "Rotating to delivery final yaw",
            6: "Requesting new points",
            7: "Fault: stuck or error",
            8: "No data received",
            9: "Hovering"
        }

        # Threshold for stale data (cycles)
        self.STALE_THRESHOLD = 30
```



```

# Default "no data" template
self.NO_DATA_STAGE = 8
self.NO_DATA_ENTRY = lambda rid: {
    'id': self.rid_label[rid],
    'time': '--:--:--.--',
    'align': 'N/A',
    'pose': '( -.--, -.--, -.--°)',
    'stage': rid * 0 + self.NO_DATA_STAGE, # always 8
    'desc': self.stage_desc[self.NO_DATA_STAGE]}

# 1. Initialize statuses and stale counters
self.statuses = {rid: self.NO_DATA_ENTRY(rid) for rid in range(1, self.robot_num
→ + 1)}
self.no_update_counts = {rid: 0 for rid in range(1, self.robot_num + 1)}

# 2. Subscribe to /rdk_x3_amr/robot_status
self.create_subscription(AMRStatus, '/rdk_x3_amr/robot_status',
    lambda msg: self.status_callback(msg, 1), 10)

# 3. Subscribe to /tello_uav/robot_status
self.create_subscription(UAVStatus, '/tello_uav/robot_status',
    lambda msg: self.status_callback(msg, 2), 10)

# 4. Timer to print periodic updates (~30 Hz)
self.create_timer(0.033, self.print_table)

def status_callback(self, msg, robot_id: int):
    # 1. Format timestamp as HH:MM:SS.ss
    ts = msg.timestamp.sec + msg.timestamp.nanosec * 1e-9
    dt = datetime.fromtimestamp(ts,
    → tz=timezone.utc).astimezone(ZoneInfo("America/Mexico_City"))
    timestamp_str = dt.strftime('%H:%M:%S.%f')[:-4] # two decimals

    # 2. Alignment?
    if hasattr(msg, 'align_yaw'):
        align_str = "True" if msg.align_yaw else "False"
    else:
        align_str = 'N/A'

    # 3. Pose
    if hasattr(msg.pose, 'position') and hasattr(msg.pose, 'orientation'):
        # Pose3D as (x,y,z, roll°, pitch°, yaw°)
        x, y, z = msg.pose.position.x, msg.pose.position.y, msg.pose.position.z
        qx, qy, qz, qw = msg.pose.orientation.x, msg.pose.orientation.y,
        → msg.pose.orientation.z, msg.pose.orientation.w
        # Get roll, pitch, yaw in degrees
        euler = tf_transformations.euler_from_quaternion((qx, qy, qz, qw))
        roll_deg, pitch_deg, yaw_deg = math.degrees(euler[0]),
        → math.degrees(euler[1]), math.degrees(euler[2])
        pose_str = f"({x:6.2f},{y:6.2f},{z:6.2f},\
            {roll_deg:7.1f}°,{pitch_deg:7.1f}°,{yaw_deg:7.1f}°)"
    else: # Pose2D as (x,y,theta°)
        x, y = msg.pose.x, msg.pose.y
        theta_deg = math.degrees(msg.pose.theta)
        pose_str = f"({x:6.2f},{y:6.2f},{theta_deg:7.1f}°)"

```



```

# 4. Stage and description
if hasattr(msg, 'task_stage'):
    stage = msg.task_stage
    desc = self.stage_desc.get(stage, "Unknown stage")
else:
    stage = 9
    desc = self.stage_desc.get(stage, "Unknown stage")

# Update status entry
self.statuses[robot_id] = {
    'id': self.rid_label.get(robot_id),
    'time': timestamp_str,
    'align': align_str,
    'pose': pose_str,
    'stage': stage,
    'desc': desc
}

# Reset stale counter
self.no_update_counts[robot_id] = 0

def print_table(self):
    # 0. Record the local "snapshot" time
    now_local = datetime.now(ZoneInfo("America/Mexico_City"))
    snap_str = now_local.strftime('%Y-%m-%d %H:%M:%S.%f')[:-4]

    # 1. Emit a line showing when we took this snapshot
    header = f"--- Robot status snapshot at local time: {snap_str} ---"
    lines = [f"{header:^112s}"]

    # 2. Fields
    fields = "| Robot ID | Timestamp | Pose2D (x,y,t°) / Pose3D  

    ↳ (x,y,z,r°,p°,y°) | Alignment? | Task Stage | Task Description  

    ↳ |"
    lines.append(fields)

    # 3. Rows for robots 1-4, guaranteed presence
    for rid in sorted(self.statuses):
        row = self.statuses[rid]
        line = (
            f"| {row['id']:<8s} | {row['time']:^11s} | {row['pose']:<49s} "  

            f"| {row['align']:^10s} | {row['stage']:^10d} | {row['desc']:<30s} |"
        )
        lines.append(line)

    # 4. Print once
    self.get_logger().info("\n" + "\n".join(lines))

    # 5. Increment counters for robots without updates
    for rid in self.no_update_counts:
        self.no_update_counts[rid] += 1
        # If a robot has gone stale, reset to NO_DATA
        if self.no_update_counts[rid] >= self.STALE_THRESHOLD:
            self.statuses[rid] = self.NO_DATA_ENTRY(rid)
            self.no_update_counts[rid] = 0

```



```
def main(args=None):
    rclpy.init(args=args)
    node = RobotStatusLogger()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

7.6.9 Integrating the Python Status Subscriber Node into the Package

Modify `setup.py` of your Python package (`s7_py_robot_task_monitoring`) to add an executable for the status subscriber node after any existing ROS 2 node entries (e.g., `cylinders_exe` and `robots_exe`):

```
entry_points={
    'console_scripts': [
        "cylinders_exe = s7_py_robot_task_monitoring.cylinders:main",
        "robot_exe = s7_py_robot_task_monitoring.robot:main",
        "robot_status_logger_exe = s7_py_robot_task_monitoring.robot_status_logger:main"
    ],
},
```

7.6.10 Summary of Python Status Subscriber Node Key Identifiers

- `robot_status_logger.py`: Name of the source file for the Python status subscriber node.
- `robot_status_logger`: Name of the Python status subscriber node as defined in the constructor of the Python class.
- `robot_status_logger_exe`: Name of the executable defined in `setup.py`.
- `/rdk_x3_amr/robot_status`: Name of the topic (of `AMRStatus.msg` type) that the Python node subscribes to get the status from the `rdk_x3_amr` client.
- `/tello_uav/robot_status`: Name of the topic (of `UAVStatus.msg` type) that the Python node subscribes to get the status from the `tello_uav` publisher.

7.7 Building the Packages and Running the Nodes

After making the above package creation and installation, editings and saving all the changes, go to the root of your workspace (e.g., `/ros2_ws`) and build the packages:

```
$ colcon build --packages-select s7_robot_network_interface
$ colcon build --packages-select s7_py_task_manager
$ colcon build --packages-select s7_py_client_robot
$ colcon build --packages-select s7_py_robot_task_monitoring
```



Then, in five terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, run the pose server node:

```
$ source install/setup.bash
$ ros2 run s7_py_task_manager task_manager_exe
```

- In **Terminal 2**, launch the robot client node:

```
$ source install/setup.bash
$ ros2 launch s7_py_client_robot client_robot_launch.py
```

- In **Terminal 3**, launch the pose sub-pub (pickup and delivery cylinders) and status sub-pub (robot status) nodes:

```
$ source install/setup.bash
$ ros2 launch s7_py_robot_task_monitoring robot_network_launch.py
```

- In **Terminal 4**, start **RViz** with the configuration file stored in the **robot_task_monitoring** package:

```
$ source install/setup.bash
$ rviz2 -d src/s7_py_robot_task_monitoring/rviz/robot_network.rviz
```

- In **Terminal 5**, run the status subscriber node to log the robot status:

```
$ source install/setup.bash
$ ros2 run s7_py_robot_task_monitoring robot_status_logger_exe
```

Additionally, open two terminals for diagnostics and introspection tools:

- **Terminal 6:** Launch the **rqt_graph** tool to view the current **ROS 2** node and topic graph:

```
$ rqt_graph
```

- **Terminal 7:** Use the **view_frames** tool from the **tf2_tools** package to visualize the current coordinate frame hierarchy (tree):

```
$ ros2 run tf2_tools view_frames
```

7.8 Practice Assignment: tf2 Library –Robot Network

In this assignment, you will run the ROS 2 robot network implemented in Python, as described in Sections 7.3, 7.4, 7.5, and 7.6. Capture evidence of successful execution using four files: three PNG images and one PDF file, and submit them in the designated assignment area on Canvas (within the ROS 2 module). The filenames must follow this format:

FirstnameLastname_evidence_sX_Y.png

where sX corresponds to the session number, and Y is the evidence number. For example, correct filenames would be:

EduardoDavila_evidence_s7_1.png,
EduardoDavila_evidence_s7_2.png,
EduardoDavila_evidence_s7_3.pdf,
EduardoDavila_evidence_s7_4.png.

7.8.1 Executing ROS 2 Robot Network and Verifying Functionality

After creating and installing the necessary packages as described in Section 7.2, editing and saving the Python nodes, and updating the CMakeLists.txt, setup.py, and package.xml files, build the packages:

```
$ colcon build --packages-select s7_robot_network_interface
$ colcon build --packages-select s7_py_task_manager
$ colcon build --packages-select s7_py_client_robot
$ colcon build --packages-select s7_py_robot_task_monitoring
```

Then, in five terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, run the pose server node:

```
$ source install/setup.bash
$ ros2 run s7_py_task_manager task_manager_exe
```

- In **Terminal 2**, launch the robot client node (AMR and UAV publishing their status and pose):

```
$ source install/setup.bash
$ ros2 launch s7_py_client_robot client_robot_launch.py
```

- In **Terminal 3**, launch the pose sub-pub (pickup and delivery cylinders) and status sub-pub (robot status) nodes:

```
$ source install/setup.bash
$ ros2 launch s7_py_robot_task_monitoring robot_network_launch.py
```

- In **Terminal 4**, start RViz with the configuration file stored in the robot_task_monitoring package:

```
$ source install/setup.bash
$ rviz2 -d src/s7_py_robot_task_monitoring/rviz/robot_network.rviz
```



- In **Terminal 5**, run the status subscriber node to log the robot status:

```
$ source install/setup.bash
$ ros2 run s7_py_robot_task_monitoring robot_status_logger_exe
```

You should observe that:

1. The `task_manager` node process the service requests from the `rdk_x3_amr` client node.
2. The `cylinder1/robot_state_publisher` and the `cylinder2/robot_state_publisher` nodes publish the corresponding cylinder `URDF` models (for visualization in `RViz`) that represents the pickup and delivery poses, respectively, for the `rdk_x3_amr` client node.
3. The `rdk_x3_amr/robot_state_publisher` node publishes the RDK X3 `AMR URDF` model and the coordinate transforms for visualization in `RViz`. Additionally, the `rdk_x3_amr` client node publishes its status and 2D pose values.
4. The `tello_uav/robot_state_publisher` node publishes the Tello `UAV URDF` model and the coordinate transforms for visualization in `RViz`. Additionally, the `tello_uav` node publishes its status and 3D pose values.
5. The `cylinders` node subscribes and publishes the pose values from the pickup and delivery cylinders for the `rdk_x3_amr` client, to broadcast its coordinate transforms for visualization in `RViz`.
6. The `robots` node subscribes and publishes the status from `rdk_x3_amr` client, its joint state values, and the overall robot geometry to broadcast its coordinate transforms for visualization in `RViz`.
7. The `robot_status_logger` node constantly prints the robot status if its subscription receives data.

7.8.2 Visualizing the ROS Graph with `rqt_graph`

To verify that all the nodes are correctly connected, use the `rqt_graph` tool. Then, in **Terminal 6**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run rqt_graph rqt_graph # Or simply: rqt_graph
```

This will visualize the `ROS 2` graph, showing topics connecting the robot network nodes, as described above.

7.8.3 Visualizing the frames with `view_frames`

To observe the current coordinate frame hierarchy, use the `image_frames` tool from the `tf2_tools` package. Then, in **Terminal 7**, execute the following command (without the need to enter the workspace directory):

```
$ ros2 run tf2_tools view_frames
```

This tool will generate a PDF file in the current directory of the terminal. The file provides a graphical representation of the current coordinate frame tree published by the `robot_state_publisher` nodes for the `AMR`, `UAV`, and cylinder `URDF` models, having the `odom` frame as the global coordinate frame and the `base_footprint` frames as the main local coordinate frame for each robot model.



7.8.4 Submission Instructions

The screenshot displays five terminal windows from a ROS 2 environment. Terminal 1 and 2 show the `task_manager` node providing pickup and delivery poses to the `tello_uav` node. Terminal 3 and 4 show the `cylinder_visualizer` node broadcasting cylinder data to the `rviz2` node. Terminal 5 shows a table of robot status snapshots at a specific time.

Robot ID	Timestamp	Pose2D (x,y,θ°)	Pose3D (x,y,z,r°,p°,y°)	Alignment?	Task Stage	Task Description
1 - AMR	21:10:11.76	(0.05, 0.76, 149.9°)	(0.04, 0.77, 149.9°)	True	1	Moving to pickup point
2 - UAV	21:10:11.78	(-1.99, -0.09, 1.30, 0.0°, -0.0°, -85.0°)	(-1.99, -0.10, 1.30, 0.0°, -0.0°, -84.0°)	N/A	9	Hovering

Figure 7.2: Example screenshot showing terminal outputs of the robot network nodes in action.

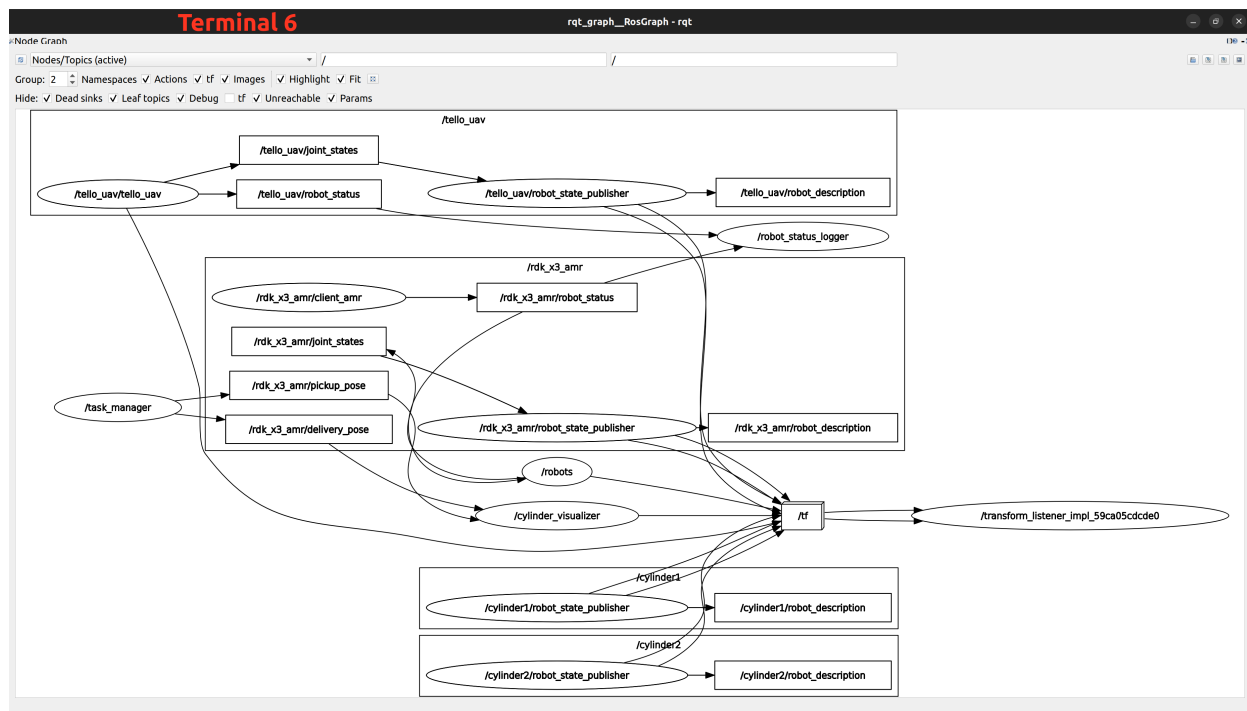


Figure 7.3: Example screenshot showing the ROS 2 graph of the robot network nodes visualized with the `rqt_graph` tool.

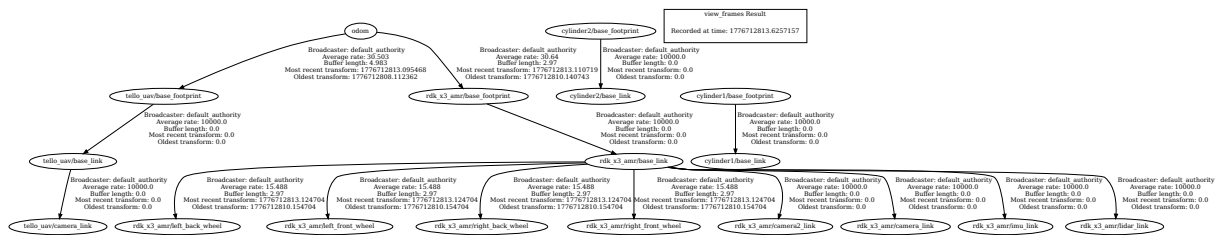


Figure 7.4: PDF showing the graphical representation of the current coordinate frame tree visualized with the `view_frames` tool.

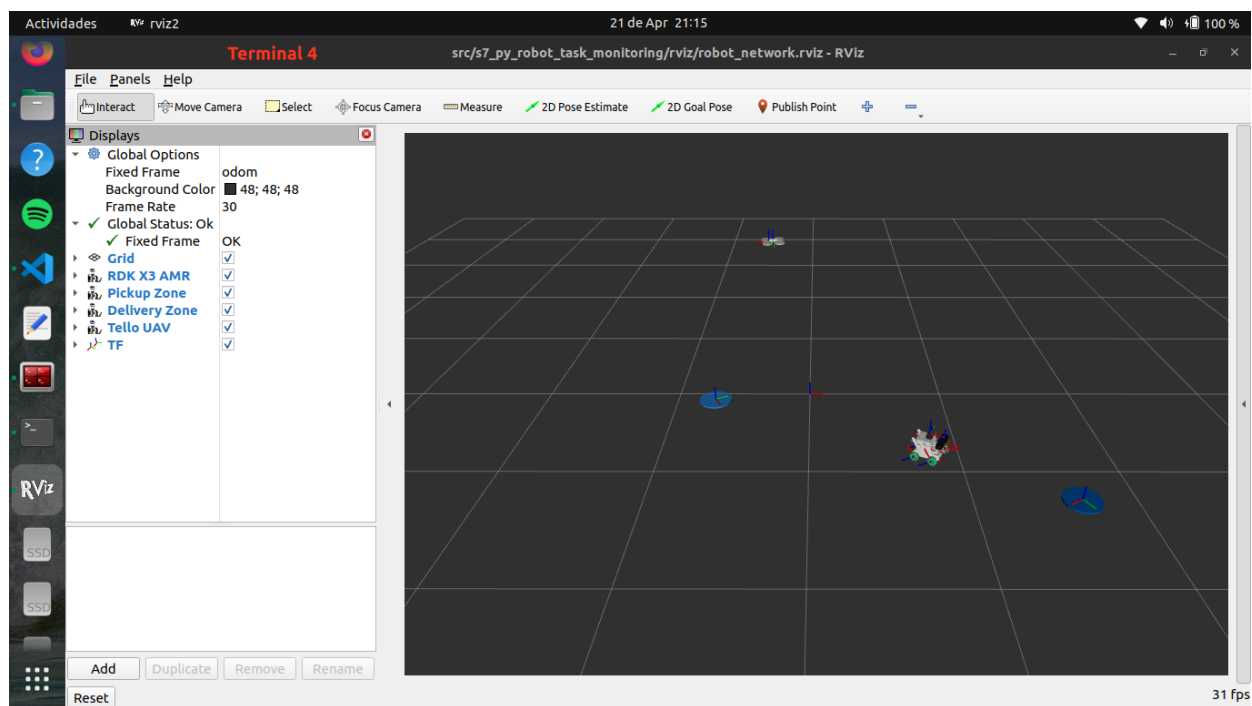


Figure 7.5: Example screenshot showing the visualization of the RDK X3 AMR, the Tello UAV, and the goal poses represented by cylinders in RViz.

1. Run the ROS 2 robot network nodes as described above.
2. As **first evidence**, capture a **screenshot showing the terminal outputs** from the publisher, subscriber, service, and client nodes, as described in Section 7.8.1. See Figure 7.2 for reference. Save the screenshot as a PNG file.
3. As **second evidence**, capture a **screenshot showing the ROS 2 graph** (from the `rqt_graph` tool), illustrating the connections among the robot network nodes: publisher, subscriber, service, and client nodes, as described in Section 7.8.2. See Figure 7.3 for reference. Save the screenshot as a PNG file.
4. As **third evidence**, locate the PDF file generated by the `view_frames` tool, from the `tf2_tools` package, as described in Section 7.8.3. See Figure 7.4 for reference. Save the evidence as a PDF or PNG file.

5. As **fourth evidence**, capture a **screenshot showing the visualization of the RDK X3 AMR and the Tello UAV in RViz**. See Figure 7.5 for reference. Save the screenshot as a PNG file.
6. Name the files following this format: `FirstnameLastname_evidence_sX_Y.png` (replace **X** with the session number and **Y** with the evidence number).
7. Submit the files as specified by the course guidelines on Canvas.

Note: Your machine's username and hostname must be visible in the screenshots to verify the authenticity of your submission, as shown in Figure 7.2. Any submission that appears copied, unclear, or altered will be considered invalid and may receive a score of 0.



CHAPTER

8

Occupancy Grid Maps – SLAM

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✳ **Abstract** - In this chapter, we will explore the fundamental concepts and practical applications of [Simultaneous Localization and Mapping \(SLAM\)](#) in the context of [Robot Operating System \(ROS\) 2](#). We will cover the workflow to create, save, and load occupancy grid maps, which are essential for enabling autonomous navigation in mobile robots. By the end of this chapter, you will have a solid understanding of how to implement the [SLAM](#) technique using [ROS 2 Humble](#) environment, and how to utilize occupancy grid maps for effective robot navigation. All the source files referenced in this chapter are available in our [ROS 2 course GitHub repository](#) , located in the `ros2_ws/src` folder, corresponding to the [ROS 2](#) packages with the `s8` prefix.

8.1 Introduction

[Simultaneous Localization and Mapping \(SLAM\)](#) is a advanced technique where the robot **simultaneously builds a map of its environment** while keeping track of its position within that map, as shown in Figure 8.1. Unlike odometry, which relies solely on internal sensor data, [SLAM](#) **incorporates external environmental data** to correct drift and improve accuracy.

Key Features of [SLAM](#):

- **Map building:** It creates a map of the environment (either a 2D grid or a 3D model) as the robot moves.

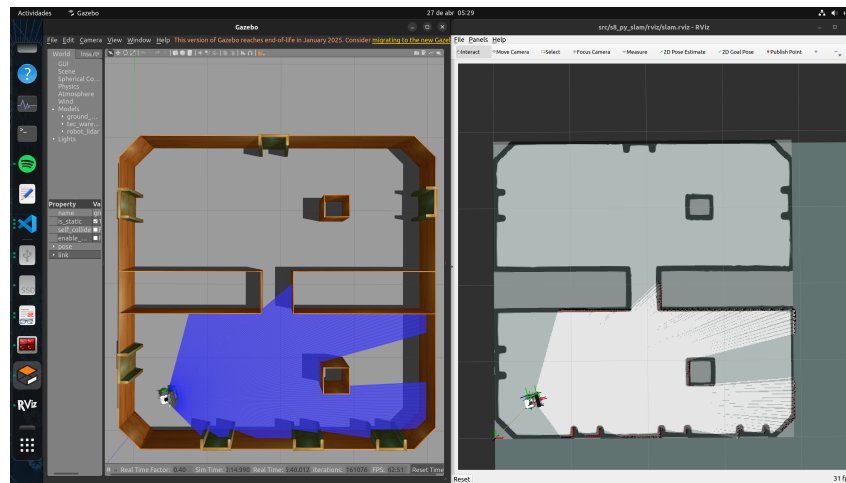


Figure 8.1: Simulation of the RDK X3 AMR in Gazebo, and its visualization in RViz building an occupancy grid map by using a LiDAR sensor.

- **Localization:** It tracks the robot's position within that map.
- **Sensor fusion:** SLAM combines data from **multiple sensors** (e.g., LiDAR, cameras, depth sensors) to estimate both the environment and the robot's location accurately.
- **Loop closure:** SLAM detects when the robot has revisited a previously seen location and corrects errors from odometry drift.

Applications:

- Used in autonomous navigation (e.g., self-driving cars, mobile robots).
- Essential for robots that need to operate in large or unstructured environments where GPS or other external positioning systems are not available.
- Common in mapping tasks, such as creating floor plans or 3D models of environments.

8.1.1 Occupancy Grid Map

In the context of ROS 2, "creating and loading occupancy grid maps" refers to the processes of generating and utilizing structured representations of an environment, which are crucial for enabling autonomous navigation in mobile robots.

Creating Grid Maps

Grid maps are two-dimensional representations of an environment, divided into a grid where each cell contains information about that specific area. There are two primary types:

1. **Occupancy Grid Maps:** These maps indicate whether a cell is occupied, free, or unknown. They are typically generated using SLAM techniques. For instance, the `slam_toolbox` package in ROS 2 Humble allows for the creation of occupancy grid maps by processing sensor data as the robot explores its environment.

2. **Multi-layered Grid Maps:** These provide more detailed information by storing multiple data layers per cell, such as elevation, surface normals, or traversability. The `grid_map` library facilitates the creation and management of such maps, which are particularly useful for robots operating in complex terrains .

Loading Grid Maps

Once a grid map is created, it needs to be loaded into the robot's navigation system for use in tasks like path planning and obstacle avoidance. In [ROS 2 Humble](#):

- The **Map Server** in the Navigation2 ([Nav2](#)) stack is responsible for loading and providing access to occupancy grid maps. It can load maps from files during startup or dynamically at runtime using the `load_map` service.
- For multi-layered grid maps, the `grid_map` library offers tools to load and manage these maps, enabling advanced navigation strategies that consider various environmental factors.

Practical Application

In a typical workflow:

1. **Map Creation:** A robot explores its environment using sensors (like LiDAR or cameras), and a [SLAM](#) algorithm processes this data to create a grid map.
2. **Map Loading:** The generated map is loaded into the robot's navigation system using the Map Server or appropriate tools from the `grid_map` library.
3. **Navigation:** The robot uses the loaded map to plan paths, avoid obstacles, and navigate effectively within the environment.

Understanding and effectively utilizing grid maps in [ROS 2 Humble](#) is essential for developing robust autonomous navigation capabilities in mobile robots.

8.2 Prerequisites

8.2.1 Workspace and Package Creation

Before getting started, make sure you are familiar with workspace and package creation, as described in Sections [2.2](#), [2.3](#), and [2.4](#).

Open a terminal and navigate to the `src` directory of your [ROS 2](#) workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src
```

Replace `ros2_ws` with the actual name of your workspace.

Next, create the following packages in your workspace:

- Python package to contain the pose publisher and occupancy grid map nodes nodes:

```
$ ros2 pkg create s8_py_slam --build-type ament_python --dependencies rclpy
→ geometry_msgs sensor_msgs tf2_ros tf_transformations nav_msgs --license
→ Apache-2.0 --description "SLAM simulation package"
```



Feel free to replace `s8_py_slam` with a name of your preference.

The parameter (`.yaml`), frame (`.pdf`), launch (`.py`), map (`.yaml` and `.pgm`), mesh (`.STL`), `RViz` (`.rviz`), `URDF` (`.urdf`), and world (`.world`) files should be organized into directories named `config`, `frames`, `launch`, `maps`, `meshes`, `rviz`, `urdf`, and `worlds`, respectively. Create these directories in the Python package:

```
$ mkdir config frames launch maps meshes rviz urdf worlds
```

8.2.2 Installing Required Packages

Before proceeding, ensure that the following library and packages are installed. Update your package list (apt repository) and install them with:

```
$ sudo apt update
$ sudo apt install ros-humble-gazebo*
$ sudo apt install ros-humble-turtlebot3*
$ sudo apt install ros-humble-slam-toolbox
$ sudo apt install ros-humble-nav2-map-server
$ sudo apt install ros-humble-nav2-lifecycle-manager
$ pip3 install backports.zoneinfo[tzdata] # For Python < 3.9 (For Foxy/Ubuntu 20.04 with
↳ Python 3.8)
$ python3 -c "import zoneinfo; print(zoneinfo)" # To confirm that 'zoneinfo' is already
↳ bundled with Python, as it is part of the standard library for Python 3.9+ (For
↳ Humble/Ubuntu 22.04 with Python 3.10)
$ pip3 install pytz # For Python 3.9+ (For Humble/Ubuntu 22.04 with Python 3.10)
```

Then, add the following sourcing command to the shell's initialization file (e.g., `.bashrc`) so that the Gazebo environment setup is automatically loaded every time a new terminal session is opened (replace `.bash` with your shell if not using bash):

```
$ echo "source /usr/share/gazebo/setup.bash" >> ~/.bashrc
$ source ~/.bashrc
$ cat ~/.bashrc # Optionally to verify the sourcing command was correctly added
```

Additionally, from the root of your workspace, use `rosdep` to check for any other missing dependencies before building:

```
$ rosdep install -i --from-path src --rosdistro humble -y
```

8.3 *setup.py*

To make accessible the parameter (`.yaml`), launch (`.py`), map (`.yaml` and `.pgm`), mesh (`.STL`), `RViz` (`.rviz`), `URDF` (`.urdf`), and world (`.world`) files we have defined in the Python package, add the data files while preserving any existing file setup, with the following lines to `setup.py`:



```
data_files=[
    ('share/ament_index/resource_index/packages',
     ['resource/' + package_name]),
    ('share/' + package_name, ['package.xml']),
    (os.path.join('share', package_name, 'config'), glob('config/*')),
    (os.path.join('share', package_name, 'launch'), glob('launch/*')),
    (os.path.join('share', package_name, 'maps'), glob('maps/*')),
    (os.path.join('share', package_name, 'meshes'), glob('meshes/*')),
    (os.path.join('share', package_name, 'rviz'), glob('rviz/*')),
    (os.path.join('share', package_name, 'urdf'), glob('urdf/*')),
    (os.path.join('share', package_name, 'worlds'), glob('worlds/*'))
],
```

8.4 package.xml

Make sure the required dependencies are defined within the `package.xml` file:

```
<depend>rclpy</depend>
<depend>sensor_msgs</depend>
<depend>geometry_msgs</depend>
<depend>nav_msgs</depend>
<depend>std_msgs</depend>
<depend>tf2_ros</depend>
<depend>tf_transformations</depend>

<!-- Added manually -->
<depend>s7_robot_network_interface</depend>
```

Note:

As suggested in Chapter 5, for packages with launch files, it is a good idea to add an `exec_depend` dependency on the `ros2launch` package in the `package.xml` file from each package that contains launch files, as well as other dependencies that were not previously defined, such as `robot_state_publisher` and `rviz2`:

```
<exec_depend>ros2launch</exec_depend>
<exec_depend>robot_state_publisher</exec_depend>
<exec_depend>rviz2</exec_depend>
```

This is not strictly necessary, but it is a good practice to ensure that all required dependencies are properly declared.

8.5 Creating a ROS 2 TF-based Pose Sub-Pub Node with Python

8.5.1 Setting Up the Python Node

Navigating to the Package Directory:



Open a terminal and navigate to your Python package directory within your ROS 2 workspace (e.g., `~/ros2_ws`):

```
$ cd ros2_ws/src/s8_py_slam/s8_py_slam
```

Replace `s8_py_slam` with your actual package name.

Creating the Python Node File:

Create a new Python file for your TF-based pose subscriber-publisher node (e.g., `pose_tf_sub_pub.py`):

```
$ touch pose_tf_sub_pub.py
```

Examine the folder contents to verify the new file has been created:

```
$ ls
```

You should now see the `pose_tf_sub_pub.py` file.

8.5.2 Editing the Python Node

Opening the Workspace in VS Code:

Navigate back to the root of your workspace:

```
$ cd ../../..
```

Then, open the workspace in Visual Studio Code:

```
$ code .
```

In VS Code, locate and open the `pose_tf_sub_pub.py` file for editing.

Implementing the Node Code:

Below is an example implementation of a ROS 2 TF-based pose subscriber-publisher node in Python:

TF-Based Pose Subscriber-Publisher Node in Python (`pose_tf_sub_pub.py`):

```
#!/usr/bin/env python3
import rclpy
from rclpy.node import Node
from tf2_ros import TransformListener, Buffer
from geometry_msgs.msg import PoseStamped
from geometry_msgs.msg import Point
from geometry_msgs.msg import Quaternion
from s7_robot_network_interface.msg import AMRStatus
import tf_transformations

class PosePublisher(Node):
    def __init__(self):
        super().__init__('pose_tf_sub_pub', namespace='rdk_x3_amr')
        self.tf_buffer = Buffer()
        self.tf_listener = TransformListener(self.tf_buffer, self)
```



```
# --- Publishers ---
# Opt 1: Publisher for a PoseStamped message
self.pose_pub = self.create_publisher(PoseStamped, '/rdk_x3_amr/pose_stamped',
    ↪ 10)
# Opt 2: Publisher for a custom AMRStatus message
self.robot_pub = self.create_publisher(AMRStatus, '/rdk_x3_amr/robot_status',
    ↪ 10)

# --- Timer for periodic updates (~15 Hz) ---
self.timer = self.create_timer(0.066, self.publish_pose)

def publish_pose(self):
    try:
        now = rclpy.time.Time()
        trans = self.tf_buffer.lookup_transform(target_frame='map',
            ↪ source_frame='base_footprint', time=now)

        # Extract position and orientation from the transform
        position = Point()
        position.x = trans.transform.translation.x
        position.y = trans.transform.translation.y
        position.z = trans.transform.translation.z
        quaternion = Quaternion()
        quaternion.x = trans.transform.rotation.x
        quaternion.y = trans.transform.rotation.y
        quaternion.z = trans.transform.rotation.z
        quaternion.w = trans.transform.rotation.w

        # Opt 1: Publish PoseStamped message
        pose_msg = PoseStamped()
        pose_msg.header.stamp = self.get_clock().now().to_msg()
        pose_msg.header.frame_id = 'map'
        pose_msg.pose.position = position # Or simply trans.transform.translation
        pose_msg.pose.orientation = quaternion # Or simply trans.transform.rotation
        self.pose_pub.publish(pose_msg)

        # Opt 2: Publish AMRStatus message
        status_msg = AMRStatus()
        status_msg.timestamp = self.get_clock().now().to_msg()
        status_msg.robot_id = 1 # Default ID for the AMR
        status_msg.task_stage = 9 # Exploring
        status_msg.pose.x = position.x # Or simply trans.transform.translation.x
        status_msg.pose.y = position.y # Or simply trans.transform.translation.y
        qx, qy, qz, qw = quaternion.x, quaternion.y, quaternion.z, quaternion.w
        # Get roll, pitch, yaw in degrees
        euler = tf_transformations.euler_from_quaternion((qx, qy, qz, qw))
        status_msg.pose.theta = euler[2]
        # status_msg.align_yaw = True
        self.robot_pub.publish(status_msg)

    except Exception as e:
        self.get_logger().warn(f'Could not transform: {e}')
```



```
def main():
    rclpy.init()
    node = PosePublisher()
    try:
        rclpy.spin(node)
    except Exception as e:
        node.get_logger().error(f'Exception caught: {e}')
    finally:
        node.destroy_node()
        rclpy.shutdown()

if __name__ == '__main__':
    main()
```

8.5.3 Integrating the Python Pose Sever Node into the Package

Modify `setup.py` of your Python package (`s8_py_slam`) to add an executable for the `TF`-based pose subscriber-publisher node:

```
entry_points={
    'console_scripts': [
        "pose_tf_sub_pub_exe = s8_py_slam.pose_tf_sub_pub:main"
    ],
},
```

8.5.4 Summary of TF-Based Pose Subscriber-Pub Node Key Identifiers

- `pose_tf_sub_pub.py`: Name of the source file for the Python `TF`-based pose subscriber-publisher node.
- `/rdk_x3_amr/pose_tf_sub_pub`: Name of the Python `TF`-based pose subscriber-publisher node as defined in the constructor of the Python class.
- `pose_tf_sub_pub_exe`: Name of the executable defined in `setup.py`.
- `/pose_stamped` and `/rdk_x3_amr/robot_status`: Name of the topics provided by the Python `TF`-based pose subscriber-publisher node.

8.6 Launch File Definition

8.6.1 Spawning the Robot

Within the `s8_py_slam/launch` directory, create (or open) a file named `spawn_robot_launch.py` with the following structure:

```
import os
from launch import LaunchDescription
from launch.actions import IncludeLaunchDescription
from launch.launch_description_sources import PythonLaunchDescriptionSource
from launch_ros.actions import Node
from ament_index_python.packages import get_package_share_directory
```



```
def generate_launch_description():
    # Path to package
    pkg_share = get_package_share_directory('s8_py_slam')

    # Path to the URDF file
    # amr_plugin or box_robot_plugin
    urdf_plugin_path = os.path.join(pkg_share, 'urdf', 'box_robot_plugin.urdf')

    # Launch file for Gazebo
    gazebo_launch = os.path.join(
        get_package_share_directory('gazebo_ros'), 'launch', 'gazebo.launch.py')

    # Path to your custom world file
    # tec_warehouse or robot_world
    world_path = os.path.join(pkg_share, 'worlds', 'tec_warehouse.world')
    return LaunchDescription([
        # 1) Start Gazebo with the specified world and verbosity
        IncludeLaunchDescription(
            PythonLaunchDescriptionSource(gazebo_launch),
            launch_arguments={
                'world': world_path,
                # 'verbose': 'true'
            }.items(),
        ),

        # 2) Spawn the robot model
        Node(
            package='gazebo_ros',
            executable='spawn_entity.py',
            arguments=[
                '-entity', 'robot_lidar',
                '-file', urdf_plugin_path,
                '-x', '0.5',
                '-y', '0.5',
                '-z', '90.0'
            ],
            output='screen'
        )
    ])

```

This launch file starts Gazebo with the following features:

- Starts Gazebo through the `gazebo.launch.py` launch file from the `gazebo_ros` package.
- Loads the **URDF** model (RDK X3 **AMR** or box robot) in the Gazebo environment by using the `spawn_entity.py` node from the `gazebo_ros` package, with the initial position at $(x = 0.5, y = 0.5, z = 90.0)$ Cartesian coordinates.
- Loads the world model (`tec_warehouse` or `robot_world`) in the Gazebo environment.

8.6.2 Displaying the Robot

Within the `s8_py_slam/launch` directory, create (or open) a file named `display_robot_launch.py` with the following structure:



```

import os
from launch import LaunchDescription
from ament_index_python.packages import get_package_share_directory
from launch_ros.actions import Node

def generate_launch_description():
    # Path to package
    pkg_share = get_package_share_directory('s8_py_slam')

    # Path to the URDF file
    # amr or box_robot_plugin
    urdf_path = os.path.join(pkg_share, 'urdf', 'box_robot_plugin.urdf')

    # Path to RViz configuration file
    # Source from src to save changes in RViz session
    rviz_config_path = os.path.join('src/s8_py_slam/', 'rviz', 'slam.rviz')

    # Read URDF files
    with open(urdf_path, 'r') as infp:
        robot_desc = infp.read()

    return LaunchDescription([
        # 1) Publish the URDF to /robot_description
        Node(
            package='robot_state_publisher',
            executable='robot_state_publisher',
            # name='robot_state_publisher',
            namespace='rdk_x3_amr',
            output='screen',
            parameters=[{'robot_description': robot_desc,
                        'use_sim_time': True
                        }],
        ),

        # 2) RViz2
        Node(
            package='rviz2',
            executable='rviz2',
            name='rviz2',
            arguments=['-d', rviz_config_path],
            output='screen',
        )
    ])

```

This launch file starts two nodes:

- The **robot** (**robot_state_publisher**) node, which publishes the **URDF** model and the transformations of either RDK X3 **AMR** or box robot.
- Starts **RViz** through the **rviz2** node from the **rviz2** package, loading the **slam.rviz** configuration file.



8.6.3 Loading a Saved Occupancy Grid Map

Within the `s8_py_slam/launch` directory, create (or open) a file named `map_server_launch.py` with the following structure:

```
import os
from ament_index_python.packages import get_package_share_directory
from launch import LaunchDescription
from launch_ros.actions import Node

def generate_launch_description():
    # Path to package
    pkg_share = get_package_share_directory('s8_py_slam')

    # Path to YAML file
    map_yaml_file = os.path.join(pkg_share, 'maps', 'map_server_params.yaml')

    return LaunchDescription([
        # Static transform from map to odom
        Node(
            package='tf2_ros',
            executable='static_transform_publisher',
            name='static_map_to_odom',
            arguments=['0', '0', '0', '0', '0', '0', 'map', 'odom'],
            output='screen'
        ),

        # Launch the lifecycle manager to manage the map_server node
        Node(
            package='nav2_lifecycle_manager',
            executable='lifecycle_manager',
            name='lifecycle_manager_map',
            output='screen',
            parameters=[{
                'use_sim_time': True,
                'autostart': True,
                'node_names': ['map_server']
            }]
        ),

        # Launch the map_server node
        Node(
            package='nav2_map_server',
            executable='map_server',
            name='map_server',
            output='screen',
            parameters=[map_yaml_file]
        ),
    ])
]
```

This launch file starts three nodes:

- A static transform publisher node from the `tf2_ros` package, which publishes a static transform from the `map` frame to the `odom` frame.



- The lifecycle manager node from the `nav2_lifecycle_manager` package, which manages the lifecycle of the `map_server` node.
- The `map_server` node from the `nav2_map_server` package, which loads a saved occupancy grid map from a YAML file and provides it to the rest of the system.

8.7 Building the Package and Running the Nodes

After making the above creations, package installations, edits and saving all the changes, go to the root of your workspace (e.g., `/ros2_ws`) and build the packages:

```
$ colcon build --packages-select s7_robot_network_interface
$ colcon build --packages-select s7_py_robot_task_monitoring
$ colcon build --packages-select s8_py_slam
$ colcon build --packages-select s8_openslam_gmapping
$ colcon build --packages-select s8_slam_gmapping
```

Then, in seven terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, spawn the robot in Gazebo:

```
$ source install/setup.bash
$ ros2 launch s8_py_slam spawn_robot_launch.py
```

- In **Terminal 2**, display the robot in `RViz`:

```
$ source install/setup.bash
$ ros2 launch s8_py_slam display_robot_launch.py
```

- In **Terminal 3**, launch the `SLAM` node:

```
$ source install/setup.bash
$ ros2 launch s8_slam_gmapping slam_gmapping.launch.py
```

Optionally, after scanning the environment, you may save the generated occupancy grid map using the `map_saver` node, as describe in Appendix D. Alternatively, you may use the pre-generated map provided in the `maps` directory of the `s8_py_slam` package, which is loaded in the next step.

- In **Terminal 4**, launch the `lifecycle_manager` and `map_server` nodes to load a previously-created occupancy grid map through the `/static_map` topic:

```
$ source install/setup.bash
$ ros2 launch s8_py_slam map_server_launch.py
```

Note: Since the `lifecycle_manager` node controls the `map_server` node, the map is published only once through the `/static_map` topic. Due to `RViz`'s asynchronous behavior, you may need to restart the nodes if the map does not appear on the first attempt in the `StaticMap` display in `RViz`. You can additionally verify successful loading by checking **Terminal 2** for a message such as:

```
[INFO] [...] [rviz2]: Trying to create a map of size 391 x 389 using 1 swatches
```



- In **Terminal 5**, run the **TF**-based pose subscriber-publisher node:

```
$ source install/setup.bash
$ ros2 run s8_py_slam pose_tf_sub_pub_exe
```

It extracts the transformations between the **map** and **base_footprint** frames from the **Transform (TF)** tree (the bus-type **/tf** topic for coordinate transforms), and publishes the robot pose as **PoseStamped** and **AMRStatus** messages.

- In **Terminal 6**, run the robot status logger node:

```
$ source install/setup.bash
$ ros2 run s7_py_robot_task_monitoring robot_status_logger_exe
```

It logs the robot status, specifically its pose, as received from the **AMRStatus** message published by the **pose_tf_sub_pub** node.

- In **Terminal 7**, preferably at a separate window, run the teleoperation node to control the robot with the keyboard:

```
$ source install/setup.bash
$ ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

Use the indicated keys to move the robot in the desired direction and velocity. As it moves, the **SLAM** node processes LiDAR data and builds the occupancy grid map, which is visualized in **RViz**.

Additionally, open two terminals for diagnostics and introspection tools:

- **Terminal 8:** Launch the **rqt_graph** tool to view the current **ROS 2** node and topic graph:

```
$ rqt_graph
```

- **Terminal 9:** Use the **view_frames** tool from the **tf2_tools** package to visualize the current coordinate frame hierarchy (tree):

```
$ ros2 run tf2_tools view_frames
```

8.8 Practice Assignment: Occupancy Grid Maps – SLAM

In this assignment, you will run the [ROS 2 SLAM](#) simulation, as described in Sections 8.5 and 8.6. Capture evidence of successful execution using three screenshots as PNG files and one PDF file, and submit them in the designated assignment area on Canvas (within the [ROS 2](#) module). The filenames must follow this format:

`FirstnameLastname_evidence_sX_Y.png`

where `sX` corresponds to the session number, and `Y` is the evidence number. For example, correct filenames would be:

`EduardoDavila_evidence_s8_1.png`,
`EduardoDavila_evidence_s8_2.png`,
`EduardoDavila_evidence_s8_3.pdf`,
`EduardoDavila_evidence_s8_4.png`.

8.8.1 Executing ROS 2 SLAM Simulation and Verifying Functionality

After creating and installing the necessary packages as described in Section 8.2, editing and saving the Python nodes, and updating the `setup.py` and `package.xml` files, build the packages:

```
$ colcon build --packages-select s7_robot_network_interface
$ colcon build --packages-select s7_py_robot_task_monitoring
$ colcon build --packages-select s8_py_slam
$ colcon build --packages-select s8_openslam_gmapping
$ colcon build --packages-select s8_slam_gmapping
```

Note: The `spawn_robot_launch.py` and `display_robot_launch.py` launch files from the `s8_py_slam` package load, by default, the `tec_warehouse` world and the box robot model (`box_robot_plugin.urdf`), to reduce computational overhead. However, you may modify these launch files to load the `robot_world` world or the RDK X3 [AMR](#) model (`amr_plugin.urdf` for `spawn_robot_launch.py` and `amr.urdf` for `display_robot_launch.py`), as described in Section 8.6. Then, in seven terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, spawn the robot in Gazebo:

```
$ source install/setup.bash
$ ros2 launch s8_py_slam spawn_robot_launch.py
```

- In **Terminal 2**, display the robot in [RViz](#):

```
$ source install/setup.bash
$ ros2 launch s8_py_slam display_robot_launch.py
```

- In **Terminal 3**, launch the [SLAM](#) node:

```
$ source install/setup.bash
$ ros2 launch s8_slam_gmapping slam_gmapping.launch.py
```

Optionally, after scanning the environment, you may save the generated occupancy grid map using the `map_saver` node, as describe in Appendix D. Alternatively, you may use the



pre-generated map provided in the `maps` directory of the `s8_py_slam` package, which is loaded in the next step.

- In **Terminal 4**, launch the `lifecycle_manager` and `map_server` nodes to load a previously-created occupancy grid map through the `/static_map` topic:

```
$ source install/setup.bash
$ ros2 launch s8_py_slam map_server_launch.py
```

Note: Since the `lifecycle_manager` node controls the `map_server` node, the map is published only once through the `/static_map` topic. Due to `RViz`'s asynchronous behavior, you may need to restart the nodes if the map does not appear on the first attempt in the `StaticMap` display in `RViz`. You can additionally verify successful loading by checking **Terminal 2** for a message such as:

```
[INFO] [...] [rviz2]: Trying to create a map of size 391 x 389 using 1 swatches
```

- In **Terminal 5**, run the `TF`-based pose subscriber-publisher node:

```
$ source install/setup.bash
$ ros2 run s8_py_slam pose_tf_sub_pub_exe
```

It extracts the transformations between the `map` and `base_footprint` frames from the `Transform (TF)` tree (the bus-type `/tf` topic for coordinate transforms), and publishes the robot pose as `PoseStamped` and `AMRStatus` messages.

- In **Terminal 6**, run the robot status logger node:

```
$ source install/setup.bash
$ ros2 run s7_py_robot_task_monitoring robot_status_logger_exe
```

It logs the robot status, specifically its pose, as received from the `AMRStatus` message published by the `pose_tf_sub_pub` node.

- In **Terminal 7**, preferably at a separate window, run the teleoperation node to control the robot with the keyboard:

```
$ source install/setup.bash
$ ros2 run teleop_twist_keyboard teleop_twist_keyboard
```

Use the indicated keys to move the robot in the desired direction and velocity. As it moves, the `SLAM` node processes LiDAR data and builds the occupancy grid map, which is visualized in `RViz`.

You should observe that:

1. The robot `URDF` model is spawned in Gazebo, inside the simulated world/environment, with a simulated LiDAR sensor.
2. The `robot_state_publisher` node publishes the robot `URDF` model and the coordinate transforms for `RViz` visualization, as simulated in Gazebo.
3. The occupancy grid map is displayed and updated through the `/map` topic, set in the `Map` display in `RViz`, as the robot explores the environment.
4. The `map_server` node correctly loads and publishes the static map through the `/static_map` topic, set in the `StaticMap` display in `RViz`.
5. The `robot_status_logger` node receives and logs pose data from the `TF`-based pose subscriber-publisher node.



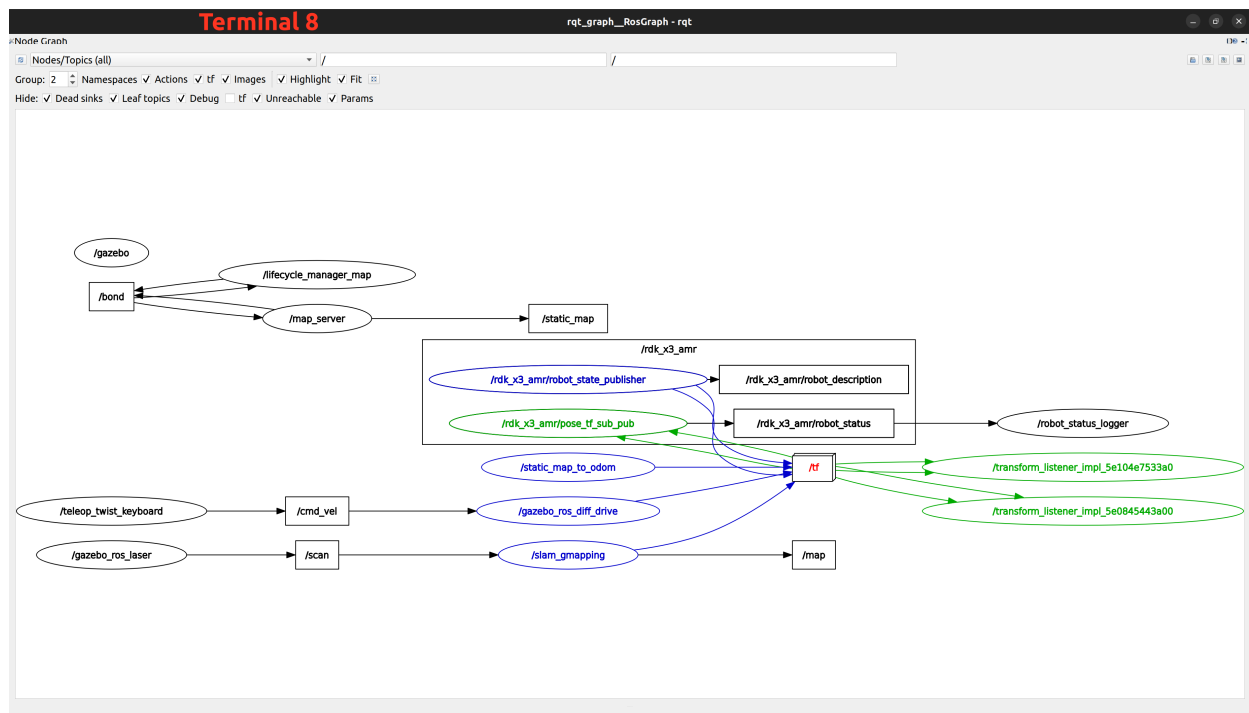


Figure 8.3: Example screenshot showing the ROS 2 graph of the SLAM simulation nodes visualized with the `rqt_graph` tool.

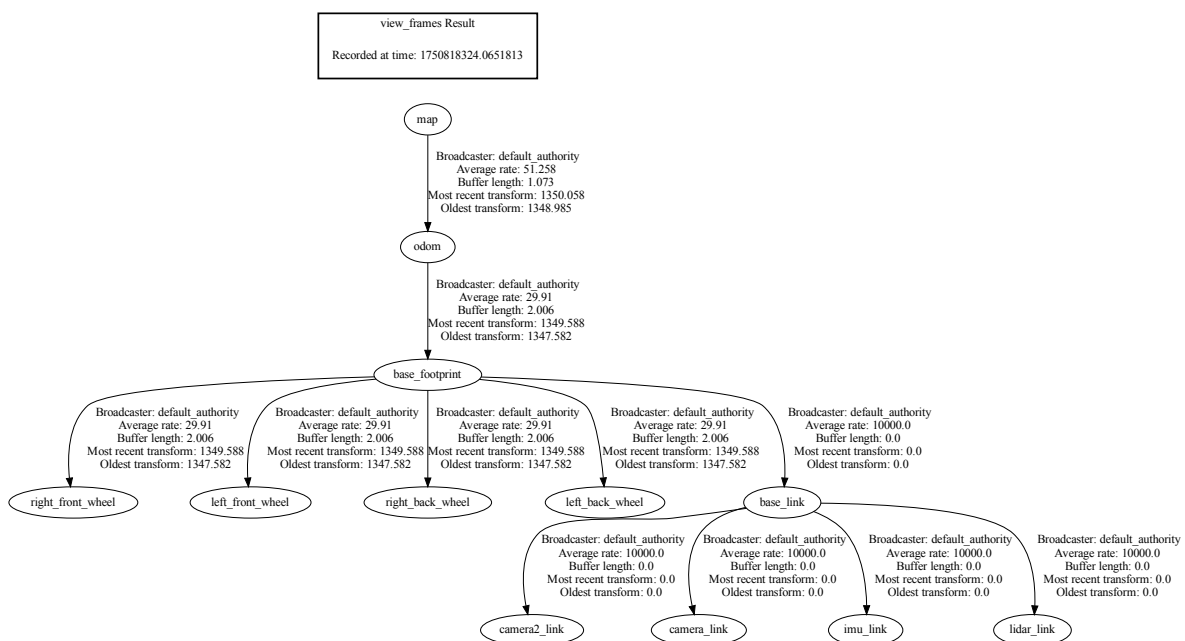


Figure 8.4: PDF showing the graphical representation of the current coordinate frame tree visualized with the `view_frames` tool.

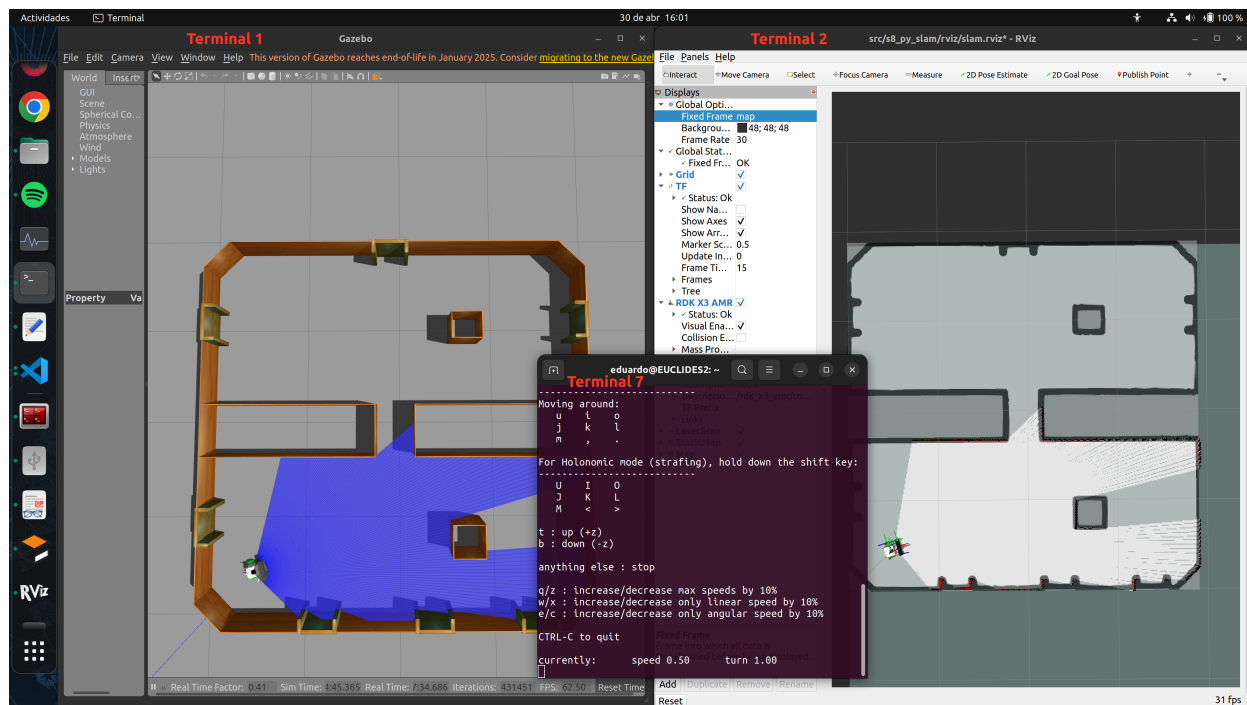


Figure 8.5: Example screenshot showing the simulation and visualization of the RDK X3 AMR in Gazebo and RViz, respectively, and the terminal where the `teleop_twist_keyboard` node is running.

1. Run the ROS 2 SLAM simulation as described above.
2. As **first evidence**, capture a **screenshot showing the terminal outputs** from the publisher, subscriber, and service nodes, as described in Sections 8.8.1. See Figure 8.2 for reference. Save the screenshot as a PNG file.
3. As **second evidence**, capture a **screenshot showing the ROS 2 graph** (from the `rqt_graph` tool), illustrating the connections among the SLAM-related nodes: publisher, subscriber, and service nodes, as described in Section 8.8.2. See Figure 8.3 for reference. Save the screenshot as a PNG file.
4. As **third evidence**, locate the PDF file generated by the `view_frames` tool, as described in Section 8.8.3. See Figure 8.4 for reference. Save the evidence as a PDF or PNG file.
5. As **fourth evidence**, capture a **screenshot showing the simulation and visualization of the robot in Gazebo and RViz**, along with the `teleop_twist_keyboard` node. See Figure 8.4 for reference.
6. As **fourth evidence**, capture a **screenshot showing the visualization of the RDK X3 AMR in RViz**. See Figure 8.5 for reference. Save the screenshot as a PNG file.
7. Name the files following this format: `FirstnameLastname_evidence_sX_Y.png` (replace `X` with the session number and `Y` with the evidence number).
8. Submit the files as specified by the course guidelines on Canvas.

Note: Your machine's username and hostname must be visible in the screenshots to verify the authenticity of your submission, as shown in Figure 8.2. Any submission that appears copied, unclear, or altered will be considered invalid and may receive a score of 0.



APPENDIX



Essential ROS 2 Command Reference

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This appendix compiles the commands utilized throughout the course for creating and configuring workspaces, packages, nodes, and using integrated tools in [Robot Operating System \(ROS\) 2](#).

A.1 System Package Management and Updates in Ubuntu

Command	Description
<code>sudo apt update</code>	Updates the list of available packages and their versions.
<code>sudo apt upgrade</code>	Installs the latest versions of all installed packages.
<code>sudo apt install <pkg_name></code>	Installs the specified package.
<code>sudo apt install ./<file>.deb</code>	Installs a package from a local <code>.deb</code> file.
<code>sudo apt install ros-humble-desktop</code>	Installs the full desktop version of ROS 2 Humble .
<code>sudo apt install ros-dev-tools</code>	Installs development tools for ROS 2 .
<code>sudo apt install ros-humble-turtlesim</code>	Installs the <code>turtlesim</code> package for ROS 2 Humble .
<code>sudo apt install python3-colcon-common-extensions</code>	Installs common extensions for <code>colcon</code> to facilitate package building.
<code>sudo apt install terminator</code>	Installs the Terminator terminal emulator.
<code>sudo snap install tree</code>	Installs the command-line utility that lists the contents of directories in a recursive, tree-like, and hierarchical format.

Table A.I: Commands for system package management and updates in Ubuntu.

A.2 ROS 2 Environment Setup

Command	Description
<code>source /opt/ros/humble/setup.bash</code>	Sets up the environment for ROS 2 Humble in the current terminal session.
<code>source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash</code>	Sets up the autocompletion for <code>colcon</code> in the current terminal session.
<code>echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc</code>	Adds the ROS 2 Humble environment setup to the <code>.bashrc</code> file for future terminal sessions.
<code>echo "source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash" >> ~/.bashrc</code>	Enables autocompletion for <code>colcon</code> in the <code>.bashrc</code> file for future terminal sessions.
<code>echo "alias rqt_service_caller='ros2 run rqt_service_caller rqt_service_caller'" >> ~/.bashrc</code>	Creates an alias for the <code>rqt_service_caller</code> tool in the <code>.bashrc</code> file.
<code>source ~/.bashrc</code>	Reloads the <code>.bashrc</code> file to apply changes without restarting the terminal.
<code>cat ~/.bashrc</code>	Displays the contents of the <code>.bashrc</code> file.
<code>gedit ~/.bashrc</code>	Opens the <code>.bashrc</code> file in the <code>gedit</code> text editor.

Table A.II: Commands for ROS 2 environment setup.

A.3 Workspace Creation and Building

Command	Description
<code>mkdir -p <ws_name>/src</code>	Creates the workspace directory and its <code>src</code> subdirectory.
<code>cd <ws_name></code>	Navigates to the workspace directory.
<code>colcon build</code>	Builds all packages in the workspace.
<code>colcon build --symlink-install</code>	Builds packages using symbolic links, useful for development purposes.
<code>source ./install/setup.bash</code>	Sets up the environment for the built workspace in the current terminal session.
<code>code .</code>	Opens the current directory in Visual Studio Code.

Table A.III: Commands for workspace creation and building in ROS 2.



A.4 Package Creation and Editing in C++ and Python

Command	Description
<code>cd <ws_name>/src</code>	Navigates to the workspace's <code>src</code> directory.
<code>ros2 pkg create <pkg_name> --build-type ament_cmake --dependencies rclcpp std_msgs --license Apache-2.0 --description "Your package description here"</code>	Creates a new C++ package with specified dependencies, license, and description.
<code>ros2 pkg create <pkg_name> --build-type ament_python --dependencies rclpy std_msgs --license Apache-2.0 --description "Your package description here"</code>	Creates a new Python package with specified dependencies, license, and description.
<code>ls</code>	Lists the files and directories in the current directory.
<code>cd ..</code>	Navigates one directory up.
<code>rosdep install -i --from-path src --rosdistro humble -y</code>	In the root of the workspace, checks for missing dependencies before building.
<code>colcon build --packages-select <pkg_name></code>	Builds only the <code><pkg_name></code> package.
<code>colcon build --packages-select <pkg_name> --symlink-install</code>	Builds the <code><pkg_name></code> package with symbolic link installation.
<code>cd src/<pkg_name></code>	Navigates to the <code><pkg_name></code> directory.
<code>code package.xml</code>	Opens the <code>package.xml</code> file in Visual Studio Code.
<code>code CMakeLists.txt</code>	Opens the <code>CMakeLists.txt</code> file in Visual Studio Code.
<code>code setup.py</code>	Opens the <code>setup.py</code> file in Visual Studio Code.
<code>cd <ws_name>/src/<pkg_name>/src</code>	Navigates to the <code>src</code> subdirectory of a C++ package.
<code>cd <ws_name>/src/<pkg_name>/<pkg_name></code>	Navigates to the <code><pkg_name></code> subdirectory of a Python package.
<code>touch <filename>.cpp</code>	Creates a new C++ source file named <code><filename></code> .
<code>touch <filename>.py</code>	Creates a new Python source file named <code><filename></code> .
<code>cd ../../..</code>	Navigates three directories up.

Table A.IV: Package creation and editing in C++ and Python.



A.5 Dev & Debug Essentials: CLI/GUI Commands

Command	Description
<code>ros2 run <pkg_name> <node_name></code>	Runs the specified node from the specified package.
<code>ros2 run <pkg_name> <node_name> <args></code>	Runs the specified node from the specified package with specified arguments.
<code>ros2 service call <service_name> <service_type> <args></code>	Calls the service <service_name> from the specified package with specified arguments.
<code>ros2 node list</code>	Lists all active nodes in the ROS 2 system.
<code>ros2 node info <node_name></code>	Displays detailed information about the specified node.
<code>ros2 topic list</code>	Lists all active topics in the ROS 2 system.
<code>ros2 topic info <topic_name></code>	Displays detailed information about the specified topic.
<code>ros2 topic echo <topic_name></code>	Echoes the messages of a specified topic.
<code>ros2 topic pub <topic_name> <msg_type> <args></code>	Publishes a message to the specified topic.
<code>ros2 topic hz <topic_name></code>	Displays the message frequency of a specified topic.
<code>ros2 interface show <pkg_name>/msg/<custom_msg_name></code>	Shows the field structure of a custom message type in the specified package.
<code>ros2 interface show <pkg_name>/srv/<custom_srv_name></code>	Shows the request/response structure of a custom service type in the specified package.
a) <code>ros2 run rqt_graph rqt_graph</code> b) <code>rqt_graph</code>	Runs the <code>rqt_graph</code> tool to visualize the ROS 2 nodes and topics.
a) <code>ros2 run rqt_service_caller rqt_service_caller</code> b) <code>rqt_service_caller</code> (after creating the alias listed in Table A.II)	Runs the <code>rqt_service_caller</code> tool to call services in a GUI.

Table A.V: Essential command-line and graphical interface commands for running nodes, interacting with topics and services, and debugging ROS 2 systems.



APPENDIX



Using the `geometry_msgs` Package in C++ and Python

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This appendix provides an overview of how to utilize the `geometry_msgs` package in both C++ and Python. This package defines standardized message types for common geometric primitives such as points, vectors, quaternions, and poses, facilitating spatial data representation and interoperability throughout the [Robot Operating System \(ROS\) 2](#) ecosystem.

B.1 Importing the Package

B.1.1 In C++

To use the `geometry_msgs` in C++, include the necessary header files in your source code:

```
#include <geometry_msgs/msg/point.hpp>
#include <geometry_msgs/msg/quaternion.hpp>
#include <geometry_msgs/msg/pose.hpp>
```

Each message type corresponds to its own header file within the `geometry_msgs/msg` directory.

B.1.2 In Python

In Python, import the required message classes as follows:

```
from geometry_msgs.msg import Point, Quaternion, Pose
```

B.1.3 Including `geometry_msgs` in Your Package Dependencies

To use the `geometry_msgs` message types in your ROS 2 package, you must explicitly declare it as a dependency in both `CMakeLists.txt` (for C++) and `package.xml` (for both C++ and Python packages):

In `CMakeLists.txt`:

```
find_package(geometry_msgs REQUIRED)

ament_target_dependencies(<node_name> geometry_msgs)
```

In `package.xml`:

```
<depend>geometry_msgs</depend>
```

B.2 Using Message Definitions

The following examples demonstrate how to create, assign, and access values for the most commonly used message types from `geometry_msgs`.

B.2.1 Point

Represents a 3D position using Cartesian coordinates `x`, `y`, and `z`.

C++

```
#include <geometry_msgs/msg/point.hpp>

geometry_msgs::msg::Point point;
point.x = 1.0;
point.y = 2.0;
point.z = 3.0;

// Accessing values
double x_value = point.x;
double y_value = point.y;
double z_value = point.z;
```

Python

```
from geometry_msgs.msg import Point

point = Point()
point.x = 1.0
point.y = 2.0
point.z = 3.0

# Accessing values
x_value = point.x
y_value = point.y
z_value = point.z
```



B.2.2 Quaternion

Represents an orientation in free space using quaternion components **x**, **y**, **z**, and **w**. This representation avoids the gimbal lock issues found in Euler angles and supports smooth interpolations.

- **x**, **y**, and **z**: Vector part of the quaternion, defining the axis of rotation.
- **w**: Scalar part, representing the amount of rotation around the axis.
- A unit quaternion (with magnitude 1) is typically required for valid orientation.

C++

```
#include <geometry_msgs/msg/quaternion.hpp>

geometry_msgs::msg::Quaternion quaternion;
quaternion.x = 0.0;
quaternion.y = 0.0;
quaternion.z = 0.0;
quaternion.w = 1.0;

// Accessing values
double w_value = quaternion.w;
```

Python

```
from geometry_msgs.msg import Quaternion

quaternion = Quaternion()
quaternion.x = 0.0
quaternion.y = 0.0
quaternion.z = 0.0
quaternion.w = 1.0

# Accessing values
w_value = quaternion.w
```

B.2.3 Pose

Combines a **position** (as a Point) and **orientation** (as a Quaternion) to define a complete 6-DOF pose in 3D space.

C++

```
#include <geometry_msgs/msg/pose.hpp>

geometry_msgs::msg::Pose pose;
pose.position.x = 1.0;
pose.position.y = 2.0;
pose.position.z = 3.0;
pose.orientation.x = 0.0;
pose.orientation.y = 0.0;
pose.orientation.z = 0.0;
pose.orientation.w = 1.0;

// Accessing values
double pos_x = pose.position.x;
double orient_w = pose.orientation.w;
```

Python

```
from geometry_msgs.msg import Pose

pose = Pose()
pose.position.x = 1.0
pose.position.y = 2.0
pose.position.z = 3.0
pose.orientation.x = 0.0
pose.orientation.y = 0.0
pose.orientation.z = 0.0
pose.orientation.w = 1.0

# Accessing values
pos_x = pose.position.x
orient_w = pose.orientation.w
```

B.3 Additional Message Types

The `geometry_msgs` package includes other message types such as `Twist`, `Accel`, and `Wrench`, each with specific fields for representing different geometric concepts. The usage pattern for these messages is similar: import the message type, create an instance, and assign or access the relevant fields accordingly.

For more detailed information on each message type, refer to the [ROS 2 official reference: geometry_msgs: Humble](#).



APPENDIX



Configuring ROS 2 Package Paths for URDF Visualizer in VS Code

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To correctly visualize URDF or Xacro files that utilize `package://` URIs (Uniform Resource Identifiers), it is essential to configure the [URDF Visualizer](#) extension in Visual Studio Code to recognize the paths of your [Robot Operating System \(ROS\) 2](#) packages. This ensures that all resources referenced in your robot description files are accurately located and rendered.

C.1 Steps to Add a ROS 2 Package Path

1. Open your project folder in Visual Studio Code.
2. Navigate to the Extensions view by clicking on the Extensions icon in the Activity Bar or pressing `Ctrl+Shift+X`.
3. Search for [URDF Visualizer](#) and ensure it is installed.
4. Click on the gear icon (Manage) next to the [URDF Visualizer](#) extension and select **Extension Settings**.
5. In the settings, locate the **Urdf-visualizer: Packages** section.
6. Click on **Edit in settings.json** to open the `settings.json` file.
7. Within the `urdf-visualizer.packages` object, add your package path without removing existing entries. You can specify:

- A relative path:

```
"package_name": "${workspaceFolder}/src/package_name"
```

- An absolute path:

```
"package_name": "/absolute/path/to/package_name"
```

8. Save the `settings.json` file and close it.
9. To visualize your URDF or Xacro file:
 - Open the file in VS Code.
 - Click on the eye icon in the top-right corner.
 - Alternatively, press `Ctrl+Shift+P` to open the Command Palette, type `URDF:Preview URDF/Xacro`, and select it.
10. If the `URDF Visualizer` is already open, click on the reload button to apply the new settings.

C.2 Additional Notes

- If your URDF or Xacro file references multiple packages, ensure all are listed in the `urdf-visualizer.packages` configuration.
- Use absolute paths if your packages are located outside the current workspace.
- The extension supports both URDF and Xacro files, provided the package paths are correctly configured.

For more detailed information, refer to the official documentation: [URDF Visualizer Extension](#).



APPENDIX



Saving and Loading Occupancy Grid Maps

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In this appendix, you will learn how to generate and save an occupancy grid map using the `map_saver` node from the `nav2_map_server` package in [Robot Operating System \(ROS\) 2](#). This process is typically performed after running a [Simultaneous Localization and Mapping \(SLAM\)](#) algorithm and fully exploring the environment with the robot. The resulting map can later be reused for localization and navigation tasks.

D.1 Steps to Save an Occupancy Grid Map

1. Ensure that your [SLAM](#) node is running (e.g., `slam_gmapping` or `slam_toolbox`), as described in [Section 8.7](#) or [Section 8.8](#), and that the robot has sufficiently explored the environment to generate a complete map.
2. Open a terminal, go to the root of your workspace (e.g., `/ros2_ws`), and source the workspace environment:

```
$ source install/setup.bash
```

3. Run the `map_saver_cli` node from the `nav2_map_server` package:

```
$ ros2 run nav2_map_server map_saver_cli -f <map_name>
```

4. This command will generate two files in the current directory:
 - `<map_name>.pgm`: the occupancy grid image.
 - `<map_name>.yaml`: the metadata file describing the map.

5. Move the generated files to the desired directory (e.g., the `maps` folder of your package):

```
$ mv <map_name>.* src/<package_name>/maps/
```

6. Verify that the files were correctly generated and stored in the target directory.

Although you can enhance the map by post-processing the `.pgm` file (e.g., using image editing software to clean up noise or improve contrast), it is generally recommended to use the original `.yaml` file and avoid modifying the values unless necessary.

D.2 Loading a Saved Map

Once the map has been generated, it can be loaded using the `map_server` node. Since it is a static map, it must be published on a topic with a different name than the default (e.g., `/static_map`). Then, it is suggested to call it through an additional parameter file (e.g., `map_server_params.yaml`) to specify the correct topic name and other essential parameters, as shown below:

```
map_server:
  ros__parameters:
    yaml_filename: "src/<package_name>/maps/<map_name>.yaml"
    topic_name: "static_map"
    frame_id: "map"
    use_sim_time: True
```

Now, in a terminal, go to the root of your workspace and ensure that your workspace environment is sourced:

```
$ source install/setup.bash
```

You can run the `map_server` node, with the specified parameter file to load the saved map, either **A)** directly through terminal commands or **B)** using a launch file (recommended).

A) Loading the Map through ROS 2 Commands in Terminals:

In two terminals, source the workspace environment in each, and proceed as follows:

- In **Terminal 1**, run the `map_server` node with the specified parameter file:

```
$ source install/setup.bash
$ ros2 run nav2_map_server map_server --ros-args --params-file
↪ src/<package_name>/maps/map_server_params.yaml
```

- In **Terminal 2**, run the lifecycle manager to activate the map server node:

```
$ source install/setup.bash
$ ros2 lifecycle set /map_server configure
$ ros2 lifecycle set /map_server activate
```

As you enter the lifecycle commands, you should observe that:

- i. The `map_server` node transitions from the **unconfigured** state to the **inactive** state after the configure command:



```
[INFO] [...] [map_server]:
map_server lifecycle node launched.
Waiting on external lifecycle transitions to activate
See https://design.ros2.org/articles/node_lifecycle.html for more information.
[INFO] [...] [map_server]: Creating
[INFO] [...] [map_server]: Configuring
[INFO] [...] [map_io]: Loading yaml file: src/<package_name>/maps/<map_name>.yaml
[INFO] [...] [map_io]: resolution: ...
[INFO] [...] [map_io]: origin[0]: ...
[INFO] [...] [map_io]: origin[1]: ...
[INFO] [...] [map_io]: origin[2]: ...
[INFO] [...] [map_io]: free_thresh: ...
[INFO] [...] [map_io]: occupied_thresh: ...
[INFO] [...] [map_io]: mode: ...
[INFO] [...] [map_io]: negate: ...
[INFO] [...] [map_io]: Loading image_file: src/<package_name>/maps/<map_name>.pgm
[INFO] [...] [map_io]: Read map src/<package_name>/maps/<map_name>.pgm: ...
```

ii. Then to the **active** state after the activate command:

```
[INFO] [...] [map_server]: Activating
[INFO] [...] [map_server]: Creating bond (map_server) to lifecycle manager.
```

iii. And for both lifecycle commands, you should see the successful state transition messages in the terminal where the lifecycle manager is running:

```
Transitioning successful
```

B) Loading the Map through a Launch File (Recommended):

Alternatively, use a launch file to start both the `lifecycle_manager` and `map_server` nodes, ensuring proper lifecycle handling. For example, you can create a launch file named `map_server_launch.py` in the `launch` directory of your package with a content similar to the launch file defined in Section 8.6.3, **making sure to include the correct path to your parameter file and adjusting the node names if necessary**. Then, run the launch file:

```
$ ros2 launch <package_name> map_server_launch.py
```

After executing the launch file, you should see the lifecycle transitions and map loading messages in the terminal where the launch file is executed, similar to the messages described in **Option A**):

```
...
[INFO] [lifecycle_manager...]: process started with pid [...]
[INFO] [map_server...]: process started with pid [...]
...
[INFO] [...] [lifecycle_manager_map]: Creating
[INFO] [...] [map_server]:
map_server lifecycle node launched.
Waiting on external lifecycle transitions to activate
See https://design.ros2.org/articles/node_lifecycle.html for more information.
```



```
[INFO] [...] [map_server]: Creating
[INFO] [...] [lifecycle_manager_map]: Creating and initializing lifecycle service
→ clients
[INFO] [...] [lifecycle_manager_map]: Starting managed nodes bringup...
[INFO] [...] [lifecycle_manager_map]: Configuring map_server
[INFO] [...] [map_server]: Configuring
[INFO] [...] [map_io]: Loading yaml file: src/<package_name>/maps/<map_name>.yaml
[INFO] [...] [map_io]: resolution: ...
[INFO] [...] [map_io]: origin[0]: ...
[INFO] [...] [map_io]: origin[1]: ...
[INFO] [...] [map_io]: origin[2]: ...
[INFO] [...] [map_io]: free_thresh: ...
[INFO] [...] [map_io]: occupied_thresh: ...
[INFO] [...] [map_io]: mode: ...
[INFO] [...] [map_io]: negate: ...
[INFO] [...] [map_io]: Loading image_file: src/<package_name>/maps/<map_name>.pgm
[INFO] [...] [map_io]: Read map src/<package_name>/maps/<map_name>.pgm: ...
[INFO] [...] [lifecycle_manager_map]: Activating map_server
[INFO] [...] [map_server]: Activating
[INFO] [...] [map_server]: Creating bond (map_server) to lifecycle manager.
[INFO] [...] [lifecycle_manager_map]: Server map_server connected with bond.
[INFO] [...] [lifecycle_manager_map]: Managed nodes are active
[INFO] [...] [lifecycle_manager_map]: Creating bond timer...
```

For both options, since the `lifecycle_manager` node controls the `map_server` node, the map is published only once through the new map topic (e.g., `/static_map`). Due to `RViz`'s asynchronous behavior, you may need to restart the nodes if the map does not appear on the first attempt in `RViz`. You can additionally verify successful loading by checking the terminal, where `RViz` was launched, for a message such as:

```
[INFO] [...] [rviz2]: Trying to create a map of size ... x ... using 1 swatches
```

D.3 Additional Notes

- The map is saved based on the current state of the `SLAM`-generated occupancy grid. Ensure the environment has been sufficiently explored before saving.
- The `.yaml` file contains important parameters such as resolution, origin, and thresholds for occupied and free space. Avoid modifying these values unless necessary.
- The `map_server` node typically publishes the map through the `/map` or `/static_map` topic, depending on the configuration.
- In addition to the observation made at the end of Section D.1, if the map does not appear in `RViz`, verify that:
 - The correct topic is selected in the Map display (e.g., either `/map` or `/static_map`).
 - The fixed frame is properly set (commonly `map`).
 - The lifecycle and map server nodes are correctly executed and activated, respectively.
- For consistent results, always use the same coordinate frame (e.g., `map`) when saving and loading maps.



APPENDIX



Security in ROS 2 Humble

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In this appendix, you will learn the essential concepts and practical steps required to implement a basic secure [Robot Operating System \(ROS\) 2](#) workspace using the `sros2` package in [ROS 2 Humble](#). The objective is not to deeply study cybersecurity, but rather to understand how [ROS 2](#) nodes can authenticate each other and exchange encrypted messages using the [Data Distribution Service \(DDS\)](#)-Security standard integrated into [ROS 2](#).

This appendix also presents a progressive methodology to experimentally compare unsecured and secured communication scenarios using packet inspection tools such as `tcpdump`.

E.1 ROS_DOMAIN_ID

[ROS 2](#) uses [DDS](#) as its communication middleware. [DDS](#) separates independent communication groups using a concept called [Domain ID](#). Nodes sharing the same `ROS_DOMAIN_ID` can discover each other and communicate, while nodes using different domain IDs remain isolated.

By default:

```
$ export ROS_DOMAIN_ID=0 # Or unset
```

However, when multiple [ROS 2](#) systems coexist on the same network (e.g., in laboratories or classrooms), it is recommended to use a different domain ID to avoid interference between independent [ROS](#) graphs.

According to the official documentation, the recommended range for most Linux systems is:

```
0 - 101
```

since [DDS](#) internally maps domain IDs to [User Datagram Protocol \(UDP\)](#) ports, and certain values may conflict with ephemeral ports used by the operating system.

In this appendix, the following configuration will be used consistently:

```
$ export ROS_DOMAIN_ID=32
```

E.2 ROS 2 Security: Essential Concepts

[ROS 2](#) Security is implemented through [DDS](#)-Security plugins provided by the underlying [DDS](#) middleware. In this course, Fast [DDS](#) has been used through:

```
$ export RMW_IMPLEMENTATION=rmw_fastrtps_cpp
```

[DDS](#)-Security provides four main capabilities:

- Authentication of participants.
- Encryption of communications.
- Integrity verification.
- Access control policies.

Security is configured through certificates, keys, governance rules, and permissions files grouped inside a [keystore](#).

E.2.1 Security Enclave

A [Security Enclave](#) defines the security identity and permissions associated with one or more [ROS 2](#) nodes. Each enclave contains:

- Certificates.
- Private keys.
- Governance rules.
- Access control permissions.

Example:

```
sros2_ws/ws_keystore/enclaves/s2_cpp_pubsub/publisher_exe
```

E.2.2 Environment Variables

Security is enabled using the following environment variables:

```
$ export ROS_SECURITY_ENABLE=true  
$ export ROS_SECURITY_STRATEGY=Enforce  
$ export ROS_SECURITY_KEYSTORE=$SROS2_WS/ws_keystore
```



Variable	Purpose
ROS_SECURITY_ENABLE	Enables DDS-Security
ROS_SECURITY_STRATEGY	Defines security behavior
ROS_SECURITY_KEYSTORE	Specifies the keystore location

Table E.I: ROS 2 Security Environment Variables

Two strategies exist:

- **Permissive**: Allows insecure fallback.
- **Enforce**: Requires valid security configuration.

In this appendix, the following strategy will be used (as suggested in [ROS 2 Humble – Setting up security](#)):

```
$ export ROS_SECURITY_STRATEGY=Enforce
```

E.3 Setting Up the Security Workspace

The security workspace contains:

- Certificates.
- Keys.
- Security enclaves.
- Governance and permissions files.

E.3.1 Creating the Workspace

Set up your workspace by creating a directory (e.g., `~/<path_to_ws>/sros2_ws`), adding the `src` folder, pasting your previously-created packages, and building the workspace:

```
$ mkdir -p sros2_ws/src
$ cd sros2_ws/src
* Paste your packages here (e.g., s2_cpp_pubsub, s2_py_pubsub)
$ cd ..
$ colcon build
```

Then source the workspace:

```
$ source install/setup.bash
```

E.3.2 Creating the Keystore

Generate the keystore:

```
$ ros2 security create_keystore ws_keystore
```

This command creates the root trust infrastructure for the secure ROS graph.



E.3.3 Creating the Enclaves

For this demonstration, only two nodes will use encryption:

- `/s2_cpp_pubsub/publisher_exe`
- `/s2_py_pubsub/subscriber_exe`

Create the enclaves:

```
$ ros2 security create_enclave ws_keystore /s2_cpp_pubsub/publisher_exe
$ ros2 security create_enclave ws_keystore /s2_py_pubsub/subscriber_exe
```

E.4 Structure of the Keystore

The generated keystore contains the certificates, keys, permissions, and governance files required by DDS-Security.

```
ws_keystore/
|-- enclaves/
|-- private/
'-- public/
```

E.4.1 Public Directory

Contains the public certificate authorities:

- `ca.cert.pem`
- `identity_ca.cert.pem`
- `permissions_ca.cert.pem`

These files are publicly shared among trusted participants.

E.4.2 Private Directory

Contains the private keys associated with the certificate authorities:

- `ca.key.pem`
- `identity_ca.key.pem`
- `permissions_ca.key.pem`

These files must remain protected because they define the trust root of the secure system.

E.4.3 Enclaves Directory

Contains the security material associated with each enclave.

Example:

```
$ tree ws_keystore/enclaves/s2_cpp_pubsub/publisher_exe/
```



Important files:

File	Purpose
<code>cert.pem</code>	Participant certificate
<code>key.pem</code>	Participant private key
<code>permissions.xml</code>	Access control policy
<code>permissions.p7s</code>	Signed permissions file
<code>governance.xml</code>	Security governance rules
<code>governance.p7s</code>	Signed governance file

Table E.II: Important Files Inside an Enclave

E.5 Multi-Machine Configuration

For secure communication between two computers, both machines must use:

- The same `ROS_DOMAIN_ID`.
- The same middleware implementation.
- The same keystore.

A suggested configuration in `~/.bashrc` is:

```
## ROS2 CONFIGURATION ##
source /opt/ros/humble/setup.bash
source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash
source /usr/share/gazebo/setup.bash

alias rqt_service_caller='ros2 run rqt_service_caller rqt_service_caller'
alias rqt_image_view='ros2 run rqt_image_view rqt_image_view'

export ROS2_WS=~/.Documents/ros2_ws # Or /home/<user>/<path_to_ws>/<ws_name>
export SROS2_WS=~/.Documents/sros2_ws # Or /home/<user>/<path_to_ws>/<ws_name>

### ROS2 NETWORK AND SECURITY ###
export ROS_DOMAIN_ID=32
export RMW_IMPLEMENTATION=rmw_fastrtps_cpp

export ROS_SECURITY_ENABLE=true
export ROS_SECURITY_STRATEGY=Enforce
export ROS_SECURITY_KEYSTORE=$SROS2_WS/ws_keystore

# export ROS_SECURITY_ENCLAVE_OVERRIDE=/s2_cpp_pubsub/publisher_exe
```

The complete `ws_keystore` directory must be copied identically to both machines. Otherwise, DDS participant authentication will fail.

E.6 Examining Network Traffic

ROS 2 security can be experimentally verified using packet inspection tools such as `tcpdump`.



Because the following configuration is used:

```
$ export ROS_DOMAIN_ID=32
```

DDS communication is mapped approximately to the following UDP port range:

```
15400 - 15500
```

E.6.1 Inspecting Discovery Packets

DDS participant discovery packets can be inspected using (for `ROS_DOMAIN_ID=32`):

```
$ sudo tcpdump -X -i any udp port 15400
```

E.6.2 Inspecting Data Packets

DDS topic communication packets can be inspected using (for `ROS_DOMAIN_ID=32`):

```
$ sudo tcpdump -X -i any udp portrange 15401-15500
```

E.6.3 Security Implications

Without security enabled:

- Topic names may appear readable.
- Message payloads may appear in plain text.
- DDS participant information may be visible.

With DDS-Security enabled:

- DDS headers remain visible.
- Payloads appear encrypted and unreadable.
- Packet sizes typically increase.

E.7 Progressive Security Demonstration

The following sequence is recommended for laboratory sessions and demonstrations, or for testing stages into a secure deployment of your workspace with DDS-Security. On each stage, after setting up the package and the security configuration, you can inspect the network traffic to observe the differences between unsecured and secured communication.

Executing the Nodes

All the four nodes, from `s2_cpp_pubsub` and `s2_py_pubsub` packages, communicate through the topic `str_topic`. The nodes should be executed with the following basic command syntax, adjusting the security arguments as needed for each stage.



1. Python Publisher (without encryption):

```
$ source install/setup.bash
$ ros2 run s2_py_pubsub publisher_exe
```

2. Python Subscriber (encrypted):

```
$ source install/setup.bash
$ ros2 run s2_py_pubsub subscriber_exe --ros-args --enclave
  → /s2_py_pubsub/subscriber_exe
```

3. C++ Publisher (encrypted):

```
$ source install/setup.bash
$ ros2 run s2_cpp_pubsub publisher_exe --ros-args --enclave
  → /s2_cpp_pubsub/publisher_exe
```

4. C++ Subscriber (without encryption):

```
$ source install/setup.bash
$ ros2 run s2_cpp_pubsub subscriber_exe
```

E.7.1 Stage 1 — Single Machine Without Security

Objective: Observe standard DDS communication without encryption.

Expected behavior:

- Nodes communicate normally.
- DDS packets contain readable payloads.
- Automatic discovery occurs.

Suggested command previous to node execution:

```
$ unset ROS_SECURITY_ENABLE
```

Display unencrypted local ROS 2 graph

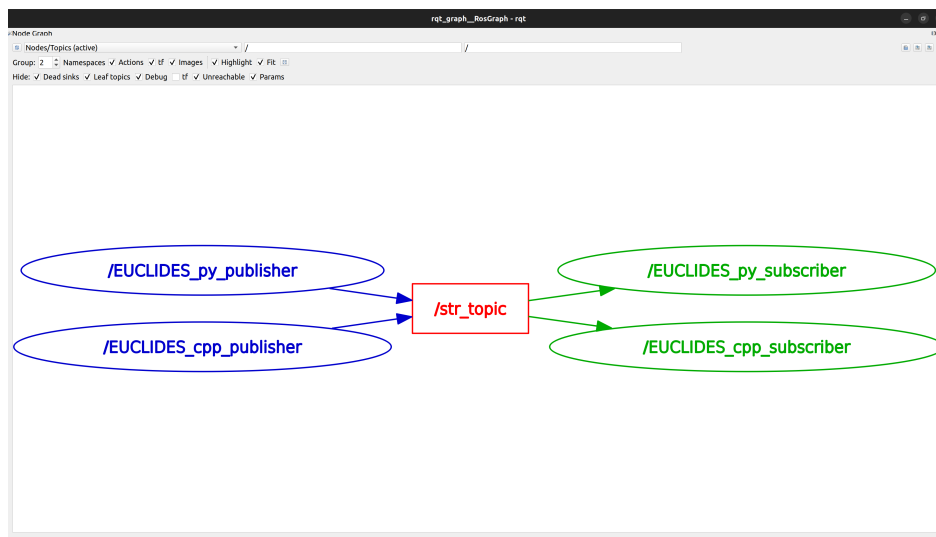


Figure E.1: ROS 2 graph in a single machine without security.

E.7.2 Stage 2 — Single Machine With Security

Objective: Enable DDS-Security locally.

Expected behavior:

- Nodes authenticate successfully.
- Payloads become encrypted.
- Shared memory transport may reduce visible UDP traffic.

Display encrypted discovery packets

```
eduardo@EUCALIDES:~/.../sros2_ws$ sudo tcpdump -X -i any udp port 15400
[sudo] contraseña para eduardo:
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
14:29:24.644686 lo In IP localhost.42843 > 239.255.0.1.15400: UDP, length 704
    0x0000: 4500 02dc 0dad 4000 0111 fa62 7f00 0001 E.....@....b....
    0x0010: efff 0001 a75b 3c28 02c8 71db 5254 5053 .....[<(.q.RTPS
    0x0020: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0030: 0901 0800 cd3b 026a f068 0324 1507 9c02 .....;j.h.$....
    0x0040: 0000 1000 0001 00c7 0001 00c2 0000 0000 .....
    0x0050: 0100 0000 0f80 1800 be8c 63ac 014d 6ef6 .....c..Mn.
    0x0060: eb88 483b ff01 01c2 0000 0000 0100 0000 ..H;.....
    0x0070: 0100 0000 0003 0000 1500 0400 0203 0000 .....
    0x0080: 1600 0400 010f 0000 5000 1000 be8c 63ac .....P.....c.
    0x0090: 014d 6ef6 eb88 483b 0000 01c1 3200 1800 .Mn...H;...2...
    0x00a0: 0100 0000 323c 0000 0000 0000 0000 0000 ....2<.....
    0x00b0: 0000 0000 c0a8 640b 3100 1800 1000 0000 .....d.1.....
    0x00c0: 333c 0000 5558 ea00 0000 0000 0000 0000 3<..UX.....
    0x00d0: 0000 0000 3100 1800 0100 0000 333c 0000 ....1.....3<..
    0x00e0: 0000 0000 0000 0000 0000 0000 c0a8 640b .....d.
    0x00f0: 0200 0800 1400 0000 0000 0000 5800 0400 .....X...
    0x0100: 3f0c ff0f 6200 2400 1d00 0000 2f73 325f ?...b.$...../s2_
    0x0110: 7079 5f70 7562 7375 622f 7375 6273 6372 py_pubsub/subscr
    0x0120: 6962 6572 5f65 7865 0000 0000 2c00 2c00 iber_exe.....,.
    0x0130: 2600 0000 656e 636c 6176 653d 2f73 325f &...enclave=/s2_
    0x0140: 7079 5f70 7562 7375 622f 7375 6273 6372 py_pubsub/subscr
    0x0150: 6962 6572 5f65 7865 3b00 0000 5900 2800 iber_exe;...Y.(.
    0x0160: 0100 0000 1100 0000 5041 5254 4943 4950 .....PARTICIP
    0x0170: 414e 545f 5459 5045 0000 0000 0700 0000 ANT_TYPE.....
    0x0180: 5349 4d50 4c45 0000 0110 c400 1400 0000 SIMPLE.....
    0x0190: 4444 533a 4175 7468 3a50 4b49 2d44 483a DDS:Auth:PKI-DH:
    0x01a0: 312e 3000 0400 0000 0c00 0000 6464 732e 1.0.....dds.
    0x01b0: 6365 7274 2e73 6e00 2300 0000 2f43 4e3d cert.sn.#.../CN=
    0x01c0: 5c2f 7332 5f70 795f 7075 6273 7562 5c2f \s2_py_pubsub\
    0x01d0: 7375 6273 6372 6962 6572 5f65 7865 0000 subscriber_exe..
```



```

0x01e0: 0e00 0000 6464 732e 6365 7274 2e61 6c67 ....dds.cert.alg
0x01f0: 6f00 0000 0d00 0000 4543 4453 412d 5348 o.....ECDSA-SH
0x0200: 4132 3536 0000 0000 0a00 0000 6464 732e A256.....dds.
0x0210: 6361 2e73 6e00 0000 0c00 0000 2f43 4e3d ca.sn...../CN=
0x0220: 7372 6f73 3243 4100 0c00 0000 6464 732e sros2CA.....dds.
0x0230: 6361 2e61 6c67 6f00 0d00 0000 4543 4453 ca.algo.....ECDS
0x0240: 412d 5348 4132 3536 0000 0000 0000 0000 A-SHA256.....
0x0250: 0210 7800 1b00 0000 4444 533a 4163 6365 ..x.....DDS:Acce
0x0260: 7373 3a50 6572 6d69 7373 696f 6e73 3a31 ss:Permissions:1
0x0270: 2e30 0000 0200 0000 0f00 0000 6464 732e .0.....dds.
0x0280: 7065 726d 5f63 612e 736e 0000 0c00 0000 perm_ca.sn.....
0x0290: 2f43 4e3d 7372 6f73 3243 4100 1100 0000 /CN=sros2CA.....
0x02a0: 6464 732e 7065 726d 5f63 612e 616c 676f dds.perm_ca.algo
0x02b0: 0000 0000 0d00 0000 4543 4453 412d 5348 .....ECDSA-SH
0x02c0: 4132 3536 0000 0000 0000 0000 0510 0800 A256.....
0x02d0: 0700 0080 0600 0080 0100 0000 .....

```

Display encrypted data packets

```

eduardo@EUCLIDES2:~/.../sros2_ws$ sudo tcpdump -X -i any udp portrange 15401-15500
[sudo] contraseña para eduardo:
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
14:32:42.287181 lo In IP localhost.42843 > localhost.15412: UDP, length 288
    0x0000: 4500 013c e384 4000 4011 582a 7f00 0001 E.<..@.X*....
    0x0010: 7f00 0001 a75b 3c34 0128 ff3b 5254 5053 .....[<4.(;RTPS
    0x0020: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0030: 3301 1400 0000 0003 a63b 9df2 675d e6ee 3.....;..g]..
    0x0040: 97fb 62d4 89b7 dd74 0c01 1400 0000 0000 ..b....t.....
    0x0050: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0060: 0e01 0c00 cb51 07d8 5578 531c 458a a73a .....Q..UxS.E...
    0x0070: 3101 1400 0000 0004 d4ed 03b9 92af e9da 1.....
    0x0080: cfe8 c8e8 c25f c7cd 3001 1000 0000 000c ....._..0.....
    0x0090: 19ad 9599 5e81 8b73 fddd 7068 3201 1400 ....^..s..ph2...
    0x00a0: ce38 7a7a 33d4 6331 30e2 7b0f 5f06 2525 .8zz3.c10.{._.%%
    0x00b0: 0000 0000 3101 1400 0000 0004 d4ed 03b9 ....1.....
    0x00c0: 92af e9da bc7e 5cdd 46d1 6c2c 3001 3c00 .....~\..F..l,0.<.
    0x00d0: 0000 0038 a179 9726 94db cd88 6873 a001 ..8.y.&....hs..
    0x00e0: 1197 9a09 f229 8116 7ca1 ae71 3427 3488 .....)|..q4'4.
    0x00f0: 9629 389d 735d 434d 73aa 344d 294b 82c8 .)8.s]CMs.4M)K..
    0x0100: cd2a acdb 2884 f285 f9b8 3538 3201 1400 .*..(....582...
    0x0110: f161 fdc3 ba9e 70c4 5f32 169a d41b af7d .a...p._2.....}
    0x0120: 0000 0000 3401 1400 7570 3935 a0a1 2904 ....4...up95...}
    0x0130: 715b b19e 20b6 3bd9 0000 0000 q[....;.....

```

E.7.3 Stage 3 — Two Machines Without Security

Objective: Observe [DDS](#) traffic through a physical network.

Expected behavior:

- [DDS](#) packets traverse Ethernet or WiFi.
- Payloads remain readable.



- Discovery occurs across machines.

Display unencrypted ROS2 graph network

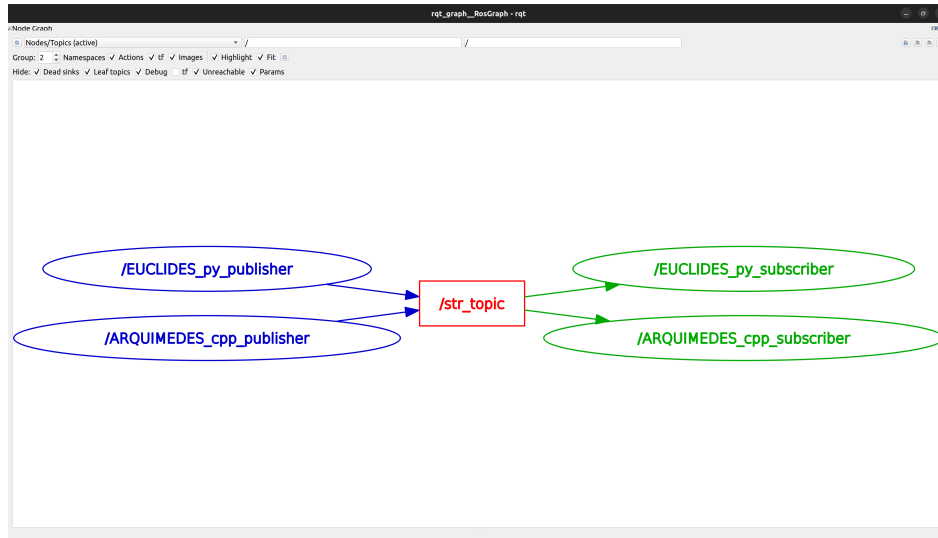


Figure E.2: ROS 2 graph in a network of machines without security.

E.7.4 Stage 4 — Two Machines With SROS2

Objective: Demonstrate authenticated and encrypted communication between two machines.

Expected behavior:

- Participants authenticate using certificates.
- Payloads become encrypted.
- Unauthorized participants cannot communicate.

Display encrypted discovery packets

```
eduardo@EUCLIDES:~/.../sros2_ws$ sudo tcpdump -X -i any udp port 15400
[sudo] contraseña para eduardo:
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
14:29:24.644686 lo In IP localhost.42843 > 239.255.0.1.15400: UDP, length 704
    0x0000: 4500 02dc 0dad 4000 0111 fa62 7f00 0001 E.....@....b....
    0x0010: efff 0001 a75b 3c28 02c8 71db 5254 5053 .....[<(.q.RTPS
    0x0020: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0030: 0901 0800 cd3b 026a f068 0324 1507 9c02 .....;j.h.$....
    0x0040: 0000 1000 0001 00c7 0001 00c2 0000 0000 .....
    0x0050: 0100 0000 0f80 1800 be8c 63ac 014d 6ef6 .....c..Mn.
    0x0060: eb88 483b ff01 01c2 0000 0000 0100 0000 ..H;.....
    0x0070: 0100 0000 0003 0000 1500 0400 0203 0000 .....
    0x0080: 1600 0400 010f 0000 5000 1000 be8c 63ac .....P.....c.
    0x0090: 014d 6ef6 eb88 483b 0000 01c1 3200 1800 .Mn...H;...2...
    0x00a0: 0100 0000 323c 0000 0000 0000 0000 0000 ....2<.....
    0x00b0: 0000 0000 c0a8 640b 3100 1800 1000 0000 .....d.1.....
    0x00c0: 333c 0000 5558 ea00 0000 0000 0000 0000 3<..UX.....
```



```

0x00d0: 0000 0000 3100 1800 0100 0000 333c 0000 ....1.....3<..
0x00e0: 0000 0000 0000 0000 0000 0000 c0a8 640b .....d.
0x00f0: 0200 0800 1400 0000 0000 0000 5800 0400 .....X...
0x0100: 3f0c ff0f 6200 2400 1d00 0000 2f73 325f ?...b.$....s2_
0x0110: 7079 5f70 7562 7375 622f 7375 6273 6372 py_pubsub/subscr
0x0120: 6962 6572 5f65 7865 0000 0000 2c00 2c00 iber_exe.....,.,.
0x0130: 2600 0000 656e 636c 6176 653d 2f73 325f &...enclave=s2_
0x0140: 7079 5f70 7562 7375 622f 7375 6273 6372 py_pubsub/subscr
0x0150: 6962 6572 5f65 7865 3b00 0000 5900 2800 iber_exe;...Y.(.
0x0160: 0100 0000 1100 0000 5041 5254 4943 4950 .....PARTICIP
0x0170: 414e 545f 5459 5045 0000 0000 0700 0000 ANT_TYPE.....
0x0180: 5349 4d50 4c45 0000 0110 c400 1400 0000 SIMPLE.....
0x0190: 4444 533a 4175 7468 3a50 4b49 2d44 483a DDS:Auth:PKI-DH:
0x01a0: 312e 3000 0400 0000 0c00 0000 6464 732e 1.0.....dds.
0x01b0: 6365 7274 2e73 6e00 2300 0000 2f43 4e3d cert.sn.#.../CN=
0x01c0: 5c2f 7332 5f70 795f 7075 6273 7562 5c2f \s2_py_pubsub\
0x01d0: 7375 6273 6372 6962 6572 5f65 7865 0000 subscriber_exe..
0x01e0: 0e00 0000 6464 732e 6365 7274 2e61 6c67 ....dds.cert.alg
0x01f0: 6f00 0000 0d00 0000 4543 4453 412d 5348 o.....ECDSA-SH
0x0200: 4132 3536 0000 0000 0a00 0000 6464 732e A256.....dds.
0x0210: 6361 2e73 6e00 0000 0c00 0000 2f43 4e3d ca.sn...../CN=
0x0220: 7372 6f73 3243 4100 0c00 0000 6464 732e sros2CA.....dds.
0x0230: 6361 2e61 6c67 6f00 0d00 0000 4543 4453 ca.algo.....ECDS
0x0240: 412d 5348 4132 3536 0000 0000 0000 0000 A-SHA256.....
0x0250: 0210 7800 1b00 0000 4444 533a 4163 6365 ..x.....DDS:Acce
0x0260: 7373 3a50 6572 6d69 7373 696f 6e73 3a31 ss:Permissions:1
0x0270: 2e30 0000 0200 0000 0f00 0000 6464 732e .0.....dds.
0x0280: 7065 726d 5f63 612e 736e 0000 0c00 0000 perm_ca.sn.....
0x0290: 2f43 4e3d 7372 6f73 3243 4100 1100 0000 /CN=sros2CA.....
0x02a0: 6464 732e 7065 726d 5f63 612e 616c 676f dds.perm_ca.algo
0x02b0: 0000 0000 0d00 0000 4543 4453 412d 5348 .....ECDSA-SH
0x02c0: 4132 3536 0000 0000 0000 0000 0510 0800 A256.....
0x02d0: 0700 0080 0600 0080 0100 0000 .....

```

Display encrypted data packets

```

eduardo@EUCLIDES2:~/.../sros2_ws$ sudo tcpdump -X -i any udp portrange 15401-15500
[sudo] contraseña para eduardo:
tcpdump: data link type LINUX_SLL2
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
14:32:42.287181 lo In IP localhost.42843 > localhost.15412: UDP, length 288
    0x0000: 4500 013c e384 4000 4011 582a 7f00 0001 E..<..@.@.X*....
    0x0010: 7f00 0001 a75b 3c34 0128 ff3b 5254 5053 ....[<4.(.;RTPS
    0x0020: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0030: 3301 1400 0000 0003 a63b 9df2 675d e6ee 3.....;..g]..
    0x0040: 97fb 62d4 89b7 dd74 0c01 1400 0000 0000 ..b....t.....
    0x0050: 0203 010f be8c 63ac 014d 6ef6 eb88 483b .....c..Mn...H;
    0x0060: 0e01 0c00 cb51 07d8 5578 531c 458a a73a ....Q..UxS.E.:
    0x0070: 3101 1400 0000 0004 d4ed 03b9 92af e9da 1.....
    0x0080: cfe8 c8e8 c25f c7cd 3001 1000 0000 000c ....._..0.....
    0x0090: 19ad 9599 5e81 8b73 fddd 7068 3201 1400 ....^..s..ph2...
    0x00a0: ce38 7a7a 33d4 6331 30e2 7b0f 5f06 2525 .8zz3.c10.{..%

```



```

0x00b0:  0000 0000 3101 1400 0000 0004 d4ed 03b9  ....1.....
0x00c0:  92af e9da bc7e 5cdd 46d1 6c2c 3001 3c00  ....~\.F.l,0.<.
0x00d0:  0000 0038 a179 9726 94db cd88 6873 a001  ...8.y.&....hs..
0x00e0:  1197 9a09 f229 8116 7ca1 ae71 3427 3488  ....)|..q4'4.
0x00f0:  9629 389d 735d 434d 73aa 344d 294b 82c8  .)8.s]CMs.4M)K..
0x0100:  cd2a acdb 2884 f285 f9b8 3538 3201 1400  .*..(.....582...
0x0110:  f161 fdc3 ba9e 70c4 5f32 169a d41b af7d  .a....p._2.....}
0x0120:  0000 0000 3401 1400 7570 3935 a0a1 2904  ....4...up95...).
0x0130:  715b b19e 20b6 3bd9 0000 0000          q[....;.....

```

E.8 Additional Notes

- Both machines must use the same `ROS_DOMAIN_ID`.
- Both machines must use identical copies of the keystore.
- The same middleware implementation should be used on all machines.
- Fast `DDS` may internally use shared memory transport when nodes execute on the same computer.
- Firewall rules may block `DDS` discovery traffic.
- For classroom or laboratory environments, it is recommended to define a fixed `ROS_DOMAIN_ID` for all students/participants/teams.

